Literature Review: Radar-Guided Camera Labeling for Maritime Dataset Construction

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1 Introduction

The autonomous surface vessel (ASV) industry has expanded rapidly over the past decade, driven by applications in maritime logistics, oceanographic research, coastal surveillance, and harbor operations. Safe autonomous navigation in these environments demands robust object detection systems capable of identifying vessels, buoys, debris, swimmers, and other obstacles across a wide range of visual conditions—from clear-sky open ocean to fog-obscured harbors and rain-swept coastal waterways. Camera-based deep learning detectors, which have achieved remarkable performance in automotive settings [1, 2], are the natural candidate for providing the high-resolution visual perception that maritime autonomy requires. However, their performance is fundamentally limited by the scale and quality of available training data. As documented extensively in the autonomous driving literature, each successive generation of benchmark datasets—KITTI [3], MS COCO [4], nuScenes [2], and the Waymo Open Dataset [1]—has enabled corresponding leaps in perception algorithm quality. The maritime domain has not benefited from an equivalent data ecosystem: the largest maritime detection datasets remain one to two orders of magnitude smaller than their automotive counterparts in labeled instances, multi-sensor scenes, and environmental diversity [5–7]. Manual annotation, which requires domain expertise and is two to four times slower per image than automotive labeling [8, 9], constitutes the principal bottleneck to closing this gap.

Radar-guided camera labeling offers a compelling alternative. The core idea is to exploit the complementary strengths of X-band marine radar and optical cameras: radar provides reliable, weather-invariant detection of surface targets with range and bearing information, while the camera provides the high-resolution imagery needed for training visual object detectors. By projecting radar detections into the camera image plane using calibrated extrinsic and intrinsic parameters, bounding box annotations can be generated automatically without human intervention. This cross-modal label transfer operates continuously during vessel operations, converting every radar-camera frame pair into a potential training example and thereby scaling dataset construction from weeks of manual effort to hours of autonomous data collection. The concept parallels the LiDAR-to-camera label transfer that has been explored in autonomous driving [10, 11], but the maritime setting introduces unique challenges: X-band marine radar produces two-dimensional plan position indicator (PPI) images rather than three-dimensional point clouds, operates at ranges of hundreds of meters to several nautical miles rather than tens to hundreds of meters, and must contend with sea clutter, multipath propagation, and platform motion that have no direct automotive analogue [12, 13].

Synthesizing the literature across the seven sections that follow, this review identifies four cross-cutting research gaps that collectively frame the open problems in radar-guided camera labeling for maritime dataset construction:

1. **X-band marine radar versus 77 GHz automotive radar transfer.** The overwhelming majority of radar-camera fusion and calibration research has been developed for 77 GHz automotive radar, which produces sparse point clouds or range-Doppler-azimuth tensors from electronically steered phased arrays at update rates of 10–20 Hz [14, 15]. X-band marine radar operates in a fundamentally different regime: mechanically rotating antennas produce 360-degree PPI images at 0.5–3 s rotation periods, with range resolutions of 5–30 m and no elevation measurement. Architectural assumptions embedded in state-of-the-art automotive fusion methods—BEV grid construction from point clouds, Doppler-based velocity estimation, frustum association

from 3D detections—do not directly transfer to the marine radar data format. Bridging this gap requires adapting both calibration procedures and fusion architectures to the PPI image modality.

2. **Wide field-of-view projection accuracy at maritime ranges.** Projecting radar detections into camera pixel coordinates is mathematically straightforward under the standard pinhole model, but maritime operating conditions amplify every source of error in the projection pipeline [14, 16]. At ranges of 500 m to 10 km, a sub-degree azimuth error in the radar-camera calibration translates to tens or hundreds of pixels of horizontal displacement. Wide field-of-view cameras, commonly used on maritime platforms for panoramic coverage, introduce nonlinear distortion that must be accurately modeled [17, 18]. The absence of elevation information in 2D marine radar creates a fundamental vertical ambiguity, and the lack of validated size priors for maritime targets at extended range makes bounding box height and width estimation unreliable. Characterizing range-dependent projection accuracy and developing robust bounding box sizing strategies for the maritime domain remain open problems.

3. **Continuous auto-labeling scalability.** Radar-projected labels are inherently noisy: angular quantization, beam spreading, sea clutter false alarms, and calibration drift all degrade annotation quality [19, 20]. The machine learning community has developed powerful tools for learning from noisy labels—including pseudo-labeling [21, 22], teacher-student frameworks [23], co-training [24], and noise-aware loss functions [25, 26]—but these methods have been validated primarily on classification tasks with synthetic noise. Their application to object detection with radar-generated bounding box noise, and their behavior under continuous self-training loops where the label quality evolves over iterations, has not been systematically investigated in the maritime context.

4. **Systematic versus random label noise.** Perhaps the most critical gap is the mismatch between the noise models assumed by existing noisy-label learning methods and the actual error structure of radar-projected labels. Standard noise models treat label corruption as independent and identically distributed (i.i.d.) across samples—symmetric, asymmetric, or instance-dependent, but always statistically independent [27, 28]. Radar projection errors, by contrast, are geometry-dependent: azimuth localization error grows with range, bounding box sizing degrades at distance, and calibration offsets bias all projections in the same direction. This systematic, spatially correlated noise structure means that standard noise-correction techniques may fail to account for the dominant error sources. Developing geometry-aware noise models that condition the noise distribution on the radar-camera geometric configuration—target range, bearing, calibration state—is an essential theoretical foundation for robust radar-guided labeling.

The remainder of this review is organized as follows. Section 2 surveys the landscape of maritime object detection datasets, quantifies the order-of-magnitude gap between maritime and automotive data resources, and motivates the need for automated labeling. Section 3 reviews the mathematical framework and practical methods for radar-camera sensor registration, with attention to the gap between the well-studied 77 GHz automotive setting and the largely unaddressed X-band marine radar domain. Section 4 traces the complete projection pipeline from radar polar coordinates to camera pixel coordinates, examines wide-FOV distortion modeling, and analyzes the error sources that limit projection fidelity at maritime ranges. Section 5 surveys automated and weakly-supervised dataset construction methods—data programming, pseudo-labeling, teacher-student training, cross-modal supervision, and noisy label learning—evaluating their applicability to radar-generated annotations. Section 6 examines radar-camera fusion architectures from the automotive domain and assesses their transferability to X-band marine radar, including the fundamental data format differences between automotive point clouds and marine PPI images. Section 7 reviews the signal processing chain underlying X-band marine radar detection, from CFAR algorithms and sea clutter models to target clustering and tracking, establishing the detection quality that drives downstream labeling accuracy. Finally, Section 8 evaluates edge computing platforms for onboard deployment of the labeling pipeline, demonstrating that continuous autonomous annotation aboard small surface vessels is computationally and energetically feasible with current hardware.

2 Maritime Object Detection Datasets and the Annotation Gap

The performance of modern object detection systems is fundamentally constrained by the scale, diversity, and quality of their training data. In the autonomous driving domain, this relationship has been demonstrated repeatedly: each successive generation of benchmark datasets—from KITTI [3] to nuScenes [2] to the Waymo Open Dataset [1]—has enabled corresponding advances in perception algorithms. Maritime autonomy, however, has not benefited from an equivalent data ecosystem. This chapter surveys the landscape of maritime object detection datasets, catalogs their scope and annotation methodology, and quantifies the order-of-magnitude gap that separates maritime data resources from their automotive counterparts. Understanding this gap is essential context for the broader literature review that follows, which examines radar-camera calibration, cross-modal projection, automated dataset construction, sensor fusion, X-band signal processing, and edge deployment—all components of a radar-guided camera labeling pipeline designed to close the maritime annotation deficit.

2.1 The Role of Large-Scale Annotated Datasets in Perception

The deep learning revolution in computer vision was enabled not only by advances in network architectures and computing hardware but also by the availability of large-scale, carefully annotated datasets. ImageNet [29], with its 1.28 million training images spanning 1,000 object categories, demonstrated that supervised pre-training on a sufficiently large corpus could yield feature representations that transferred effectively to downstream tasks. The PASCAL VOC challenge [30] established standardized evaluation protocols that allowed meaningful comparison across detection methods, while MS COCO [4] raised the bar with 330,000 images, 1.5 million object instances, and dense annotations including bounding boxes, segmentation masks, and captions. In the autonomous driving domain, this progression continued: KITTI [3] introduced multi-sensor benchmarks with LiDAR, stereo cameras, and GPS/IMU; nuScenes [2] scaled to 1,000 scenes with full 360-degree sensor coverage and 1.4 million 3D bounding box annotations; and the Waymo Open Dataset [1] reached 12.6 million LiDAR box labels and 11.8 million camera box labels across 1,150 driving scenes.

The impact of dataset scale on detector performance is well documented. Zhang et al. [9] note that large-scale, multi-scenario training data is an indispensable prerequisite for practical deployment of deep learning-based detection systems, and that the maritime domain lacks a widely accepted standardized benchmark of comparable scale. This observation motivates a systematic examination of what maritime datasets do exist, what they contain, and where the critical deficiencies lie.

2.2 A Catalog of Maritime Object Detection Datasets

Maritime detection datasets have emerged from three distinct operational contexts: shore-based surveillance systems monitoring harbors and coastal waters, onboard sensors carried by unmanned surface vehicles (USVs) or autonomous surface vehicles (ASVs), and multi-modal platforms integrating radar, LiDAR, or other active sensors alongside cameras. A fourth category encompasses specialized datasets that employ infrared, hyperspectral, or stereo imaging. The following subsections organize the literature accordingly.

2.2.1 Shore-Based Visual Datasets

Shore-based maritime datasets are typically collected from fixed camera installations overlooking harbors, canals, or coastal waterways. Their primary applications are maritime surveillance, vessel traffic monitoring, and port security.

MarDCT (2015). The Maritime Detection, Classification, and Tracking benchmark [31] was among the earliest publicly available maritime video datasets. Collected by the ARGOS automated surveillance system operating continuously in Venice, Italy, MarDCT comprises 6,742 images and 16 video sequences at 800×240 resolution. Ground truth is provided as bounding boxes for boat detection. While pioneering in its public availability, the dataset's small scale and limited resolution restrict its utility for training modern deep detectors.

Singapore Maritime Dataset (SMD) (2017). Prasad et al. [32] released the Singapore Maritime Dataset alongside a comprehensive survey of video processing for maritime object detection. The dataset contains 81 high-definition (1920×1080) video sequences captured from both onshore (32 sequences) and onboard (4 sequences) viewpoints in Singapore waters, using visual-optical and near-infrared sensors. Annotations cover 10 maritime object classes across diverse environmental conditions including haze, rain, sunrise, midday, and nighttime. SMD established an important early benchmark for evaluating detection and tracking algorithms in tropical maritime environments.

SeaShips (2018). Shao et al. [33] constructed a large-scale, precisely annotated dataset from approximately 10,080 real-world video segments captured by a deployed coastline surveillance system. SeaShips contains 31,455 images covering six common vessel types: ore carrier, bulk cargo carrier, general cargo ship, container ship, fishing boat, and passenger ship. Images were carefully selected to represent variations in scale, hull visibility, illumination, viewpoint, background complexity, and occlusion. Each image is annotated with ship type labels and high-precision bounding boxes. At the time of its release, SeaShips was among the largest publicly available maritime detection datasets and remains widely used as a benchmark.

MCShips (2020). Zheng and Zhang [34] introduced MCShips, a dataset of 14,709 annotated images collected via web crawlers and search engines, spanning 13 fine-grained ship categories (6 warship types and 7 civilian vessel types). Unlike surveillance-derived datasets, MCShips captures a wide diversity of imaging conditions, perspectives, and resolutions found in publicly available online imagery. Each image is annotated with bounding boxes and fine-grained class labels, supporting both detection and classification tasks.

ABOShips (2021). Iancu et al. [35] constructed the ABOShips dataset from video sequences recorded in the Finnish Archipelago using a camera mounted on a waterbus. The dataset comprises 9,880 images containing 41,967 annotated object instances across 9 vessel categories plus seamarks and miscellaneous floaters. A two-stage annotation process was employed: initial manual labeling followed by automated tracking (CSRT tracker) to identify and correct labeling inconsistencies. Benchmark evaluations with Faster R-CNN, R-FCN, SSD, and EfficientDet established baseline performance levels, with Faster R-CNN (Inception-ResNet v2 backbone) achieving the highest overall accuracy.

MCMOD (2023). Sun et al. [36] released the Multi-Category Maritime Object Detection dataset, consisting of 16,166 images with 98,590 annotated objects across 10 maritime categories including fishing vessels, speedboats, engineering ships, cargo ships, yachts, sailboats, buoys, rafts, cruise ships, and an "others" class. Three onshore cameras operating continuously in Hainan, China, captured high-resolution (1920 × 1080) video under varied conditions including fog, rain, and evening illumination. The cameras' rotation and zoom capabilities produced diverse scenes, and the dataset's emphasis on small-scale distant targets addresses a persistent challenge in maritime detection.

KOLOMVERSE (2024). Nanda et al. [5] introduced the Korea Open Large-Scale Image Dataset for Object Detection in the Maritime Universe, representing the largest publicly available maritime detection dataset to date. Drawn from 5,845 hours of video data captured across 21 territorial waters of South Korea, KOLOMVERSE contains 186,419 4K-resolution (3840 × 2160) images with 732,039 annotated object instances across five classes: ship (393,936), buoy (34,080), fishnet buoy (95,815), lighthouse (60,362), and wind farm (147,846). The dataset is partitioned into training (149,175), validation (18,643), and test (18,601) splits. KOLOMVERSE's scale and resolution represent a significant step forward for maritime detection, though its five-class taxonomy and shore-based perspective limit its applicability to fine-grained classification and onboard perception tasks.

2.2.2 USV/ASV-Mounted Datasets

Datasets collected from the perspective of an unmanned or autonomous surface vehicle present unique challenges not captured by shore-based imagery: low camera elevation above the waterline, dynamic horizon orientation due to wave-induced platform motion, frequent spray and lens contamination, and the need to detect obstacles at close range for collision avoidance.

MaSTr1325 (2019). Bovcon et al. [8] released a maritime semantic segmentation training dataset containing 1,325 diverse images captured over two years by a USV operating in the Gulf of Koper, Slovenia. Each image is annotated at pixel level with three semantic classes: sea, sky, and environment (encompassing all obstacles and shoreline). While the three-class taxonomy is coarse, the dataset was instrumental in training early deep segmentation models for USV obstacle detection. The annotation procedure required approximately twenty minutes per image, underscoring the labor intensity of pixel-level maritime labeling.

FloW (2021). Cheng et al. [37] presented FloW, the first dataset targeting floating waste detection in inland waterways, accepted at ICCV 2021. FloW comprises two subdatasets: FloW-Img, an image collection with bounding box annotations for floating debris, and FloW-RI, a multimodal sub-dataset containing synchronized millimeter-wave radar data and camera images. The inclusion of radar data makes FloW one of the earliest maritime-adjacent datasets to provide cross-modal supervision signals. Baseline experiments demonstrated that floating waste detection remains challenging due to small target size, strong water-surface reflections, and bankside clutter.

MassMIND (2023). Nirgudkar et al. [38] contributed the Massachusetts Maritime Infrared Dataset, containing 2,916 long-wave infrared (LWIR) images captured over two years in and around Boston Harbor. The dataset is segmented using both instance and semantic segmentation into seven classes, making it the first maritime dataset to provide fine-grained LWIR segmentation labels for both living (humans, marine wildlife) and non-living (vessels, structures) obstacles. Evaluation with UNet, PSPNet, and DeepLabv3 architectures established baseline segmentation performance for thermal maritime imagery.

LaRS (2023). Žust et al. [39] released the first maritime panoptic obstacle detection benchmark, accepted at ICCV 2023. LaRS contains over 4,000 per-pixel labeled keyframes with nine preceding temporal context frames each, totaling more than 40,000 frames. Annotations span 8 "thing" classes, 3 "stuff" classes, and 19 global scene attributes. The dataset features the largest diversity among maritime datasets in recording locations (lakes, rivers, and seas), scene types, obstacle classes, and acquisition conditions. An online evaluation server with results from 27 segmentation methods provides a standardized benchmark. LaRS represents the current state of the art for USV-perspective semantic and panoptic segmentation evaluation.

WaterScenes (2024). Zhang et al. [40] introduced WaterScenes, the first multi-task 4D radar-camera fusion dataset for autonomous driving on water surfaces. Collected by a USV equipped with a 4D millimeter-wave radar and a monocular camera, the dataset comprises 54,120 synchronized image-radar frame pairs covering more than 200,000 annotated objects. Annotations support five tasks: object detection, instance segmentation, semantic segmentation, free-space segmentation, and waterline segmentation. Environmental diversity spans daytime, nightfall, and nighttime conditions; normal, dim, and strong lighting; sunny, overcast, rainy, and snowy weather; and four waterway types (river, lake, canal, moat). WaterScenes is notable for being the first maritime dataset to provide both pixel-level camera annotations and point-level radar annotations, enabling direct evaluation of radar-camera fusion algorithms in the maritime domain.

SeePerSea (2024). Jeong et al. [41] released the first publicly accessible multi-modal perception dataset for in-water obstacle detection from ASVs, featuring synchronized LiDAR point clouds and RGB images. Collected over four years across locations in the United States, Barbados, and South Korea, SeePerSea contains 11,561 annotated frames with navigation-oriented three-class labels (ship, buoy, other). The dataset is significant as the first maritime LiDAR-camera dataset with object labels across both modalities, supporting evaluation of cross-modal fusion for aquatic obstacle detection.

2.2.3 Multi-Modal Radar/LiDAR/Camera Datasets

The most recent generation of maritime datasets integrates multiple active and passive sensing modalities, reflecting the recognition that robust maritime perception requires sensor fusion. These datasets are also the most directly relevant to radar-guided camera labeling, as they provide synchronized radar and camera data that can serve as ground truth for evaluating cross-modal projection and automatic annotation pipelines.

Pohang Canal Dataset (2023). Chung et al. [42] released a comprehensive multi-modal maritime dataset collected along a 7.5 km route through the Pohang Canal, inner and outer ports, and near-coastal areas in South Korea. The sensor suite includes stereo cameras, an infrared camera, an omnidirectional camera, three LiDARs, a marine radar, GPS, and an attitude heading reference system. Data were collected under diverse weather and visibility conditions. The Pohang Canal Dataset is among the most sensor-rich maritime collections available, though it is primarily designed for navigation and SLAM rather than object detection, and its annotation density for detection tasks is limited.

M^{2}SODAI (2023). Jang et al. [43] presented a multi-modal maritime object detection dataset combining RGB and hyperspectral image sensors, accepted at NeurIPS 2023 Datasets and Benchmarks track. The dataset provides synchronized aerial image pairs with bounding box annotations across multiple maritime object categories. Hyperspectral imaging (HSI), which captures over 100 spectral channels in the visible and near-infrared range, enables extraction of material-specific signatures that can discriminate maritime objects from sea-surface clutter even when visual appearance is ambiguous. M^{2}SODAI is the first maritime dataset to incorporate hyperspectral sensing, addressing the false-positive challenge inherent in single-modality visual detection over dynamic water surfaces.

PoLaRIS (2024). Choi et al. [6] released the PoLaRIS (Pohang Canal Object Detection and Tracking Dataset in Maritime Environments) dataset, accepted at ICRA 2025. PoLaRIS provides approximately 360,000 labeled images from RGB and thermal infrared cameras, along with point-wise annotations from radar and LiDAR. It is the first dataset offering multi-modal annotations specifically tailored to maritime object detection and tracking, including tracking ground truth for dynamic objects with bounding boxes projected across modalities. The dataset includes annotations for objects as small as 10×10 pixels, reflecting the prevalence of small, distant targets in maritime environments. With approximately 190,000 detection-grade bounding box annotations, PoLaRIS represents the most densely annotated multi-sensor maritime dataset available.

MOANA (2024). Jang et al. [7] introduced the Multi-Radar Dataset for Maritime Odometry and Autonomous Navigation Application, integrating short-range LiDAR, medium-range W-band radar, and long-range X-band radar into a unified multi-scale sensing framework. The dataset comprises 7 sequences collected from diverse coastal regions with varying navigation difficulty. MOANA is the first maritime dataset to combine multiple radar frequency bands, enabling research on multi-scale maritime perception from close-range obstacle detection to long-range situational awareness. Its primary focus is odometry estimation, SLAM, and place recognition rather than object detection, and the limited number of sequences constrains its statistical power for training data-hungry detection models.

2.2.4 Multi-Spectral and Specialized Datasets

Several datasets address maritime perception through non-standard imaging modalities or specialized acquisition configurations.

VAIS (2015). Zhang et al. [44] released the Visible and Infrared Ship dataset, the first publicly available collection of paired visible and thermal infrared maritime imagery. VAIS contains over 1,000 paired RGB-IR images across six vessel categories (merchant, sailing, passenger, medium-other, tug, and small), all acquired from pier-side vantage points. The simultaneous dual-band acquisition enables research on cross-spectral recognition and day-night vessel classification, though the dataset's modest scale limits its applicability for training deep networks.

MARVEL (2016). Gundogdu et al. [45] constructed a large-scale maritime vessel image dataset by harvesting approximately 2 million user-uploaded images and metadata from an online ship-spotting community. Images are categorized into 26 superclasses constructed from 109 fine-grained vessel type labels, with additional attributes including vessel identity, photograph category, and year of construction. MARVEL supports vessel classification, verification, retrieval, and recognition tasks. However, the dataset's web-sourced provenance means images are captured under uncontrolled conditions without standardized annotations, and no bounding box or segmentation labels are provided, limiting its utility for detection tasks.

MODS (2022). Bovcon et al. [46] released a USV-oriented object detection and obstacle segmentation benchmark containing approximately 81,000 stereo images synchronized with onboard IMU measurements, with over 60,000 annotated objects. MODS addresses two major perception tasks—maritime object detection and maritime obstacle segmentation—and introduces a standardized evaluation protocol that measures detection accuracy in terms meaningful for practical USV navigation. Benchmark results from 19 state-of-the-art detection and segmentation methods provide a comprehensive performance landscape for USV perception.

2.3 Comprehensive Dataset Comparison

Table 1 presents a systematic comparison of the maritime datasets surveyed in this chapter. The table reveals several patterns: (1) dataset scale has grown substantially over the past decade, from thousands to hundreds of thousands of images; (2) the majority of datasets remain single-modality visual collections; (3) multi-modal datasets with radar or LiDAR are a recent development (2023–2024) and remain small by comparison; and (4) annotation granularity varies widely, from image-level class labels to pixel-level panoptic segmentation.

Several observations emerge from this comparison. First, KOLOMVERSE [5] is the only maritime dataset to exceed 100,000 detection-annotated images, yet it covers only five object classes from a shore-based perspective. Second, WaterScenes [40] is the only dataset providing both dense camera annotations and radar point annotations from a vessel-mounted platform, making it uniquely relevant for radar-camera fusion research. Third, the multi-sensor datasets most suitable for evaluating radar-guided labeling—Pohang [42], PoLaRIS [6], and MOANA [7]—either lack dense detection annotations or comprise only a handful of sequences.

2.4 The Automotive Benchmark: Scale and Infrastructure

To contextualize the maritime annotation gap, it is instructive to examine the scale of automotive perception benchmarks that have driven progress in self-driving vehicle perception. Table 2 summarizes the key automotive datasets.

The automotive ecosystem benefits from a virtuous cycle: large datasets enable better models, which attract more investment, which funds further data collection and annotation. The Waymo Open Dataset alone contains 12.6 million LiDAR 3D bounding box labels with tracking identifiers across 1,150 scenes, each spanning 20 seconds of driving

Table 1: Comprehensive comparison of maritime object detection datasets. Annotation types: BBox = bounding box, Seg = semantic segmentation, Inst = instance segmentation, Pan = panoptic segmentation, PW = point-wise, Trk = tracking ID.

Dataset	Year	Sensors	Images	Objects	Classes	Annotation
MarDCT [31]	2015	RGB	6,742	—	Boat	BBox
VAIS [44]	2015	RGB + IR	2,000	1,242	6	BBox
MARVEL [45]	2016	RGB (web)	2,000,000	—	26	Class label
SMD [32]	2017	RGB + NIR	81 vid	—	10	BBox (video)
SeaShips [33]	2018	RGB	31,455	40,077	6	BBox
MaSTr1325 [8]	2019	RGB + IMU	1,325	—	3	Seg (pixel)
MCShips [34]	2020	RGB (web)	14,709	14,709	13	BBox
ABOShips [35]	2021	RGB	9,880	41,967	9+	BBox
FloW [37]	2021	RGB + mmW radar	2,000+	6,000+	Waste	BBox + PW
MODS [46]	2022	Stereo + IMU	81,000	60,000+	Multi	BBox + Seg
MassMIND [38]	2023	LWIR	2,916	—	7	Seg + Inst
LaRS [39]	2023	RGB	40,000+	—	8+3	Pan (pixel)
M^{2}SODAI [43]	2023	RGB + HSI	5,764+	—	Multi	BBox
Pohang [42]	2023	Multi (7+)	—	—	—	Nav. labels
MCMOD [36]	2023	RGB	16,166	98,590	10	BBox
KOLOMVERSE [5]	2024	RGB	186,419	732,039	5	BBox
WaterScenes [40]	2024	RGB + 4D radar	54,120	200,000+	Multi	BBox + Seg + PW
PoLaRIS [6]	2024	RGB+IR+LiDAR+R	360,000	∼190k	Multi	BBox + PW + Trk
SeePerSea [41]	2024	RGB + LiDAR	11,561	—	3	BBox + PW
MOANA [7]	2024	Multi-radar+L	7 seq.	—	—	Nav. labels

Table 2: Major automotive perception benchmark datasets for comparison with maritime equivalents.

Dataset	Year	Sensors	Images	Labels	Classes	Annotation
ImageNet [29]	2012	RGB	1,281,167	—	1,000	Class label
KITTI [3]	2012	Stereo+LiDAR	14,999	200,000+	8	3D BBox
MS COCO [4]	2014	RGB	330,000	1,500,000	80	BBox + Seg
nuScenes [2]	2020	6C+5R+1L	1,400,000	1,400,000	23	3D BBox + Trk
Waymo [1]	2020	5C+5L	—	12,600,000	4	3D BBox + Trk

with well-synchronized and calibrated LiDAR and camera data [1]. nuScenes provides 1.4 million 3D bounding box annotations across 1,000 scenes with full 360-degree sensor coverage including six cameras, five radars, and one LiDAR [2]. The automotive radar-camera datasets RADIATE [47] and the Oxford Radar RobotCar Dataset [48] further demonstrate the automotive community's investment in multi-modal data collection.

2.5 Quantifying the Maritime–Automotive Gap

Table 3 presents a direct quantitative comparison between the best-resourced maritime and automotive datasets across several key metrics. The gaps are stark and consistent across all dimensions.

Table 3: Order-of-magnitude gap between automotive and maritime dataset resources across key metrics. Gap factor is computed as the ratio of automotive to maritime values.

Metric	**Automotive Best**	**Maritime Best**	**Gap Factor**
Detection images (single dataset)	1,281,167 (ImageNet)	186,419 (KOLOMVERSE)	$\sim 7\times$
3D bbox labels (single dataset)	12,600,000 (Waymo)	$\sim$190,000 (PoLaRIS)	$\sim 66\times$
Multi-sensor scenes	2,150 (Waymo+nuScenes)	7 (MOANA)	$\sim 307\times$
Annotation instances (total)	>15,000,000	$\sim$1,300,000	$\sim 12\times$
Sensor modalities per dataset (max)	12 (nuScenes)	7+ (Pohang/PoLaRIS)	$\sim 2\times$
Detection classes (max, single)	80 (COCO)	13 (MCShips)	$\sim 6\times$

The most critical gap is in multi-sensor data volume. The Waymo Open Dataset provides 1,150 fully annotated multi-sensor scenes, and nuScenes adds another 1,000, for a combined total exceeding 2,000 scenes with synchronized LiDAR, camera, and radar data. By comparison, MOANA [7]—the most comprehensive multi-radar maritime dataset—comprises only 7 sequences. Even aggregating all maritime multi-modal datasets (Pohang, PoLaRIS, MOANA, WaterScenes, FloW, SeePerSea), the total number of independently collected multi-sensor sequences remains well below 100. This approximately 300-fold gap in multi-sensor scene diversity represents perhaps the most consequential deficiency for developing robust maritime perception systems, as sensor fusion algorithms require diverse training examples to generalize across the wide range of environmental conditions encountered at sea.

The 3D bounding box label gap is similarly severe. Waymo's 12.6 million LiDAR-derived 3D labels dwarf the approximately 190,000 detection-grade annotations in PoLaRIS [6], yielding a gap factor of roughly 66. This disparity is compounded by the fact that maritime 3D labels are almost exclusively derived from 2D bounding boxes (sometimes augmented with range information), whereas automotive 3D labels benefit from dense LiDAR point clouds that enable precise 3D cuboid fitting.

Aggregating across all maritime datasets surveyed in this chapter, the total number of detection-grade annotated maritime images (those with bounding box or segmentation annotations) is approximately 700,000. This figure is dominated by KOLOMVERSE at

186,419, followed by PoLaRIS at approximately 360,000 (though many of these are repetitive keyframes from overlapping sequences), MODS at 81,000, WaterScenes at 54,120, SeaShips at 31,455, and the remaining datasets collectively contributing under 100,000. For comparison, MS COCO alone provides 330,000 annotated images with 1.5 million object instances [4], and the Waymo Open Dataset encompasses over 200,000 annotated LiDAR frames [1].

2.6 Annotation Cost and Quality Challenges

The maritime annotation gap reflects not merely a shortage of collection infrastructure but also the disproportionate difficulty and cost of annotating maritime imagery compared to terrestrial scenes.

2.6.1 Domain Expertise Requirements

Maritime object annotation demands specialized knowledge that general-purpose crowd workers typically lack. Distinguishing between vessel types—cargo ships, tankers, fishing trawlers, naval vessels, recreational craft—requires familiarity with hull forms, superstructure configurations, and operational context that is not readily acquired without maritime training. Zhang et al. [9] emphasize that domain-specific expertise is a prerequisite for constructing high-quality maritime training data, noting that misclassification at the labeling stage propagates directly into detector training as systematic label noise. The ABOShips annotation pipeline [35], which required a second pass using visual tracking to identify and correct initial labeling errors, illustrates the additional quality-control overhead inherent in maritime annotation.

2.6.2 Small and Distant Targets

Maritime scenes routinely contain targets at ranges of several kilometers, appearing as clusters of only a few pixels in the image. MCMOD [36] explicitly addresses the prevalence of small-scale objects, and PoLaRIS [6] includes annotations for objects as small as 10×10 pixels. Annotating such targets is time-consuming and error-prone: annotators must zoom to high magnification to delineate object boundaries, and the low signal-to-noise ratio for distant targets increases inter-annotator disagreement. The annotation time per image for maritime datasets is correspondingly high; Bovcon et al. [8] report approximately 20 minutes per image for pixel-level semantic segmentation of MaSTr1325, compared to typical rates of 5–10 minutes for road-scene segmentation in automotive datasets.

2.6.3 Ambiguous Boundaries and Visual Clutter

Water-surface conditions create annotation ambiguities that are largely absent from terrestrial scenes. Vessel wakes, bow waves, and spray obscure hull boundaries; sun glare and specular reflections from the water surface produce bright artifacts that can be confused with objects; and atmospheric effects (haze, fog, rain) reduce contrast between objects and background. These factors increase the variance of bounding box placement across annotators. Additionally, partially submerged objects, overlapping vessels at marinas, and the lack of fixed spatial reference points (unlike lane markings or curbs in driving scenes) further complicate boundary delineation.

2.6.4 Class Imbalance and Long-Tail Distribution

Maritime traffic exhibits severe class imbalance. In KOLOMVERSE [5], ships account for 393,936 of 732,039 instances (53.8%), while buoys contribute only 34,080 (4.7%). Similar distributions appear in other datasets: small recreational craft dominate in coastal waters, while large commercial vessels are relatively rare. This long-tail distribution means that even large datasets may contain too few examples of safety-critical minority classes (e.g., kayaks, swimmers, semi-submerged debris) to train reliable detectors.

2.7 Domain-Specific Perception Challenges

Beyond annotation difficulty, the maritime domain presents perception challenges that necessitate larger and more diverse training sets than might be required for a comparably complex terrestrial task.

2.7.1 Environmental Variability

Maritime environments exhibit extreme variability across temporal, meteorological, and geographic dimensions. Sea state ranges from mirror-calm to storm conditions with multi-meter waves; lighting spans from direct tropical sun (with intense specular reflections) to overcast polar twilight; visibility ranges from unlimited to near-zero in fog, rain, or snow. WaterScenes [40] explicitly captures variation across daytime, nightfall, and nighttime conditions; sunny, overcast, rainy, and snowy weather; and four distinct waterway types. KOLOMVERSE [5] samples from 21 territorial waters to achieve geographic diversity. Despite these efforts, no single maritime dataset approaches the environmental coverage that automotive datasets achieve through crowd-sourced collection across entire nations and seasons.

2.7.2 Horizon and Background Complexity

The maritime horizon, while seemingly simple, creates distinctive detection challenges. Objects transitioning from below to above the horizon change dramatically in background context (dark water to bright sky or vice versa). Coastal backgrounds introduce shoreline structures, breakwaters, and moored vessels that create complex clutter patterns. In narrow waterways, such as those captured in the Pohang Canal Dataset [42], the visual scene more closely resembles an urban canyon than open ocean, requiring detectors to handle both maritime and semi-terrestrial visual contexts.

2.7.3 Scale and Aspect Ratio Diversity

The range of object scales in maritime scenes exceeds that in typical driving scenarios. A single image may contain both a large vessel filling much of the frame and distant targets comprising fewer than 100 pixels. Moreover, the aspect ratios of maritime objects vary more widely than those of vehicles, pedestrians, and cyclists: a container ship viewed broadside may have an aspect ratio exceeding 8:1, while a small buoy is approximately 1:1. This scale diversity challenges both anchor-based and anchor-free detection architectures, which must maintain sensitivity across several orders of magnitude in object area.

2.7.4 Temporal Dynamics and Motion Patterns

Wave-induced platform motion on USV-mounted cameras produces rapid and largely unpredictable changes in camera orientation, leading to motion blur, horizon oscillation, and intermittent spray occlusion. The MODS dataset [46] and LaRS [39] both capture these dynamics through temporally consecutive frame sequences. Additionally, the relative motion between an observing platform and maritime targets differs qualitatively from automotive scenarios: closing rates can be very slow (vessels approaching head-on at a few knots) or very fast (high-speed craft), and lateral apparent motion is dominated by the observer's heading changes rather than lane-based geometry.

2.8 Summary and Research Gaps

This chapter has surveyed 20 maritime object detection datasets spanning a decade of development (2015–2024), organized by collection platform and sensor modality. Several structural factors account for the persistent annotation gap between maritime and automotive domains:

1. **Limited collection infrastructure.** Automotive datasets benefit from fleets of sensor-equipped vehicles operating on public roads; maritime data collection requires access to vessels, coastal installations, or USVs that are less widely available and more expensive to operate.

2. **Fewer research groups.** The autonomous driving community encompasses dozens of major industrial laboratories (Waymo, Cruise, Argo AI, Mobileye, Baidu Apollo) and hundreds of academic groups, each contributing data and benchmarks. The maritime perception community is substantially smaller, with fewer organizations possessing both the sensing infrastructure and the annotation resources to produce large-scale datasets.

3. **Environmental diversity.** Maritime environments span open ocean, coastal waters, rivers, lakes, canals, and harbors, each with distinct visual characteristics, traffic patterns, and object taxonomies. Achieving representative coverage requires data collection across many geographically and climatically distinct locations, a logistical challenge that compounds the per-image annotation cost.

4. **Annotation difficulty.** As documented in Section 2.6, the combination of small distant targets, ambiguous boundaries, domain expertise requirements, and severe class imbalance makes maritime annotation two to four times slower per image than automotive annotation. At typical annotation service rates, the cost of manually labeling a maritime dataset to Waymo-equivalent scale (12 million bounding boxes) would be prohibitive for most research groups.

Against this backdrop, several emerging approaches offer pathways to narrow the gap:

1. **Radar-guided camera labeling.** By projecting radar detections into the camera frame, bounding box annotations can be generated automatically without human intervention. The availability of WaterScenes [40], PoLaRIS [6], and MOANA [7] as multi-modal reference datasets provides initial validation targets for such pipelines. The noise characteristics of radar-projected labels are systematic (arising from calibration and projection geometry) rather than random, which suggests that domain-specific denoising strategies may be required.

2. **AIS-assisted annotation.** The Automatic Identification System (AIS) provides vessel identity, position, heading, and speed for cooperative maritime traffic. Fusing AIS tracks with camera imagery enables automatic association of detected objects with vessel metadata, facilitating both bounding box generation (via projected AIS positions) and fine-grained class labeling (via vessel type codes transmitted in AIS messages).

3. **Active learning and selective annotation.** Rather than annotating all collected frames uniformly, active learning strategies can identify the most informative frames for human annotation, maximizing detection improvement per annotation dollar. The temporal redundancy inherent in maritime video (vessels move slowly relative to frame rate) suggests that aggressive frame selection could substantially reduce annotation volume without sacrificing model performance.

4. **Cross-modal label transfer.** Beyond radar-to-camera projection, cross-modal transfer encompasses LiDAR-to-camera (as explored in SeePerSea [41]), thermal-to-visible (as enabled by VAIS [44] and PoLaRIS [6]), and hyperspectral-to-RGB (as proposed in M^{2}SODAI [43]). Each modality pair offers distinct advantages: radar provides range and velocity; LiDAR provides precise 3D geometry; thermal provides illumination-invariant detection; and hyperspectral provides material discrimination. A comprehensive automated labeling framework may ultimately combine multiple cross-modal signals to produce annotations that exceed the quality achievable by any single modality alone.

The remainder of this literature review examines each of these technical components in detail, beginning with radar-camera sensor registration and calibration (Section 3), progressing through radar-to-camera projection and bounding box generation (Section 4), and extending to automated dataset construction methodologies (Section 5), radar-camera fusion (Section 6), X-band signal processing (Section 7), and edge computing for real-time deployment (Section 8).

3 Radar-Camera Sensor Registration and Calibration Methods

The preceding chapter established that maritime object detection datasets remain orders of magnitude smaller than their automotive counterparts, and that manual annotation constitutes the principal bottleneck to scaling these collections. A radar-guided labeling pipeline—one that projects radar detections into camera imagery to generate bounding box annotations automatically—offers a compelling path toward closing this gap. However, the fidelity of any such cross-modal labeling system is bounded above by the accuracy of the geometric registration between the radar and camera coordinate frames. An error of even a few pixels in the projected radar centroid propagates directly into label noise, degrading downstream detector training. Precise sensor registration is therefore not merely a desirable system property but a strict prerequisite for the entire radar-guided labeling paradigm.

This chapter surveys the methods by which radar and camera sensors are brought into geometric correspondence. The discussion begins with the mathematical framework that underlies all extrinsic calibration formulations, proceeds through the three dominant families of calibration techniques—target-based, targetless, and motion-based—and then addresses the complementary problems of intrinsic camera calibration and temporal synchronization. Throughout, attention is drawn to the gap between the well-studied 77 GHz automotive radar domain and the largely unaddressed X-band marine radar setting that motivates the present work.

3.1 Mathematical Framework for Extrinsic Calibration

The spatial relationship between any two rigidly mounted sensors is described by six degrees of freedom (6DOF): three translational components and three rotational components. In the context of radar-camera calibration, the objective is to recover the rigid-body transformation that maps points expressed in the radar coordinate frame into the camera coordinate frame [16]. This transformation is parameterized by a 3×3 rotation matrix $\mathbf{R} \in SO(3)$ and a 3×1 translation vector $\mathbf{t} \in \mathbb{R}^3$.

3.1.1 Rotation Representation

The rotation matrix $\mathbf{R}$ is an orthonormal matrix satisfying $\mathbf{R}^\top \mathbf{R} = \mathbf{I}$ and $\det(\mathbf{R}) = +1$, encoding the orientation of the radar frame with respect to the camera frame. Explicitly, $\mathbf{R}$ can be decomposed into three successive rotations about the coordinate axes, parameterized by the Euler angles (ϕ, θ, ψ) corresponding to roll, pitch, and yaw, respectively:

$$\mathbf{R} = \mathbf{R}_z(\psi)\, \mathbf{R}_y(\theta)\, \mathbf{R}_x(\phi), \tag{1}$$

where

$$\mathbf{R}_x(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi & \cos\phi \end{bmatrix}, \quad \mathbf{R}_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}, \quad \mathbf{R}_z(\psi) = \begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{2}$$

Alternative rotation parameterizations—axis-angle, unit quaternions, and rotation vectors—are often preferred in optimization to avoid gimbal lock and to simplify constraint enforcement [49]. In particular, the unit quaternion representation $\mathbf{q} = (q_w, q_x, q_y, q_z)$ with $\|\mathbf{q}\| = 1$ is widely used in hand-eye calibration formulations [50].

3.1.2 Homogeneous Transformation

The rotation and translation are combined into a single 4×4 homogeneous transformation matrix:

$$\mathbf{T}_c^r = \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix} \in SE(3), \tag{3}$$

where the superscript and subscript denote the source (radar) and target (camera) frames. A point $\mathbf{P}_r = [x_r,\, y_r,\, z_r,\, 1]^\top$ expressed in homogeneous radar coordinates is transformed into the camera frame as:

$$\mathbf{P}_c = \mathbf{T}_c^r\, \mathbf{P}_r = \mathbf{R}\, \mathbf{P}_r^{(3)} + \mathbf{t}, \tag{4}$$

where $\mathbf{P}_r^{(3)}$ denotes the inhomogeneous (three-element) coordinate vector. Once transformed into the camera frame, the point is projected onto the image plane via the camera intrinsic matrix $\mathbf{K}$:

$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \mathbf{K}\, [\mathbf{R} \mid \mathbf{t}]\, \mathbf{P}_r, \qquad \mathbf{K} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}, \tag{5}$$

where s is a scalar depth factor, (f_x, f_y) are the focal lengths in pixel units, and (c_x, c_y) is the principal point. This projection model forms the basis of reprojection error metrics used throughout calibration optimization [51].

3.1.3 Implications for Marine Radar

For automotive 77 GHz radar, each detection typically provides range, azimuth, and Doppler velocity, but elevation information may be absent or coarsely resolved [52]. Marine X-band radar presents an even more constrained geometry: the mechanically rotating antenna produces a plan-position indicator (PPI) display that records range and azimuth only, with no elevation whatsoever [13]. Consequently, the transformation in Equation (4) is underdetermined unless an elevation assumption is imposed—typically that all targets lie at sea level—reducing the 3D problem to a 2D-to-2D homography or a constrained 3D projection with a fixed height prior.

3.2 Target-Based Calibration Methods

Target-based calibration methods rely on purpose-built calibration artifacts whose positions and reflective properties are known a priori. These approaches yield the highest absolute accuracy but require controlled experimental conditions and cooperative placement of calibration objects.

3.2.1 Corner Reflectors and Radar-Visible Targets

Trihedral corner reflectors are the standard calibration target for radar systems. Their large and predictable radar cross section (RCS) ensures reliable detection at known positions, providing precise range-azimuth anchor points in the radar frame [13]. When co-located with a visually identifiable marker—a painted panel, an LED array, or a distinctive geometric pattern—the same physical location can be identified in both sensor modalities, yielding a set of correspondence pairs for extrinsic parameter estimation.

Peršić et al. [16] developed a multi-target calibration system for a 77 GHz radar–LiDAR–camera suite that exploits the RCS characteristics of corner reflectors to resolve the data association problem. By deploying multiple reflectors at varying ranges and azimuths, the authors obtained sufficient correspondences to solve for the full 6DOF extrinsic transformation using nonlinear least-squares optimization. The reported reprojection accuracy was 2.2 pixels RMSE in the horizontal direction and 3.0 pixels RMSE in the vertical direction, demonstrating that sub-3-pixel accuracy is attainable with careful target placement and sufficient geometric diversity among the calibration points.

3.2.2 Homography-Based Approaches

Sugimoto et al. [53] proposed an early radar-camera registration method based on planar homography estimation. Under the assumption that both radar detections and camera observations lie on a common ground plane, the relationship between radar coordinates (r, θ) and pixel coordinates (u, v) reduces to a 3×3 homography matrix $\mathbf{H}$:

$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \mathbf{H} \begin{bmatrix} x_r \\ y_r \\ 1 \end{bmatrix}, \tag{6}$$

where (x_r, y_r) are the Cartesian radar coordinates on the ground plane. This formulation requires a minimum of four non-collinear point correspondences and is estimated robustly using RANSAC [54]. While the planar assumption is well-suited to flat roadway environments, it also applies naturally to the maritime setting where targets occupy a nominally planar sea surface. The principal limitation is that elevated structures—ship superstructures, masts, bridge decks—violate the planarity constraint and introduce systematic projection errors at close range.

3.2.3 Checkerboard Patterns for Camera Intrinsics

Target-based calibration of the camera intrinsic parameters is a prerequisite for accurate extrinsic estimation. Zhang's flexible calibration technique [51] uses images of a planar checkerboard pattern captured at multiple orientations to solve for the focal lengths, principal point, and lens distortion coefficients. The method minimizes the reprojection error between detected corner locations and their model-predicted positions via Levenberg–Marquardt optimization. This technique has become the de facto standard in both academic research and industrial practice, and is implemented in widely used libraries such as OpenCV. In the context of radar-camera systems, camera intrinsic calibration is performed independently and its results are treated as fixed during the subsequent extrinsic calibration stage [2, 3].

3.3 Targetless Calibration Methods

Target-based methods, while accurate, require physical access to the sensing environment and cooperative placement of calibration artifacts. These requirements are impractical for deployed maritime systems where sensors are permanently mounted on vessels or coastal infrastructure. Targetless calibration methods instead exploit naturally occurring correspondences in the sensor data streams.

3.3.1 Trajectory Alignment

When both sensors independently track moving objects over time, the resulting trajectories can be aligned to recover the inter-sensor transformation. The Kabsch algorithm [55] solves for the optimal rotation between two sets of corresponding 3D points by minimizing the root-mean-square deviation. Its extension by Umeyama [56] additionally recovers a uniform scale factor, yielding a similarity transformation. In the radar-camera context, radar tracks provide range-azimuth trajectories while camera tracks provide pixel-domain trajectories. Under a known camera model, the pixel trajectories can be back-projected into bearing-only rays, and the correspondence between radar and camera tracks is established through temporal co-occurrence and motion consistency.

Wang et al. [57] formalized this approach as an online track-to-track association framework for automotive radar-camera calibration. By continuously matching radar detections to camera-tracked objects using a joint cost function over spatial proximity and velocity consistency, the authors accumulate correspondences over time and solve for the extrinsic parameters using a Perspective-n-Point (PnP) formulation [58]. The online formulation permits continuous recalibration, compensating for mechanical drift and thermal deformation—properties that are particularly relevant for shipboard installations subject to vibration and thermal cycling.

3.3.2 Mutual Information and Feature-Based Methods

Lin et al. [59] proposed a targetless radar-camera calibration method that maximizes the mutual information between radar measurements and camera features. By treating the calibration problem as an information-theoretic alignment task, the method avoids explicit feature detection or correspondence establishment. Instead, it searches over the space of extrinsic parameters to find the transformation that maximizes the statistical dependence between the projected radar data and the image content. This approach is particularly robust to the sparse and noisy nature of radar point clouds, which often frustrate traditional feature matching techniques.

3.3.3 Deep Learning Approaches

The application of deep neural networks to sensor calibration has grown rapidly in recent years. Schneider et al. [60] introduced RegNet, a convolutional neural network that predicts the extrinsic calibration between a LiDAR and a camera from a single pair of depth and color images. The network is trained on synthetically perturbed calibrations and learns to regress the six extrinsic parameters directly. While originally developed for LiDAR-camera pairs, the RegNet architecture has been adapted to radar-camera calibration by substituting radar-derived depth or occupancy representations for the LiDAR depth image [61].

More recently, end-to-end calibration networks such as RC-AutoCalib [62] have been proposed specifically for the radar-camera modality pair. These networks ingest raw radar tensors and camera images and output the extrinsic transformation without requiring intermediate detection or tracking stages. A comprehensive review by Peng et al. [61] categorizes deep learning calibration methods along axes of supervision level (fully supervised, self-supervised, unsupervised), input representation (point cloud, bird's-eye view, range-Doppler), and output parameterization (Euler angles, quaternions, direct matrix). The review observes that deep learning calibration methods typically achieve accuracy within 1–5 pixels of target-based methods on automotive benchmarks, but require large volumes of diverse training data that are not yet available for the maritime domain.

3.4 Motion-Based (Ego-Motion) Calibration

Motion-based calibration exploits the ego-motion of the sensor platform itself rather than the motion of external targets. By observing the same platform motion from two different sensor viewpoints, the relative transformation between the sensors can be inferred.

3.4.1 The Hand-Eye Calibration Formulation

The classical formulation of motion-based calibration originates in robotics, where the problem of determining the transformation between a camera mounted on a robot end-effector (the "eye") and the end-effector itself (the "hand") was first posed by Shiu and Ahmad and subsequently solved in closed form by Tsai and Lenz [63]. The hand-eye equation takes the form:

$$\mathbf{AX} = \mathbf{XB}, \tag{7}$$

where $\mathbf{A}$ and $\mathbf{B}$ are the motions observed by the two sensors between successive time steps, and $\mathbf{X}$ is the unknown inter-sensor transformation. Given n motion pairs $\{(\mathbf{A}_i, \mathbf{B}_i)\}_{i=1}^{n}$, the system is overdetermined for $n \geq 2$ (with non-parallel rotation axes) and can be solved via least-squares methods.

3.4.2 Dual Quaternion Solution

Daniilidis [50] reformulated the hand-eye problem using dual quaternions, which encode rotation and translation in a unified algebraic structure. The dual quaternion representation enables a simultaneous, closed-form solution for both $\mathbf{R}$ and $\mathbf{t}$, avoiding the sequential rotation-then-translation estimation of earlier methods. The resulting algorithm is computationally efficient, requiring only a singular value decomposition (SVD) of a $6n \times 8$ matrix, and is provably optimal under isotropic Gaussian noise. This formulation has been adopted in several subsequent radar-camera calibration systems [16] and remains the preferred closed-form solution when sufficient motion diversity is available.

3.4.3 Continuous-Time Batch Estimation

A limitation of the classical hand-eye formulation is that it assumes discrete, synchronized motion observations from both sensors. In practice, radar and camera operate at different frame rates and are rarely hardware-synchronized. Furgale et al. [64] addressed this challenge by proposing a continuous-time batch estimation framework in which sensor trajectories are represented as temporal basis spline (B-spline) curves. The continuous-time representation decouples the calibration problem from sensor frame rates: observations from asynchronous sensors are related through the common spline representation, and both temporal offsets and spatial extrinsics are estimated jointly. This approach is particularly relevant for maritime systems where the radar antenna rotation period (typically 1–3 seconds for X-band marine radars) is substantially longer than the camera frame interval, creating large temporal gaps between radar and camera observations of the same scene.

3.5 Intrinsic Camera Calibration

Accurate intrinsic calibration is a prerequisite for meaningful extrinsic estimation, as errors in the camera model propagate directly into the projection of radar coordinates onto the image plane. Zhang's method [51] recovers the intrinsic parameter vector $\boldsymbol{\xi} = (f_x, f_y, c_x, c_y, k_1, k_2, p_1, p_2, \ldots)$, where k_i are radial distortion coefficients and p_i are tangential distortion coefficients. The calibration procedure captures m images of a planar checkerboard at varied orientations and minimizes the total reprojection error:

$$\min_{\boldsymbol{\xi},\, \{\mathbf{R}_j, \mathbf{t}_j\}_{j=1}^{m}} \sum_{j=1}^{m} \sum_{i=1}^{n} \left\| \mathbf{u}_{ij} - \hat{\mathbf{u}}(\boldsymbol{\xi},\, \mathbf{R}_j,\, \mathbf{t}_j,\, \mathbf{X}_i) \right\|^2, \tag{8}$$

where $\mathbf{u}_{ij}$ is the observed pixel location of the i-th checkerboard corner in the j-th image and $\hat{\mathbf{u}}(\cdot)$ is its predicted location under the camera model. The optimization is initialized via a closed-form homography decomposition and refined iteratively.

For wide field-of-view cameras commonly used in maritime surveillance, lens distortion can exceed 20 pixels at the image periphery. Higher-order distortion models (e.g., the rational model with six coefficients or fisheye models) may be necessary to achieve sub-pixel reprojection accuracy across the full field of view [51]. Failure to adequately model distortion introduces spatially varying bias in the projected radar coordinates, which is particularly problematic for targets at extreme bearing angles near the edges of the camera's field of view.

3.6 LiDAR-Camera Calibration: Transferable Concepts

The LiDAR-camera calibration literature is substantially more mature than its radar-camera counterpart, and many techniques developed for LiDAR-camera registration are directly transferable or adaptable to the radar-camera problem. Li et al. [65] provide a comprehensive survey that organizes LiDAR-camera calibration methods into four categories: target-based, feature-based, motion-based, and learning-based. This taxonomy maps cleanly onto the radar-camera methods discussed above.

Target-based LiDAR-camera methods typically use planar boards or polygonal cutouts that produce strong edges in both the point cloud and the image [3, 66]. The analogous radar target is the corner reflector, though its point-like radar signature provides less geometric structure than a planar board in a LiDAR scan. Feature-based LiDAR-camera methods exploit edge correspondences, line features, or statistical correlations between depth discontinuities and image gradients; these techniques assume dense point clouds that are not available from radar. Motion-based and learning-based methods are more readily transferable because they operate on trajectory-level or representation-level abstractions that are less sensitive to point cloud density.

The KITTI benchmark [3] and nuScenes dataset [2] have served as standard evaluation platforms for LiDAR-camera calibration. Both provide factory-calibrated extrinsic parameters as ground truth, enabling quantitative comparison of calibration algorithms. No analogous benchmark exists for radar-camera calibration in either the automotive or maritime domain, complicating the evaluation of proposed methods.

3.7 Temporal Synchronization Challenges

Even a geometrically perfect extrinsic calibration degrades rapidly if the two sensor data streams are not temporally aligned. At maritime vessel speeds of 5–15 knots and typical ranges of 0.5–5 km, a timing offset of 100 ms produces a target displacement of 0.26–0.77 m, which may translate into several pixels of projection error depending on range and camera resolution.

3.7.1 Asynchronous Sensor Architectures

Radar and camera systems rarely share a common clock. Automotive 77 GHz radars typically output detections at 10–20 Hz, while cameras operate at 15–30 fps. In the nuScenes system [2], hardware triggering and a central timing bus achieve sub-millisecond synchronization across all sensors. Such infrastructure is unavailable in typical maritime installations, where the radar interface provides PPI sweeps at the antenna rotation rate (20–48 rpm for X-band marine radars) and the camera free-runs at its own frame rate [13].

3.7.2 The Rotating Antenna Problem

A challenge unique to mechanically scanned marine radar is that the antenna sweeps through 360° over a rotation period of 1.25–3 s. Unlike a phased-array automotive radar that illuminates its entire field of view quasi-simultaneously, a rotating marine radar observes different azimuth sectors at different instants within each sweep. A camera frame captured at time t corresponds to a single azimuth sector of the radar sweep, not to the full PPI image. Naive association of a complete PPI scan with a single camera frame introduces bearing-dependent timing errors that worsen for targets at azimuths far from the instantaneous antenna pointing direction at time t.

Li and Pollefeys [67] studied temporal calibration for asynchronous sensor pairs and demonstrated that temporal offsets can be estimated jointly with spatial extrinsics by exploiting motion consistency constraints. The continuous-time framework of Furgale et al. [64] extends this idea by representing sensor trajectories as continuous functions of time, enabling interpolation between discrete observations. For maritime radar-camera systems, such continuous-time methods offer a principled solution to the rotating antenna synchronization problem, provided that the antenna angle encoder signal is available to timestamp individual radar returns within each sweep.

3.8 Maritime-Specific Challenges

The vast majority of radar-camera calibration research targets the automotive domain, where 77 GHz millimeter-wave radar is the standard sensor. Transferring these methods to the maritime setting introduces several challenges arising from fundamental differences in sensor characteristics, platform dynamics, and operational environment.

3.8.1 X-Band Marine Radar Versus 77 GHz Automotive Radar

Table 4 summarizes the key differences between X-band marine radar and 77 GHz automotive radar that affect calibration methodology and achievable accuracy.

Table 4: Comparison of X-band marine radar and 77 GHz automotive radar characteristics relevant to calibration.

Parameter	**X-Band Marine Radar**	**77 GHz Automotive Radar**
Operating frequency	9.3–9.5 GHz	76–81 GHz
Wavelength	∼32 mm	∼3.9 mm
Maximum range	50–130 km	0.1–0.3 km
Range resolution	5–50 m	0.15–0.5 m
Azimuth beamwidth	1.2°–5.2°	1°–15°
Elevation information	None (2D PPI)	Limited or none
Scan mechanism	Mechanical rotation	Electronic beam steering
Scan rate	20–48 rpm	10–20 Hz (full FOV)
Output format	PPI image (polar)	Detection list (range, azimuth, Doppler)
Doppler capability	Typically none	Yes (FMCW)
Typical mounting height	15–40 m (ship mast)	0.3–0.8 m (vehicle bumper)

Several of these differences have direct consequences for calibration. The absence of elevation data in X-band PPI radar means that the full 3D-to-2D projection model (Equation (5)) is underdetermined without an elevation prior. The coarse range resolution (meters to tens of meters) and azimuth resolution ($> 1^\circ$) limit the precision of radar-side point localization, placing an intrinsic floor on achievable reprojection accuracy that is typically far above the sub-pixel precision attainable with camera corner detection

[13, 68]. The PPI output format requires centroid extraction from extended radar returns, introducing additional localization uncertainty compared to the point-detection output of automotive FMCW radar [52].

3.8.2 Platform Dynamics and Mounting Geometry

Maritime platforms are subject to continuous six-degree-of-freedom motion—surge, sway, heave, roll, pitch, and yaw—driven by sea state. On vessels, this motion can exceed $\pm 15^\circ$ in roll and $\pm 5^\circ$ in pitch under moderate sea conditions [12]. Such motion invalidates any static calibration unless the calibration is referenced to a stabilized frame or compensated in real time using inertial measurements.

Furthermore, the large physical separation between maritime radar and camera installations introduces substantial lever-arm effects. A marine radar antenna mounted on a mast 30 m above the waterline and a camera mounted at bridge level 20 m above the waterline are separated by a vertical baseline of 10 m and potentially a horizontal offset of several meters. Platform rotation amplifies these offsets: a 5° roll about the ship's longitudinal axis displaces the radar antenna tip by $30 \times \sin(5^\circ) \approx 2.6$ m laterally, while the camera moves by $20 \times \sin(5^\circ) \approx 1.7$ m. The differential displacement of approximately 0.9 m introduces time-varying extrinsic errors that must be compensated.

Coastal surveillance installations on fixed infrastructure (piers, towers, buildings) avoid vessel motion but introduce their own challenges. The radar antenna is often mounted on a dedicated mast structure separated from the camera by tens of meters, and the installation geometry is dictated by structural constraints rather than calibration convenience. Han et al. [69] describe a coastal radar-camera system for autonomous surface vehicle navigation in which the radar and camera are co-located on a fixed tower, but note that thermal expansion and wind-induced vibration can shift the extrinsic calibration by several pixels over the course of a day.

3.8.3 Environmental Factors

The maritime environment presents clutter and propagation conditions that have no automotive analogue. Sea clutter from wave returns can produce false radar detections that, if mistakenly associated with camera features, corrupt calibration correspondences [12]. Multipath propagation over the sea surface creates ghost targets at incorrect ranges. Atmospheric ducting in the marine boundary layer bends radar beams, causing anomalous propagation that shifts the apparent azimuth and range of targets [13]. On the camera side, sun glare reflected off the water surface can saturate image regions and prevent feature detection. These environmental factors collectively reduce the number of reliable sensor correspondences available for calibration and increase the importance of robust estimation techniques such as RANSAC [54].

3.9 Summary and Research Gaps

This chapter has surveyed the three principal families of radar-camera calibration methods—target-based, targetless, and motion-based—along with the supporting infrastructure of intrinsic camera calibration, temporal synchronization, and the transferable insights from LiDAR-camera calibration research. The mathematical framework for extrinsic calibration is well-established: the 6DOF rigid-body transformation in $SE(3)$ and its projection through the camera intrinsic model provide a sound theoretical basis for all methods reviewed.

Within the automotive domain, each calibration family has demonstrated practical viability. Target-based methods achieve sub-3-pixel reprojection accuracy [16], online targetless methods enable continuous recalibration [57], and continuous-time batch estimation methods handle asynchronous sensor architectures [64]. Deep learning approaches offer the prospect of fully automatic calibration without specialized targets or motion requirements [60, 62].

However, the literature reveals a pronounced gap at the intersection of these methods and the X-band marine radar domain. The key observations are as follows:

1. **No established X-band marine radar to camera calibration method exists.** The overwhelming majority of published radar-camera calibration work uses 77 GHz automotive radar. The handful of maritime radar-camera studies [69, 70] address narrow operational scenarios and do not provide general-purpose calibration frameworks.

2. **The 2D PPI output of marine radar lacks elevation information**, rendering standard 3D-to-2D projection models underdetermined. A principled treatment of the elevation ambiguity—whether through sea-surface planarity assumptions, homographic models, or probabilistic elevation priors—has not been established for X-band radar-camera systems.

3. **Temporal synchronization between mechanically rotating radar and free-running cameras** presents a unique challenge that has no direct automotive analogue. The continuous-time estimation frameworks of Furgale et al. [64] provide a theoretical solution, but their application to marine radar with per-spoke timestamping has not been demonstrated.

4. **Platform dynamics on vessels** introduce time-varying extrinsic parameters that violate the static calibration assumption underlying most existing methods. While inertial compensation is conceptually straightforward, its integration with radar-camera calibration for maritime platforms has received limited attention.

5. **No benchmark dataset with ground-truth calibration exists** for X-band marine radar-camera systems, precluding quantitative comparison of competing approaches. The KITTI [3] and nuScenes [2] benchmarks serve this role for automotive LiDAR and radar, but no maritime equivalent is available.

6. **Coarse spatial resolution of X-band radar** places an intrinsic lower bound on calibration accuracy that is substantially higher than the sub-pixel precision achievable with automotive radar. Characterizing this bound and understanding its implications for downstream labeling quality is an open problem.

These gaps collectively indicate that developing a calibration methodology specifically tailored to X-band marine radar-camera systems is a necessary precondition for the radar-guided camera labeling pipeline that motivates this review. The following chapter addresses the next link in that pipeline: the projection of calibrated radar detections into camera pixel coordinates and the generation of bounding box annotations from those projections.

4 Radar-to-Camera Projection and Bounding Box Generation

Once the extrinsic and intrinsic parameters linking a radar and a camera have been established through calibration (Section 3), the next challenge is to exploit those parameters operationally: given a radar detection expressed in polar coordinates, compute the corresponding pixel location in the camera image and generate a bounding box suitable for training an object detector. This projection pipeline constitutes the central algorithmic bridge between calibrated sensor geometry and usable training labels, and its fidelity determines whether the resulting annotations are accurate enough to replace or supplement manual labeling. The present section traces the complete mathematical chain from radar measurement space through world, camera, and pixel coordinate systems; examines the lens distortion models required for wide field-of-view (FOV) maritime cameras; surveys the bird's-eye-view (BEV) and frustum-based association strategies that the autonomous-driving community has developed; and analyzes the error sources that limit projection accuracy, with particular attention to the three-dimensional ambiguity inherent in two-dimensional radar data and the additional complications introduced by the maritime environment.

4.1 Coordinate Transformation Pipeline

The transformation from a radar detection to a camera pixel proceeds through four sequential coordinate conversions: polar to Cartesian, Cartesian radar frame to world frame, world frame to camera frame, and camera frame to pixel coordinates. Each conversion introduces its own parameterization and potential error sources.

4.1.1 Polar to Cartesian Conversion

Marine radar and many automotive radar systems report detections in polar form as a tuple (r, θ, ϕ), where r is the slant range to the target, θ is the azimuth angle measured from the boresight, and ϕ is the elevation angle [13]. The conversion to a local Cartesian frame centered on the radar antenna is given by

$$x_r = r \cos\theta \cos\phi, \tag{9}$$
$$y_r = r \sin\theta \cos\phi, \tag{10}$$
$$z_r = r \sin\phi. \tag{11}$$

For conventional two-dimensional marine radar operating in the X-band, the elevation angle ϕ is not measured; the antenna fan beam illuminates a broad vertical sector and the return is collapsed onto the horizontal plane, effectively forcing $\phi = 0$ and yielding only the range–azimuth pair (r, θ) [12]. This collapse creates the elevation ambiguity that will be discussed at length in Section 4.7. Three-dimensional and four-dimensional imaging radars resolve targets in elevation as well, providing an explicit ϕ measurement that significantly simplifies the downstream projection [71].

4.1.2 Radar-Frame to World-Frame Transformation

The Cartesian radar coordinates $\mathbf{P}_r = [x_r,\, y_r,\, z_r]^\top$ must be expressed in a common world (or platform) frame before they can be related to the camera. Denoting the rotation and translation of the radar with respect to the world frame as $\mathbf{R}_r$ and $\mathbf{t}_r$, the world-frame point is

$$\mathbf{P}_w = \mathbf{R}_r \, \mathbf{P}_r + \mathbf{t}_r. \tag{12}$$

In practice, the platform frame is often chosen to coincide with one of the sensors so that one set of extrinsic parameters reduces to identity. The nuScenes dataset, for example, defines its ego-vehicle frame at the rear axle and provides per-sensor extrinsic matrices that encode the rigid-body transformation from each sensor to that frame [2].

4.1.3 World-Frame to Camera-Frame Transformation

Given the camera extrinsic parameters—rotation $\mathbf{R}$ and translation $\mathbf{t}$ from the world frame to the camera frame—a world point $\mathbf{P}_w$ is mapped to camera coordinates by

$$\mathbf{P}_c = \mathbf{R} \, \mathbf{P}_w + \mathbf{t}, \tag{13}$$

where $\mathbf{P}_c = [X_c, \, Y_c, \, Z_c]^\top$ and $Z_c > 0$ for points in front of the camera. The combined transformation from radar to camera is therefore

$$\mathbf{P}_c = \mathbf{R}\big(\mathbf{R}_r \, \mathbf{P}_r + \mathbf{t}_r\big) + \mathbf{t}. \tag{14}$$

If both sensor extrinsics are referenced to the same world frame, the radar-to-camera transformation can be written compactly in homogeneous coordinates as $\mathbf{T}_{c \leftarrow r} = \mathbf{T}_{c \leftarrow w} \, \mathbf{T}_{w \leftarrow r}$, where each $\mathbf{T}$ is a 4×4 matrix [51].

4.1.4 Pinhole Camera Model and Full Projection

The mapping from camera-frame three-dimensional coordinates to pixel coordinates is governed by the pinhole model. Letting $\mathbf{K}$ denote the 3×3 intrinsic matrix,

$$\mathbf{K} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}, \tag{15}$$

where f_x and f_y are the focal lengths in pixel units and (c_x, c_y) is the principal point, the projection is

$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \mathbf{K} \begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix}, \tag{16}$$

with $s = Z_c$ the projective depth [51]. Solving for the pixel coordinates explicitly,

$$u = f_x \frac{X_c}{Z_c} + c_x, \tag{17}$$

$$v = f_y \frac{Y_c}{Z_c} + c_y. \tag{18}$$

Combining the extrinsic and intrinsic transformations, the full projection of a world point to pixel coordinates is captured by the 3×4 projection matrix

$$\mathbf{P} = \mathbf{K} \, [\mathbf{R} \mid \mathbf{t}], \tag{19}$$

so that $s \, \tilde{\mathbf{u}} = \mathbf{P} \, \tilde{\mathbf{P}}_w$, where tildes denote homogeneous coordinate vectors. This factorization is the standard camera model that underpins virtually all radar-to-camera projection methods in the literature [2, 72, 73].

4.2 Lens Distortion Models

The ideal pinhole model of Equation (16) assumes rectilinear projection, which breaks down for real lenses—particularly the wide-FOV optics preferred in maritime surveillance for their panoramic coverage [74]. Accurately modeling lens distortion is essential for projecting radar detections to the correct pixel location, especially near the image periphery where distortion is most severe.

4.2.1 Brown–Conrady Model

The most widely adopted distortion model is the Brown–Conrady formulation, which decomposes distortion into radial and tangential components [51]. Let (x_n, y_n) be the normalized (undistorted) image coordinates, obtained by dividing the camera-frame point by Z_c and subtracting the principal point offset:

$$x_n = \frac{X_c}{Z_c}, \qquad\qquad y_n = \frac{Y_c}{Z_c}. \tag{20}$$

Define $r^2 = x_n^2 + y_n^2$. The distorted normalized coordinates (x_d, y_d) are then

$$x_d = x_n\big(1 + k_1 r^2 + k_2 r^4 + k_3 r^6\big) + 2\,p_1\,x_n\,y_n + p_2\big(r^2 + 2\,x_n^2\big), \tag{21}$$

$$y_d = y_n\big(1 + k_1 r^2 + k_2 r^4 + k_3 r^6\big) + p_1\big(r^2 + 2\,y_n^2\big) + 2\,p_2\,x_n\,y_n, \tag{22}$$

where k_1, k_2, k_3 are radial distortion coefficients and p_1, p_2 are tangential (decentering) distortion coefficients. The distorted pixel coordinates are recovered as

$$u_d = f_x\,x_d + c_x, \tag{23}$$

$$v_d = f_y\,y_d + c_y. \tag{24}$$

This model is the default in the OpenCV calibration toolbox, which implements Zhang's flexible calibration procedure [51], and is used by the nuScenes calibration pipeline [2]. It is generally adequate for lenses with FOV below approximately 100°, beyond which the polynomial expansion becomes poorly conditioned and higher-order terms are required [17].

4.2.2 Kannala–Brandt Wide-FOV Model

For cameras with FOV exceeding 100°—fisheye and ultra-wide-angle lenses increasingly deployed on maritime platforms for panoramic situational awareness—the Brown–Conrady polynomial struggles to capture the severe barrel distortion. Kannala and Brandt [17] proposed a generic model that parameterizes the mapping from the angle of incidence θ_i (the angle between the incoming ray and the optical axis) to the image radius r_d as an odd-order polynomial:

$$r_d = k_1\,\theta_i + k_2\,\theta_i^3 + k_3\,\theta_i^5 + k_4\,\theta_i^7 + k_5\,\theta_i^9, \tag{25}$$

where $\theta_i = \operatorname{atan2}\big(\sqrt{x_n^2 + y_n^2},\,1\big)$. The distorted coordinates are then

$$x_d = r_d\,\frac{x_n}{\sqrt{x_n^2 + y_n^2}}, \qquad\qquad y_d = r_d\,\frac{y_n}{\sqrt{x_n^2 + y_n^2}}. \tag{26}$$

This angle-based formulation naturally handles the compression of the image toward the periphery that characterizes equidistant, equisolid, and stereographic projection models. Kannala and Brandt demonstrated calibration residuals below 0.4 pixels for lenses with 180° FOV [17]. The model has been adopted in several autonomous-driving calibration toolboxes and is the recommended distortion model for the fisheye cameras used in extended nuScenes variants.

4.2.3 Scaramuzza Omnidirectional Model

Scaramuzza et al. [18] introduced a complementary approach for omnidirectional and catadioptric cameras in which the mapping between a three-dimensional point and its image location is described by a polynomial in the image radius. Specifically, the model fits the relationship between the z-component of the viewing ray and the image radius ρ:

$$z = a_0 + a_1\,\rho + a_2\,\rho^2 + a_3\,\rho^3 + a_4\,\rho^4, \tag{27}$$

where the polynomial coefficients are determined from a calibration grid observed at multiple orientations. This formulation is particularly useful for mirror-based omnidirectional systems and for lenses that exceed 180° FOV, situations where the Kannala–Brandt model may require impractically many terms. The OCamCalib toolbox implementing this model has been widely used in mobile robotics [18].

4.2.4 Maritime Relevance of Wide-FOV Distortion Modeling

Maritime surveillance installations typically require cameras with FOV of 90° or greater to cover the full horizon from a fixed mounting point [74]. Xu et al. [74] observed that failure to account for lens distortion in maritime radar-camera fusion can produce projection errors exceeding 20 pixels at the image periphery for a 100° FOV lens, corresponding to displacement of several meters at ranges of a few hundred meters. The choice of distortion model therefore has direct consequences for the quality of radar-projected bounding boxes: an inadequate model will systematically shift projected centroids away from true target locations, producing biased training labels that degrade detector performance. As a practical guideline, the Brown–Conrady model suffices for rectilinear lenses below 100° FOV, the Kannala–Brandt model should be employed for 100°–180° fisheye lenses, and the Scaramuzza model is indicated for omnidirectional or catadioptric systems [17, 18].

4.3 Bird's-Eye-View Projection

An alternative to projecting radar detections into the perspective image is to fuse both modalities in a shared bird's-eye-view (BEV) representation, which removes perspective foreshortening and provides a metrically regular grid well suited to range and bearing reasoning.

4.3.1 Inverse Perspective Mapping

The simplest BEV construction is inverse perspective mapping (IPM), which assumes a flat ground plane at a known height h below the camera. Under this assumption, every pixel in the image can be back-projected to a unique ground-plane point via the homography

$$\mathbf{H} = \mathbf{K}\,[\mathbf{r}_1\ \ \mathbf{r}_2\ \ \mathbf{t}], \tag{28}$$

where $\mathbf{r}_1$ and $\mathbf{r}_2$ are the first two columns of the rotation matrix and $\mathbf{t}$ is the translation to the ground plane [51]. IPM is computationally inexpensive but fails wherever the flat-ground assumption is violated—a critical limitation in maritime settings where the sea surface is neither rigid nor planar.

4.3.2 Learned BEV Representations: BEVFusion

Liu et al. [75] introduced BEVFusion, a framework that projects multi-view camera features into a unified BEV grid using a learned view transformer and then fuses them with LiDAR or radar BEV features through concatenation followed by convolutional processing. A key contribution was the identification of the BEV pooling operation as a computational bottleneck; by precomputing and caching the projection indices, BEVFusion achieves a roughly 40× reduction in latency compared to naive implementations, enabling real-time operation. On the nuScenes detection benchmark, camera–LiDAR BEVFusion attained 72.9% NDS, demonstrating that BEV fusion can match or exceed perspective-space methods while providing a representation that naturally accommodates radar point clouds, which are inherently top-down [75].

4.3.3 RCS-Aware BEV Encoding: RCBEVDet

Lin et al. [76] extended the BEV fusion paradigm with RCBEVDet, which incorporates the radar cross section (RCS) as an additional feature channel in the BEV encoder. The authors argue that RCS carries implicit information about target size and reflectivity that complements the spatial information from range and azimuth. RCBEVDet introduces a RadarBEVNet module that processes the raw radar point cloud—including range, azimuth, radial velocity, and RCS—into a dense BEV feature map via pillar-based encoding. On nuScenes, RCBEVDet achieved 55.1% NDS with camera-plus-radar inputs alone (no LiDAR), establishing a new state of the art for radar-camera BEV fusion [76]. The explicit use of RCS is particularly relevant for maritime applications where vessel RCS varies over several orders of magnitude depending on vessel size, aspect angle, and superstructure geometry [13].

4.4 Frustum-Based Radar–Camera Association

Rather than constructing a shared BEV grid, an alternative strategy projects each radar detection into the camera image and associates it with image-space detections or features within a frustum defined by the detection's range and angular uncertainty. This family of methods is especially natural when the goal is to generate or refine per-object bounding boxes.

4.4.1 CenterFusion

Nabati and Qi [72] proposed CenterFusion, which takes the output of a center-based image detector (CenterNet [77]) and enriches each detected object with radar features. Radar points are projected onto the image plane and associated with detected centers by a pillar-expansion and frustum-association mechanism: a radar detection at range r is expanded into a vertical frustum of width determined by the radar azimuth resolution, and all image detections whose centers fall within the frustum are associated with that radar point. The associated radar features—range, radial velocity, and RCS—are concatenated to the image feature map and passed to secondary regression heads that predict refined depth, velocity, and three-dimensional bounding box attributes. On nuScenes, CenterFusion improved the NDS of the camera-only CenterNet baseline by 12.8 percentage points (from 33.8% to 46.6%), with particularly large gains in velocity estimation accuracy [72]. The frustum-association idea is directly applicable to radar-guided label generation: instead of associating radar points with existing image detections, one can project radar points to seed candidate bounding boxes in regions of the image where no prior detection exists.

4.4.2 GRIF Net

Kim et al. [73] introduced the Gated Region of Interest Fusion (GRIF) network, which performs feature-level fusion within regions of interest (RoIs) rather than at the full-image level. Given a radar detection projected into the image, GRIF Net extracts a local image feature patch and a corresponding radar feature vector, then combines them through a gating mechanism that learns to weight the contribution of each modality. The gating is important because radar and camera provide complementary but sometimes conflicting information: the radar may localize a target in range while the camera resolves its lateral extent, and the gate learns to trust each modality where it is most reliable. GRIF Net demonstrated improved 3D detection performance over camera-only and naive concatenation baselines, particularly under adverse weather conditions where camera features degrade but radar remains robust [73].

4.4.3 Radar Region Proposal Network (RRPN)

An earlier contribution by Nabati and Qi [78] showed that radar detections can serve as highly efficient region proposals for two-stage object detectors. In the Radar Region Proposal Network (RRPN), each radar point projected onto the image defines a set of anchor boxes with dimensions estimated from the radar range and an assumed object size prior. These radar-derived proposals replace the exhaustive search of Selective Search or the learned RPN of Faster R-CNN [79], achieving a roughly 100× speedup in proposal generation while maintaining comparable recall. The key insight is that radar detections are already sparsely distributed on objects of interest, so using them as proposal seeds avoids the computational waste of evaluating thousands of background proposals [78]. This approach bears a direct resemblance to the radar-guided labeling pipeline, where radar detections similarly seed candidate annotation regions.

4.4.4 CRAFT: Polar Coordinate Association

Kim et al. [80] proposed CRAFT (Camera-Radar Association Framework for multi-modal 3D object deTection), which performs radar–camera association in polar coordinates rather than in Cartesian or image space. By representing both radar detections and image features in a range–azimuth grid, CRAFT avoids the information loss incurred by projecting three-dimensional radar measurements onto the two-dimensional image plane. The polar-coordinate association allows the network to exploit the natural structure of radar data—targets at similar range and bearing are grouped together regardless of their pixel separation—and to handle the non-uniform angular sampling of automotive radar. On nuScenes, CRAFT achieved 41.1% mAP, a significant improvement over prior radar-camera methods that rely on Cartesian or image-space association [80].

4.5 Range-Dependent Bounding Box Sizing

Projecting a radar detection yields a point, not a bounding box. Generating a usable annotation from that point requires estimating the spatial extent of the target in the image—a fundamentally underdetermined problem because radar provides range and bearing but generally not object dimensions. Several approaches have been explored to bridge this gap.

4.5.1 Challenges of Extent Estimation from Radar

The core difficulty is that radar angular resolution is typically much coarser than the angular subtent of the target. An X-band marine radar with 1.8° azimuth beamwidth cannot resolve the lateral extent of a vessel at most operational ranges [13]. Even if multiple resolution cells span the target, the mapping from radar extent to image extent is nonlinear and depends on the target's orientation, range, and the camera projection geometry. Furthermore, the radar measurement corresponds to a three-dimensional scattering center—not necessarily the geometric centroid of the object—so the projected point may not coincide with the center of the image bounding box [72].

4.5.2 DBSCAN Clustering with Range-Adaptive Parameters

When multiple radar returns are available from a single target—as is common for large vessels or at short range—density-based clustering algorithms can group detections into object-level clusters whose spatial extent informs the bounding box dimensions. The DBSCAN algorithm and its spatiotemporal extension ST-DBSCAN [81, 82] are widely used for this purpose. Schumann et al. [83] demonstrated that range-adaptive clustering parameters are essential: the minimum number of points and the neighborhood radius ϵ must scale with range to account for the angular divergence of the radar beam, which causes close targets to produce dense clusters and distant targets to produce sparse ones. A typical adaptation sets $\epsilon(r) = \epsilon_0 + \alpha\, r$, where ϵ_0 is a minimum neighborhood size and α is a range-scaling coefficient tied to the radar angular resolution [83].

4.5.3 RCS-Based Size Estimation

The radar cross section provides a complementary cue for estimating target size. While the relationship between RCS and physical dimensions is complicated by aspect-angle dependence, multipath, and constructive or destructive interference, broad correlations exist: a container ship at X-band may present an RCS of 10,000–100,000 m^2, while a small pleasure craft returns 1–100 m^2 [13]. Several authors have proposed using RCS as a soft prior for bounding box dimensions. Wang et al. [84] showed that incorporating RCS alongside range into a regression model improves range-dependent object size estimation compared to range-only baselines. The logarithmic nature of RCS variation motivates the use of $\log(\sigma_{\mathrm{RCS}})$ as a feature, which compresses the dynamic range and provides a roughly linear relationship with the logarithm of target length for broadside aspects [13].

4.5.4 Height Extension for Missing Elevation

When the radar provides only range and azimuth (no elevation), the projected bounding box has a well-defined horizontal center and width but an indeterminate vertical position and height. A common heuristic is to assume that the target rests on a known reference surface (the sea surface in maritime applications) and extends upward by an amount determined by a class-specific height prior [78]. Denoting the assumed height of the reference surface below the camera as h_0 and the expected target height as H, the vertical extent of the bounding box in pixels can be approximated as

$$\Delta v \approx f_y \frac{H}{Z_c}, \tag{29}$$

where f_y is the vertical focal length and Z_c is the depth to the target. This approximation is adequate at long range where perspective distortion is mild but becomes inaccurate at short range or for targets displaced vertically from the assumed surface. Singh et al. [85] proposed a depth-conditioned height regression that replaces the fixed prior with a learned mapping from radar range to bounding box height, trained on a small set of manually annotated examples.

4.6 Projection Error Analysis

Understanding the sources and magnitudes of projection error is essential for characterizing the quality of radar-generated bounding boxes and for designing noise-mitigation strategies in the downstream training pipeline.

4.6.1 Sources of Projection Error

Four primary error sources affect the accuracy of radar-to-camera projection:

(i) **Calibration error.** Residual inaccuracy in the estimated extrinsic and intrinsic parameters propagates through the projection equations. A translational offset of δt in the extrinsic calibration produces a pixel displacement of approximately $f\,\delta t/Z_c$, which grows linearly with the focal length and inversely with range [51].

(ii) **Distortion model mismatch.** If the lens distortion model does not match the true distortion, the undistortion step introduces spatially varying errors that are largest at the image periphery [17]. As noted in Section 4.2, using a Brown–Conrady model for a fisheye lens can produce errors exceeding 20 pixels [74].

(iii) **Temporal misalignment.** Radar and camera typically operate at different frame rates and are not hardware-synchronized, so a radar detection may correspond to a camera frame acquired at a slightly different time. For a target moving at velocity v with a temporal offset Δt, the positional error is $v\,\Delta t$. At 10 m/s and $\Delta t = 50$ ms, the displacement is 0.5 m, which at 100 m range and $f_x = 1000$ pixels produces a 5-pixel error [72]. Hardware triggering or interpolation-based alignment is therefore essential for moving targets.

(iv) **Three-dimensional ambiguity.** When the radar does not measure elevation, the projection must assume a vertical position. Errors in this assumption translate directly into vertical pixel errors, as described in Section 4.7.

4.6.2 Reprojection Error Metrics

The standard metric for evaluating projection accuracy is the Mean Absolute Reprojection Error (MARE), defined as

$$\text{MARE} = \frac{1}{N}\sum_{i=1}^{N}\left\|\mathbf{u}_i^{\text{proj}} - \mathbf{u}_i^{\text{gt}}\right\|_2, \tag{30}$$

where $\mathbf{u}_i^{\text{proj}}$ is the projected pixel location of the i-th radar detection and $\mathbf{u}_i^{\text{gt}}$ is the corresponding ground-truth pixel location determined from manual annotation or a higher-fidelity sensor [72]. For bounding box quality, the Intersection-over-Union (IoU) between the projected box and the ground-truth box is used; IoU thresholds of 0.5 and 0.7 are conventional in the object detection community [4].

4.6.3 Range-Dependent Error Characteristics

Projection error is not uniform across range. At close range, angular errors in calibration and radar measurement translate to large pixel displacements because the target subtends many pixels. At long range, the target occupies few pixels and small absolute errors can still produce poor IoU. Sengupta et al. [14] conducted a comprehensive analysis of radar-to-camera projection accuracy on the nuScenes dataset and reported a mean reprojection error of approximately 1.5 m at 200 m range, corresponding to roughly 8 pixels at a typical automotive focal length. The error was decomposed into a range-invariant calibration component and a range-dependent component dominated by radar angular uncertainty, with the latter scaling linearly with range [14]. In the maritime domain, where operational ranges extend to several nautical miles, the angular component dominates, and the coarser azimuth resolution of X-band marine radar (typically 1–2° versus 1–2° for automotive radar but at much greater range) exacerbates the problem [12].

4.7 The Three-Dimensional Ambiguity Problem

The most fundamental limitation of two-dimensional radar for camera projection is the absence of elevation information. Because the radar antenna integrates returns over a broad vertical beamwidth, the measured range and azimuth define a vertical line in three-dimensional space rather than a point. The projected pixel location depends on which point along that line is assumed, and any error in the assumed elevation maps directly to a vertical pixel displacement.

4.7.1 Elevation Uncertainty in 2D and 3D Radar

For a conventional marine radar with a fan beam spanning $\pm 12.5°$ in elevation [13], a target at range r could lie anywhere in a vertical band of height $2\,r\,\tan(12.5°) \approx 0.44\,r$. At 1 km, this corresponds to a 440 m vertical ambiguity—clearly unacceptable for precise camera projection. In practice, the assumption that maritime targets are located at the sea surface collapses this ambiguity to the waterline-to-masthead extent of the vessel, but even this reduced uncertainty can span tens of meters for large ships.

Three-dimensional radar systems that measure elevation, such as the 77 GHz automotive imaging radars discussed by Meyer and Kuschk [71], reduce the elevation ambiguity to the elevation resolution of the antenna, typically 2–5°. At 100 m, a 3° elevation resolution constrains the target's height to within approximately 5 m, which is often sufficient for accurate camera projection in automotive scenarios.

4.7.2 4D Imaging Radar

Four-dimensional imaging radar, which resolves targets simultaneously in range, azimuth, elevation, and Doppler, offers a further reduction in the three-dimensional ambiguity. These radars use large virtual antenna arrays to achieve elevation resolution below 2°, providing point clouds with hundreds or thousands of points per frame that approach LiDAR in spatial density [71]. The availability of per-point elevation measurements allows each radar return to be projected to a specific pixel rather than to a vertical column, eliminating the dominant source of projection uncertainty for two-dimensional systems. However, four-dimensional imaging radar is not yet available in the X-band marine form factor relevant to the present work.

4.7.3 Cross-Attention Approaches: CramNet

When elevation data is unavailable, learning-based methods can partially recover three-dimensional information by exploiting image context. Qian et al. [86] proposed a multi-view approach in which radar and camera features are associated along the epipolar line corresponding to the radar detection's bearing, effectively marginalizing over the unknown elevation. The cross-attention mechanism learns to weight image features at different heights based on their semantic content, assigning high weight to the image region that most likely corresponds to the radar target. This approach can be interpreted as a soft version of the frustum association used in CenterFusion [72], where instead of a hard frustum boundary, a learned attention distribution determines the association.

4.8 Maritime-Specific Considerations

While the projection mathematics are universal, the maritime environment introduces several practical complications that are absent or less severe in the automotive domain.

4.8.1 Non-Planar Surface and Wave Action

The sea surface is neither rigid nor planar. Wave action causes targets to heave, pitch, and roll, perturbing their three-dimensional positions on timescales of seconds. For the sensor platform itself—whether a fixed pier, a buoy, or a vessel—wave-induced motion shifts the camera and radar from their calibrated positions, invalidating the static extrinsic assumption. Inertial measurement units (IMUs) can compensate for platform motion at the cost of additional calibration between the IMU and sensor frames [87]. For target motion, the sea state imposes a stochastic displacement that adds to the projection error budget.

4.8.2 Extended Operational Ranges

Maritime radar-camera systems operate at ranges of hundreds of meters to several nautical miles, far exceeding the 50–200 m typical of automotive scenarios [74]. At these ranges, targets occupy very few pixels, bounding boxes are small, and the tolerance for

Table 5: Summary of radar-to-camera projection and association methods.

Method	Year	Association Space	Key Contribution / Performance
RRPN [78]	2019	Image plane	Radar-seeded region proposals; $\sim$100$\times$ faster than Selective Search
GRIF Net [73]	2020	RoI features	Gated modality weighting within projected RoIs
CenterFusion [72]	2021	Image plane (frustum)	Frustum-based association with CenterNet; +12.8 NDS on nuScenes
RODNet [84]	2021	Radar RF images	Object detection directly on radar heatmaps; camera-supervised training
RadarNet [83]	2020	BEV	Range-adaptive DBSCAN clustering; learned radar-to-3D-box regression
BEVFusion [75]	2023	BEV grid	Unified BEV pooling with 40$\times$ latency reduction; 72.9% NDS
CRAFT [80]	2023	Polar coordinates	Polar-space association; 41.1% mAP on nuScenes
RCBEVDet [76]	2024	BEV grid (RCS-aware)	RCS as BEV feature channel; 55.1% NDS (camera + radar)

projection error is correspondingly tighter in relative terms. An absolute reprojection error of 2 m may be acceptable at 50 m (corresponding to a few percent of the bounding box width for a passenger car) but is catastrophic at 2000 m, where a 10 m vessel subtends only a handful of pixels. This range asymmetry motivates range-dependent evaluation and possibly range-stratified training strategies [14].

4.8.3 AIS Integration for Vessel Identification

The Automatic Identification System (AIS), mandated for vessels above 300 gross tons under the International Maritime Organization's SOLAS convention, broadcasts vessel identity, type, dimensions, heading, and speed at regular intervals. When AIS tracks are available, they provide ground-truth vessel dimensions that can inform bounding box sizing far more precisely than RCS-based heuristics [74]. An AIS-reported vessel length of 200 m, combined with the projected range from radar, determines the expected angular subtent and hence the bounding box width in pixels. The fusion of radar, camera, and AIS data therefore constitutes a triple-sensor approach to label generation: radar provides range and bearing for projection, AIS provides dimensions and class identity for bounding box sizing, and the camera provides the image in which the annotation is placed [74, 87].

Table 5 summarizes the major projection and association methods reviewed in this section, together with their coordinate spaces, key contributions, and reported performance figures where available.

4.9 Summary and Research Gaps

This section has traced the complete mathematical pipeline from a radar detection in polar coordinates through world, camera, and pixel coordinate systems, establishing the projection equations that form the core of any radar-guided labeling system. The key findings and open gaps can be summarized as follows.

Mature projection mathematics, immature maritime application. The coordinate transformation chain—polar to Cartesian, rigid-body extrinsic transformation, pinhole projection with distortion correction—is well understood and standardized in the computer vision and autonomous-driving communities [2, 51]. However, its application to maritime X-band radar-camera systems has received limited attention, and the error characteristics at maritime ranges (500 m to several nautical miles) are not well quantified.

Wide-FOV distortion modeling is essential but under-studied for maritime radar-camera systems. The Kannala–Brandt [17] and Scaramuzza [18] models provide adequate tools for wide-FOV cameras, but their integration into maritime radar projection pipelines has not been systematically evaluated. In particular, the interaction between distortion modeling errors and range-dependent bounding box sizing is an unexplored source of systematic label noise.

BEV fusion is powerful but assumes a flat ground plane. Methods such as BEVFusion [75] and RCBEVDet [76] achieve state-of-the-art performance on automotive benchmarks by projecting features into a BEV grid, but the flat-ground assumption underlying BEV construction is violated in maritime environments with wave action. Adapting BEV methods to a non-planar, time-varying reference surface is an open problem.

Frustum association methods are directly applicable to radar-guided labeling. CenterFusion [72], GRIF Net [73], RRPN [78], and CRAFT [80] demonstrate that radar detections can be projected into the image to generate or refine bounding box proposals. These methods have been validated exclusively on the nuScenes automotive benchmark, and their extension to maritime targets—which differ in size, aspect ratio, and range distribution—remains untested.

The 3D ambiguity is the dominant error source for 2D radar. Without elevation information, the vertical position and extent of the projected bounding box must be assumed, producing systematic errors that degrade both localization and classification. Four-dimensional imaging radar resolves this ambiguity but is not available in the X-band marine form factor [71]. For two-dimensional marine radar, the combination of a sea-surface height assumption, AIS-reported vessel dimensions, and learned height regression represents the most promising mitigation strategy, though it has not been demonstrated end-to-end.

Bounding box sizing from radar remains an open problem. RCS-based size estimation [84], range-adaptive clustering [83], and depth-conditioned regression [85] each address part of the problem, but none provides a complete solution. The maritime domain offers the additional resource of AIS data, which has not been integrated into any published radar-camera bounding box generation pipeline [74].

Range-dependent error analysis is needed. The $\sim$1.5 m mean reprojection error reported for automotive scenarios at 200 m [14] provides a useful baseline, but maritime applications require characterization over a much wider range envelope (50 m to 5000 m). The interplay between radar angular resolution, camera focal length, and distortion model accuracy at extended range is an important direction for future work.

In summary, the projection and bounding box generation pipeline reviewed in this section provides the necessary mathematical and algorithmic machinery for converting radar detections into camera-space annotations. The principal challenge for maritime application lies not in the projection equations themselves but in the error sources that are amplified by the maritime operating environment: extended ranges, the absence of elevation data in conventional marine radar, wide-FOV distortion, wave-induced motion, and the lack of validated size priors for maritime targets. Addressing these challenges is essential for realizing a practical radar-guided labeling system.

5 Automated and Weakly-Supervised Dataset Construction Methods

The radar-to-camera projection pipeline described in Section 4 produces bounding box annotations by transforming radar detections into camera pixel coordinates. While this pipeline provides a scalable mechanism for generating training labels without human annotators, the resulting labels are inherently imperfect. Radar detections are subject to angular quantization, range-dependent beam spreading, multipath propagation, sea clutter false alarms, and calibration offsets—each of which introduces spatial error into the projected bounding boxes. The central methodological challenge, therefore, is not merely producing these labels but determining how to use them effectively: how can a camera-based object detector be trained from noisy, radar-generated supervision in a manner that yields performance competitive with models trained on manually curated annotations?

This question sits at the intersection of several active research areas in machine learning, including data programming, semi-supervised learning, teacher-student frameworks, cross-modal supervision, noisy label learning, and knowledge distillation. Each of these fields offers partial solutions, but none directly addresses the specific noise characteristics of radar-projected labels—noise that is systematic rather than random, geometry-dependent rather than class-dependent, and correlated across frames rather than independently distributed. This section surveys each of these research threads, evaluates their applicability to the radar-guided labeling problem, and identifies the theoretical and practical gaps that remain.

5.1 Data Programming and Programmatic Weak Supervision

The data programming paradigm, introduced by Ratner et al. [20], formalizes the idea that domain experts can encode their knowledge as *labeling functions*—simple, heuristic programs that each provide noisy votes on the correct label for a data point. Rather than requiring every labeling function to be individually accurate, the framework models their collective agreements and disagreements through a generative probabilistic model that estimates the true latent labels. Ratner et al. demonstrated that ensembles of twenty or more labeling functions could produce training sets yielding classifiers within a few percentage points of those trained on hand-labeled data, while requiring 2.8 times less labeling time. The key theoretical insight was that correlations among labeling functions—which arise naturally when heuristics share common failure modes—must be explicitly modeled to avoid overconfident label estimates.

The Snorkel system [88] extended data programming into a practical end-to-end framework. Snorkel introduced a noise-aware generative model that jointly estimates labeling function accuracies and correlations, producing probabilistic training labels that can be used directly with any discriminative model. On benchmark tasks spanning text classification, medical record annotation, and image labeling, Snorkel achieved an average improvement of 45.5% over the majority-vote baseline for combining labeling functions, while enabling dataset creation at rates 2.8 times faster than manual annotation. Critically, Snorkel demonstrated that the generative model could recover from systematic biases in individual labeling functions, provided those biases were not perfectly correlated across all functions.

The data programming framework is directly relevant to radar-guided camera labeling. A radar detection projected into camera coordinates can be viewed as a labeling function that votes on the presence and location of an object in the image. Additional labeling functions could include temporal consistency checks (does the detection persist across frames?), size plausibility constraints (is the projected bounding box consistent with expected vessel dimensions at the estimated range?), and clutter likelihood estimates (does the radar return exhibit characteristics of sea clutter?). The Snorkel framework provides a principled mechanism for combining these heterogeneous signals. However, a significant limitation is that standard data programming treats each data point as receiving independent label votes, whereas radar projections produce spatially structured labels—a calibration offset biases all projections in the same direction, not randomly. Extending the generative model to account for such spatially correlated error structures remains an open problem.

5.2 Pseudo-Labeling and Self-Training

Pseudo-labeling, introduced by Lee [21], is among the simplest semi-supervised learning strategies: a model trained on a small labeled set generates predictions on unlabeled data, and the high-confidence predictions are treated as ground truth for subsequent training rounds. Despite its simplicity, pseudo-labeling has proven remarkably effective. Lee demonstrated that converting soft model outputs to hard pseudo-labels above a confidence threshold τ functions as an entropy regularization mechanism, encouraging the model to produce sharper decision boundaries. The method requires no architectural modifications and integrates naturally with standard training pipelines.

Rosenberg et al. [89] provided an early application of self-training to object detection, demonstrating that iterative pseudo-labeling could improve detector performance when only a fraction of training images carried annotations. Their key finding was that self-training benefited from conservative confidence thresholds—using only the most confident predictions as pseudo-labels reduced the risk of error propagation, at the cost of slower training set expansion. This conservatism-accuracy tradeoff remains a central design consideration in modern self-training systems.

The Noisy Student framework of Xie et al. [22] transformed pseudo-labeling into a state-of-the-art training methodology. The approach trains a teacher model on labeled data, generates pseudo-labels for a large unlabeled corpus, then trains a student model (equal or larger in capacity) on the combined labeled and pseudo-labeled data with aggressive noise injection—including data augmentation, dropout, and stochastic depth. The student becomes the new teacher, and the process iterates. On ImageNet, Noisy Student achieved 88.4% top-1 accuracy, a substantial improvement over the supervised baseline, while also demonstrating dramatic improvements in robustness: the resulting models showed significantly reduced error rates on ImageNet-C (corrupted images), ImageNet-P (perturbation sequences), and ImageNet-A (adversarial examples). The critical insight was that noise injection during student training forced the student to learn representations more robust than those of its teacher, enabling progressive improvement across iterations.

FixMatch [90] unified consistency regularization and pseudo-labeling into a single, elegant framework. The method generates pseudo-labels from weakly augmented unlabeled images, then trains the model to match those pseudo-labels when processing strongly augmented versions of the same images. Only pseudo-labels above a confidence threshold contribute to the loss. On CIFAR-10, FixMatch achieved 94.93% accuracy using only 250 labeled examples—a result that approached the fully supervised baseline with four orders of magnitude fewer labels. The method's effectiveness stems from exploiting the model's own consistency as a signal for label quality, an idea with natural extensions to the cross-modal setting.

Caine et al. [91] applied pseudo-labeling to three-dimensional object detection in the autonomous driving domain, using the Waymo Open Dataset. Their work demonstrated that pseudo-labels generated by a high-capacity LiDAR-based detector could be used to train smaller, more efficient models, and that careful handling of confirmation bias—the tendency for pseudo-labels to reinforce the teacher's errors—was essential for maintaining detection quality across training iterations. This work is particularly relevant to the maritime setting, where a radar-based detection system (analogous to LiDAR in driving) provides initial labels for training a camera-based detector.

For radar-guided camera labeling, the self-training paradigm offers a natural iterative refinement strategy. An initial camera detector can be trained on radar-projected labels, and its high-confidence predictions can then supplement or replace the noisiest radar labels in subsequent training rounds. However, standard pseudo-labeling assumes that the initial labels are noisy but unbiased—that errors are approximately symmetric around the true label. Radar projection errors are not symmetric: calibration offsets produce consistent directional biases, and range-dependent beam spreading causes systematic box size errors that grow with distance. Applying naive self-training to such systematically biased initial labels risks amplifying, rather than correcting, these errors.

5.3 Teacher-Student Frameworks for Semi-Supervised Detection

Teacher-student frameworks extend the pseudo-labeling paradigm by maintaining two coupled models: a teacher that generates pseudo-labels and a student that learns from them. The critical design choice is the mechanism by which the teacher is updated, which determines the stability and quality of the pseudo-label signal.

The Mean Teacher framework of Tarvainen and Valpola [23] introduced the exponential moving average (EMA) mechanism for teacher updates. Rather than copying the student's weights directly, the teacher's parameters θ_T are updated as a running average: $\theta_T \leftarrow \alpha\theta_T + (1-\alpha)\theta_S$, where θ_S are the student parameters and α is a smoothing coefficient (typically 0.999). This averaging produces a teacher that is more stable than the student at any individual training step, yielding pseudo-labels with lower variance. On semi-supervised image classification benchmarks, Mean Teacher substantially outperformed both the Π-model and temporal ensembling baselines, demonstrating that teacher stability was more important than teacher accuracy for effective semi-supervised learning.

STAC (Self-Training with Augmentation-driven Consistency) [92] adapted the teacher-student paradigm specifically for object detection. The approach first trains a detector on labeled data, generates pseudo-labels for unlabeled images using the trained model, filters these pseudo-labels by confidence, and then retrains the detector on both labeled and pseudo-labeled data with strong augmentation applied to the pseudo-labeled images. On COCO, STAC demonstrated approximately two times the data efficiency of fully supervised training, meaning that a detector trained with STAC on N labeled images and M unlabeled images performed comparably to a supervised detector trained on roughly $2N$ labeled images. This result established that unlabeled data, when properly leveraged through pseudo-labeling, could substitute for expensive manual annotation.

Unbiased Teacher [93] addressed a fundamental limitation of prior teacher-student methods: the pseudo-labels generated by a teacher trained on limited labeled data are biased toward the class distribution of the labeled set, causing the student to inherit and amplify these biases. The solution was twofold. First, the teacher's classification head was calibrated using focal loss to reduce over-confidence on frequent classes. Second, the EMA update mechanism was applied throughout training rather than only at initialization, allowing the teacher to continuously benefit from the student's exposure to unlabeled data. On COCO, Unbiased Teacher achieved a 6.8 mAP improvement over the supervised baseline when using only 1% of available labels, with gains persisting at higher labeling fractions. The debiasing mechanism is particularly relevant when radar pseudo-labels exhibit class imbalance, as may occur when certain vessel types or sizes are more reliably detected by radar than others.

Soft Teacher [94] introduced a soft weighting mechanism for pseudo-labels rather than the hard confidence thresholding used in prior work. Instead of discarding pseudo-labels below a confidence threshold, Soft Teacher assigned each pseudo-label a weight proportional to the teacher's confidence, allowing uncertain but potentially informative labels to contribute to training with reduced influence. Additionally, the method employed a box jittering strategy to refine pseudo-label localization: multiple jittered versions of each pseudo-box were scored by the teacher, and the resulting scores were used to estimate a reliability weight for the box regression target. On COCO, Soft Teacher achieved state-of-the-art semi-supervised detection performance, demonstrating that soft weighting was strictly superior to hard thresholding for pseudo-label selection.

The teacher-student paradigm is architecturally well-suited to radar-guided camera labeling. The radar projection pipeline serves as an initial "teacher" that provides coarse bounding box annotations, and a camera detector trained on these annotations can serve as a refined teacher for subsequent student models. The EMA mechanism provides a natural smoothing of the noisy radar-derived supervision signal. However, existing teacher-student frameworks assume that pseudo-label noise decreases monotonically as training progresses—the teacher improves over time, producing better pseudo-labels for the student. With radar-projected labels, certain error modes (calibration offsets, range-dependent scaling) are fixed properties of the sensor geometry and cannot be corrected by model training alone. This creates a noise floor below which the teacher-student loop cannot improve without explicit correction of the underlying geometric errors.

5.4 Cross-Modal Supervision

Cross-modal supervision exploits the complementary strengths of different sensor modalities to generate training labels for one modality using detections from another. In the autonomous driving domain, the most common instantiation uses LiDAR point clouds—which provide accurate three-dimensional geometry but are expensive to deploy at scale—to generate training labels for camera-based detectors. The key advantage of cross-modal supervision over standard pseudo-labeling is that the teacher and student operate on fundamentally different data representations, reducing the risk of confirmation bias: a camera detector cannot simply memorize the visual features that triggered a radar detection, because radar and camera respond to different physical phenomena.

Hoffmann et al. [11] developed a differentiable warping module that projects radar features into the camera image plane, enabling end-to-end training of radar-camera fusion networks. While the primary application was fusion rather than label transfer, the differentiable projection mechanism demonstrated that geometric correspondences between radar and camera could be learned jointly with detection features, providing a pathway for correcting systematic projection errors during training. The approach achieved improved detection performance on the nuScenes dataset, particularly for distant and occluded objects where radar provides range information that is unavailable from monocular images alone.

Kim et al. [95] introduced AISail, which uses Automatic Identification System (AIS) transponder data—rather than radar—as a cross-modal supervision source for maritime camera detectors. AIS provides vessel identity, position, heading, and speed, which can be projected into camera coordinates using known camera calibration parameters. AISail demonstrated that automated AIS-camera annotation could produce training sets sufficient for competitive maritime object detection, achieving detection performance within a few percentage points of models trained on manual annotations. The AIS system provides highly accurate position information (GPS-grade, typically under 10 meters), but covers only vessels equipped with AIS transponders, which excludes small recreational craft, kayaks, debris, and other objects of interest. The radar-guided approach complements AIS by detecting all radar-reflective objects regardless of transponder status.

The cross-modal supervision paradigm highlights an important asymmetry between radar and camera perception. Radar provides geometric supervision: it tells the camera system *where* objects are in physical space. Camera provides appearance supervision: it reveals *what* objects look like and enables fine-grained spatial delineation. This complementarity is analogous to the LiDAR-camera relationship in autonomous driving, but with fundamentally different error characteristics. LiDAR provides centimeter-accurate three-dimensional point clouds; X-band marine radar provides two-dimensional (range, bearing) detections with angular resolution on the order of several degrees and range resolution on the order of meters. Consequently, radar-to-camera projected labels are substantially noisier than LiDAR-to-camera projected labels, necessitating more aggressive noise mitigation strategies. Furthermore, radar detections in the maritime environment are contaminated by sea clutter returns, multipath reflections from the water surface, and sidelobe detections—error sources that have no direct analog in the LiDAR-camera setting.

5.5 Learning with Noisy Labels

The noisy label learning literature provides the theoretical foundations most directly relevant to training camera detectors from radar-generated annotations. This subsection reviews the taxonomy of label noise, the major algorithmic approaches to noise-robust training, and—critically—the ways in which radar projection noise departs from the assumptions underlying existing methods.

5.5.1 Noise Taxonomy

Song et al. [19] provide a comprehensive survey of learning with noisy labels, establishing a taxonomy of noise types. *Symmetric noise* (also called uniform noise) corrupts each label with equal probability to any other class, independent of the true label and the instance. This is the simplest model and the most commonly studied, but it is rarely realistic. *Asymmetric noise* (also called class-dependent noise) models corruption probabilities that depend on the true class but not on the specific instance—for example, "ship" might be more frequently mislabeled as "buoy" than as "bridge." This model is captured by a noise transition matrix $T \in \mathbb{R}^{K\times K}$, where $T_{ij} = P(\tilde{y} = j \mid y = i)$ gives the probability that true class i is observed as noisy class j. *Instance-dependent noise* allows the corruption probability to depend on both the true class and the specific input features, modeling situations where some examples are inherently more difficult to label correctly than others. Instance-dependent noise is the most realistic but also the most difficult to model and correct.

5.5.2 Foundational Approaches

Natarajan et al. [27] established the theoretical foundation for learning with noisy labels by deriving unbiased risk estimators for binary classification under symmetric and asymmetric noise. They showed that if the noise rates ρ_+ and ρ_- (the probabilities of positive-to-negative and negative-to-positive label flips, respectively) are known, then the standard empirical risk on noisy labels can be corrected to yield an unbiased estimate of the clean risk. The corrected loss takes the form:

$$\tilde{\ell}(f(x), \tilde{y}) = \frac{(1 - \rho_{-\tilde{y}})\ell(f(x), \tilde{y}) - \rho_{\tilde{y}}\ell(f(x), -\tilde{y})}{1 - \rho_+ - \rho_-} \tag{31}$$

where ℓ is the base loss function and $\tilde{y}$ is the observed noisy label. While this result applies to classification, the insight that noise correction requires knowledge of the noise structure—specifically, the transition probabilities—generalizes to all noisy label settings.

Patrini et al. [25] extended this approach to multi-class deep learning by proposing two loss correction strategies based on the noise transition matrix T. The *forward correction* modifies the network's output probabilities: $\tilde{p}(y \mid x) = T^\top p(y \mid x)$, accounting for the fact that the observed label distribution is a noisy version of the true distribution. The *backward correction* modifies the loss function directly, scaling the per-class losses by the inverse noise transition matrix T^{-1}. Patrini et al. proposed estimating T from the data itself by examining the network's predictions on a clean validation set or by using the network's own confident predictions as proxies for clean labels. On CIFAR-10 and CIFAR-100 with synthetic noise, both correction methods substantially outperformed uncorrected training, with forward correction being more numerically stable.

The Generalized Cross Entropy (GCE) loss of Zhang and Sabuncu [26] provides a noise-robust alternative to the standard cross entropy loss without requiring explicit noise estimation. GCE is defined as:

$$L_q(f(x), y) = \frac{1 - f(x)_y^q}{q} \tag{32}$$

where $f(x)_y$ is the predicted probability for the observed label y and $q \in (0, 1]$ is a hyperparameter that interpolates between the mean absolute error ($q \to 0$, maximally noise-robust but slow to converge) and the standard cross entropy ($q \to 1$, fast convergence but noise-sensitive). The theoretical analysis showed that GCE is noise-tolerant under symmetric noise for any $q < 1$, and that moderate values ($q \approx 0.7$) provide a good tradeoff between noise robustness and convergence speed. The appeal of GCE is its simplicity: it requires no noise estimation and can be applied as a drop-in replacement for cross entropy.

5.5.3 Sample Selection and Peer Network Approaches

Co-teaching [96] takes a fundamentally different approach to noisy labels by leveraging the observation that two networks initialized differently will disagree on which examples are noisy. Co-teaching maintains two networks that are trained simultaneously; in each mini-batch, each network selects the examples with the smallest loss (which are most likely to have clean labels) and feeds them to its peer network for gradient updates. The fraction of examples selected is gradually reduced according to a schedule that estimates the noise rate. On CIFAR-10 with 45% symmetric noise, co-teaching maintained accuracy above 70%, whereas standard training collapsed to near-random performance. The method's effectiveness stems from the "small-loss" heuristic: deep networks tend to learn clean patterns before memorizing noisy ones, so examples with small loss early in training are disproportionately likely to have correct labels.

5.5.4 Noisy Labels in Object Detection

While most noisy label research focuses on image classification, Chadwick and Newman [10] specifically addressed the problem of training object detectors with noisy annotations. Working in the autonomous driving domain, they demonstrated that detector architectures (Faster R-CNN, SSD) are sensitive to both label noise in class assignments and localization noise in bounding box coordinates. Localization noise proved particularly damaging: even moderate perturbations to bounding box coordinates (5–10% of box dimensions) caused significant mAP degradation. Their proposed solution involved noise-aware loss functions that down-weighted the contribution of likely-noisy annotations, identified by inconsistency between the detector's predictions and the provided labels.

Li et al. [97] further investigated noisy labels in object detection, demonstrating that the two components of detection loss—classification loss and localization loss—respond differently to label noise. Classification noise can be partially mitigated by standard techniques (label smoothing, mixup), but localization noise requires spatial-aware corrections. They proposed a noise-adaptive training strategy that separately estimates and corrects for classification and localization noise, achieving improved detection performance on datasets with synthetically injected annotation errors.

5.5.5 The Critical Gap: Systematic Geometry-Dependent Noise

The noise models described above—symmetric, asymmetric, and even instance-dependent—share a common assumption that is violated by radar-projected labels: they treat label noise as approximately stationary, uncorrelated across examples, or at most correlated with instance difficulty. Radar projection noise has none of these properties. Instead, it exhibits four distinctive characteristics that distinguish it from the noise regimes studied in the existing literature.

Geometric bias from calibration offsets. Extrinsic calibration errors between the radar and camera coordinate systems produce consistent directional biases in all projected bounding boxes. A one-degree error in the radar-camera azimuthal alignment, for example, shifts every projected bounding box by a number of pixels that varies smoothly with the object's position in the image. This error is deterministic, not stochastic: the same calibration offset biases every detection in the same direction. Standard noise transition matrices cannot capture this spatial structure because they model class-conditional corruption probabilities, not coordinate-dependent spatial shifts.

Range-dependent scaling. X-band marine radar has an angular resolution on the order of several degrees, determined by the antenna beamwidth. A target that subtends a fraction of one beamwidth at close range may subtend multiple beamwidths at long range, but the bearing quantization remains constant. Consequently, the angular uncertainty in the projected bounding box center scales linearly with range in image coordinates (for small angles), while the bounding box size uncertainty depends on both range and the physical size of the target. At ranges beyond several kilometers, the projected bounding box may be displaced by tens or hundreds of pixels from the true object center. This range dependence means that the noise characteristics vary systematically across the dataset: nearby objects receive relatively accurate labels, while distant objects receive increasingly noisy ones. No existing noise model accounts for this structured, continuously varying noise level.

Multipath and sea clutter correlations. Multipath propagation off the sea surface creates phantom radar returns that appear at predictable locations relative to genuine targets, particularly at low grazing angles. Sea clutter false alarms, while individually stochastic, exhibit spatial and temporal correlations—clutter is concentrated at short ranges, intensifies in higher sea states, and produces returns that persist over multiple radar scans. These correlated false positives generate systematic label errors (bounding boxes where no object exists) that are clustered in specific image regions rather than uniformly distributed. Standard noisy label methods assume that false positives are rare and randomly located; the concentrated, persistent nature of clutter-induced false labels violates this assumption.

Temporal correlation within tracks. Radar track-level processing associates detections across multiple scans into persistent tracks, each characterized by a smoothed trajectory. All detections within a single track share the same sensor biases—the same calibration offset, similar multipath characteristics, and correlated measurement errors from the tracking filter. When these tracked detections are projected into consecutive camera frames, they produce temporally correlated label errors: a bounding box that is shifted left in frame t will also be shifted left in frames $t-1$ and $t+1$. Training on such temporally correlated labels without accounting for the correlation risks teaching the detector to reproduce the systematic bias rather than learning the true object locations.

Table 6 summarizes the relationship between standard noise models and the characteristics of radar projection noise. The instance-dependent noise model of Cheng et al. [28] comes closest to capturing the structured nature of radar noise, but even this model treats noise as a function of input features (image content) rather than sensor geometry (range, bearing, calibration state). What is needed is a *geometry-dependent* noise model that parameterizes the noise transition as a function of the radar-camera geometric configuration—the extrinsic calibration, the object's range and bearing, and the sea state—rather than the visual content of the image.

Table 6: Comparison of standard label noise models with radar projection noise characteristics. Standard models assume stochastic, instance-level noise, while radar projection noise is deterministic, spatially structured, and temporally correlated.

Noise Property	**Symmetric / Asymmetric**	**Instance-Dependent**	**Radar Projection**
Corruption source	Random flip / class confusion	Instance difficulty	Sensor geometry
Spatial structure	None (i.i.d.)	Feature-dependent	Range/bearing-dependent
Temporal correlation	None	None assumed	Strong (track-level)
Bias direction	Unbiased (symmetric) or class-biased	Sample-dependent	Calibration-determined
Localization effect	Not modeled	Not modeled	Primary error mode
Noise rate variation	Constant or class-constant	Varies per instance	Varies with range

5.6 Knowledge Distillation

Knowledge distillation, introduced by Hinton et al. [98], provides a framework for transferring learned representations from a large, high-capacity teacher model to a smaller, more efficient student model. The key mechanism is training the student to match the teacher's *soft* output distributions (softmax probabilities at elevated temperature) rather than hard labels. Soft targets carry richer information than one-hot labels because they encode the teacher's uncertainty and inter-class similarity structure—what Hinton et al. termed "dark knowledge." On image classification benchmarks, distillation from a large ensemble teacher to a single smaller student recovered most of the ensemble's performance at a fraction of the computational cost.

Li et al. [99] extended knowledge distillation to the noisy label setting, demonstrating that a student trained to match a noise-aware teacher's soft outputs could outperform both the teacher and a student trained directly on the noisy hard labels. The soft targets acted as an implicit noise filter: the teacher's uncertainty on mislabeled examples produced softer output distributions that reduced the student's exposure to incorrect supervision. The "knowledge distillation from noisy labels" framework showed that the distillation process itself provided noise robustness, even without explicit noise modeling.

In the radar-guided labeling context, knowledge distillation offers a two-stage training strategy. A first-stage model is trained on radar-projected labels, inevitably absorbing some of the systematic noise. This model then serves as a teacher, generating soft pseudo-labels for the same images. A second-stage student model, trained on the soft labels, benefits from the teacher's implicit filtering of noise: objects that the radar labeled but the teacher learned to doubt receive soft, uncertain labels, while consistently detected objects receive confident labels. This soft-label transfer mechanism is particularly well-suited to handling the spatially varying noise of radar projections, as the teacher naturally learns to be less confident on distant or peripheral objects where radar projection errors are largest.

5.7 Co-Training and Multi-View Learning

The co-training framework of Blum and Mitchell [24] provides a theoretical foundation for learning from multiple complementary views of data. The central theorem states that if two feature sets (views) are conditionally independent given the class label, and each view is individually sufficient for classification, then a small number of labeled examples suffices to learn an accurate classifier by iteratively labeling unlabeled examples with one view and training the other. Blum and Mitchell demonstrated this on web page classification, using page content and hyperlink structure as two conditionally independent views, achieving significant accuracy improvements from co-training with as few as twelve initial labeled examples.

Radar and camera constitute a natural multi-view pair for maritime object detection. Radar provides geometric features—range, bearing, Doppler velocity, radar cross section—while the camera provides appearance features—texture, color, shape, spatial context. These two feature sets are approximately conditionally independent given the presence and identity of a maritime target: the radar cross section of a vessel depends on its material composition and physical geometry, while its camera appearance depends on lighting, viewing angle, and surface texture. Their failure modes are also complementary: radar detects objects in fog, darkness, and rain but suffers from angular ambiguity and sea clutter; cameras provide precise spatial delineation and object classification but fail in low visibility and lack direct range measurement.

The co-training paradigm suggests an iterative algorithm in which a radar-based detector labels images for training a camera detector, and the camera detector's confident predictions label data for refining the radar tracker. This bidirectional supervision can progressively improve both modalities. However, the conditional independence assumption is violated when environmental conditions simultaneously affect both sensors—for example, heavy rain degrades both radar clutter characteristics and camera image quality—and the theoretical guarantees of co-training weaken accordingly. Characterizing the degree of view dependence between radar and camera in maritime conditions, and developing co-training variants that are robust to partial view dependence, remain open research questions.

5.8 Active Learning for Selective Verification

Active learning provides a complementary strategy to fully automated labeling by selectively identifying the most informative examples for human annotation. Brust et al. [100] developed active learning strategies specifically for deep object detection, proposing acquisition functions that quantify the informativeness of unlabeled images based on the detector's uncertainty. Their approach prioritized images containing objects with high classification entropy or large disagreement between multiple detection proposals, demonstrating that actively selected annotation budgets could achieve equivalent detection performance to random selection with substantially fewer labeled images.

In the radar-guided labeling pipeline, active learning serves a verification rather than annotation function. The vast majority of radar-projected labels can be used directly for training, but a small subset—identified by high radar-camera disagreement, anomalous projection geometry, or extreme range—can be routed to human annotators for verification or correction. This targeted human-in-the-loop approach preserves the scalability of automated labeling while providing quality control at the points of greatest uncertainty. The selection criterion for verification candidates should incorporate radar-specific uncertainty estimates: detections at the edge of the radar field of view, single-scan detections without track confirmation, and projections that fall near the image periphery where lens distortion is maximal are all candidates for preferential human review.

5.9 Quality Metrics for Automatically Generated Labels

Evaluating the quality of automatically generated labels requires metrics that capture both the overall fidelity of the label set and the specific error characteristics that affect downstream model training. Standard metrics include intersection over union (IoU) between generated and reference bounding boxes, label precision (the fraction of generated labels that correspond to real objects), label recall (the fraction of real objects that receive generated labels), and the aggregate noise rate (the fraction of labels that are incorrect or significantly mislocalized).

For radar-projected labels, these aggregate metrics are insufficient because they obscure the systematic structure of the noise. A label set with 70% mean IoU may perform very differently depending on whether the 30% error is uniformly distributed or concentrated at long range. Radar-specific quality metrics should therefore include: (1) systematic bias, measured as the mean signed displacement between projected bounding box centers and reference annotations, decomposed into horizontal and vertical components; (2) range-dependent accuracy profiles, reporting IoU and localization error as functions of the radar-estimated target range; (3) false alarm spatial density, characterizing the distribution of radar-generated labels that do not correspond to real objects, particularly in clutter-prone regions; and (4) temporal consistency, measuring the frame-to-frame stability of projected labels for tracked objects. These metrics provide the diagnostic information needed to select appropriate noise mitigation strategies and to predict the downstream impact of label noise on detector training.

5.10 Summary and Research Gaps

This section has surveyed the major methodological paradigms for training models from imperfect supervision: data programming, pseudo-labeling, teacher-student frameworks, cross-modal supervision, noisy label learning, knowledge distillation, co-training, and active learning. Each paradigm offers tools applicable to the radar-guided camera labeling problem, but each also carries assumptions that are violated by the specific noise characteristics of radar-projected labels. The most critical gaps are as follows.

First, **no existing noise model captures geometry-dependent systematic noise**. The noisy label learning literature has progressed from symmetric to asymmetric to instance-dependent noise models, but none of these parameterizes noise as a function of sensor geometry—the calibration state, target range, and bearing that determine the error characteristics of radar-projected labels. Developing a geometry-aware noise transition model that conditions the noise distribution on the radar-camera geometric configuration would provide the theoretical foundation for principled noise correction.

Second, **maritime cross-modal labeling remains largely unexplored**. The AI-Sail system [95] demonstrates the feasibility of cross-modal maritime annotation using AIS data, but radar-to-camera label transfer—which covers a broader class of targets including non-transponder-equipped vessels and floating debris—has not been systematically studied. The noise characteristics of radar-projected labels in the maritime domain (sea clutter, multipath, range-dependent beam spreading) differ fundamentally from those of

LiDAR-projected labels in the automotive domain, and maritime-specific noise mitigation strategies are needed.

Third, **the combination of co-training with noisy label learning is an open problem**. Co-training assumes clean labels generated by each view for the other; noisy label learning assumes imperfect labels but typically from a single source. Radar-guided camera labeling naturally combines both challenges—labels from a complementary view that are also noisy—but no existing framework integrates co-training's multi-view leverage with noisy label learning's noise correction in a unified approach.

Fourth, **active learning for selective verification of cross-modal labels is underexplored**. While active learning for object detection is well-studied [100], its application to the specific problem of selecting which automatically generated labels to route for human verification—using sensor-specific uncertainty estimates rather than model-based uncertainty alone—has received little attention. A radar-specific acquisition function that accounts for range-dependent noise, clutter likelihood, and projection geometry could enable highly efficient human oversight of the automated labeling pipeline.

Finally, **the interaction between systematic label noise and self-training dynamics is poorly understood**. Self-training and teacher-student methods assume that iterative refinement reduces noise over time. When the initial labels contain systematic biases (from calibration errors) and spatially correlated errors (from clutter patterns), the dynamics of error propagation through self-training iterations may diverge from the well-studied case of random noise. Theoretical and empirical characterization of this interaction is essential for designing robust training pipelines for radar-guided camera labeling systems.

6 Radar-Camera Fusion for Maritime Object Detection

While Chapter 4 examined automated labeling approaches that leverage radar and camera data in semi-coupled workflows, the fundamental architecture enabling both detection and annotation systems is the radar-camera fusion network itself. These fusion architectures serve dual purposes: they are the production systems for real-time object detection in autonomous platforms, and simultaneously, they provide the prediction engines that drive iterative refinement and pseudo-label generation in automated annotation pipelines. Understanding the state-of-the-art in radar-camera fusion is therefore essential not only for detecting maritime vessels but also for constructing the labeled datasets required to train such detectors. The rapid advances in automotive radar-camera fusion, driven by autonomous driving research and supported by large-scale datasets like nuScenes [2], RADIATE [101], and View of Delft [102], have produced sophisticated architectures achieving detection performance comparable to or exceeding LiDAR-based systems. However, maritime environments present fundamentally different sensor characteristics, environmental conditions, and object dynamics that challenge direct transfer of automotive fusion paradigms. This chapter surveys the taxonomy of fusion approaches, analyzes key architectural innovations from the automotive domain, examines emerging maritime-specific fusion methods, and evaluates the transferability of automotive architectures to X-band marine radar systems used in vessel detection.

6.1 Fusion Architecture Taxonomy

Radar-camera fusion methods can be taxonomized according to the processing stage at which heterogeneous sensor data is combined. This classification reveals fundamental trade-offs between complementarity preservation, computational efficiency, and robustness to sensor failures [15]. The three primary categories are early fusion, late fusion, and mid-level fusion, each with distinct architectural characteristics and performance implications.

Early fusion architectures merge raw or minimally pre-processed data from both modalities at the lowest level of the processing pipeline. In radar-camera systems, this typically involves projecting radar detections or range-Doppler-azimuth tensors into the image plane and concatenating them with RGB channels to form multi-channel input tensors. CRF-Net [103] exemplifies this approach by creating a four-channel input combining camera RGB and radar range information, processing the fused representation through a shared encoder backbone. Early fusion maximizes the network's capacity to learn cross-modal correlations at fine spatial scales, potentially capturing subtle complementarities such as radar reflections corresponding to specific visual textures. However, this approach suffers from several limitations: it requires precise spatial alignment between modalities, making it sensitive to calibration errors; it commits to a fixed fusion strategy that cannot adapt to varying sensor reliability; and it provides limited interpretability regarding which modality contributes to specific predictions. In maritime applications, where radar and camera data exhibit vastly different spatial resolutions and coverage patterns, early fusion proves particularly challenging.

Late fusion architectures adopt the opposite strategy, processing each modality through independent detection pipelines and combining predictions only at the output stage. This decoupled approach enables modality-specific architectures optimized for each sensor's characteristics and provides natural robustness to sensor failures, as one modality can compensate when the other is degraded. Fusion at the decision level typically employs Kalman filters, weighted score combination, or non-maximum suppression across modalities to produce final detections. Late fusion is conceptually simple and allows incremental addition of new sensors without retraining entire networks. However, it fails to exploit complementary features during the critical feature extraction phase, potentially missing opportunities for one modality to disambiguate difficult cases in the other. Performance is fundamentally limited by the weakest modality, as fusion occurs too late to guide feature learning based on cross-modal correspondences.

Mid-level fusion, also termed feature-level fusion, has emerged as the dominant paradigm in contemporary radar-camera architectures. These methods extract intermediate feature representations from each modality through dedicated encoders, then combine features through concatenation, attention mechanisms, or learned fusion modules before proceeding to detection heads. This approach balances the complementarity exploitation of early fusion with the modularity and robustness of late fusion. The field has converged toward Bird's Eye View (BEV) representations as the canonical fusion space, transforming both camera perspective view and radar polar coordinates into a common metric ground plane [75, 76, 104]. BEV fusion provides natural alignment between modalities, explicitly represents spatial relationships critical for autonomous navigation, and facilitates multi-frame temporal integration. The key architectural challenge in mid-level fusion is designing effective cross-modal attention or gating mechanisms that enable each modality to adaptively weight the other's contributions based on contextual reliability [73, 105].

6.2 Automotive Radar-Camera Fusion Architectures

The automotive domain has produced a rich landscape of radar-camera fusion architectures, driven by the dual imperatives of safety-critical perception and cost-effective sensor suites. These methods have established architectural patterns and performance benchmarks that inform maritime fusion system design, even as domain-specific adaptations remain necessary.

6.2.1 Early and Multi-Level Fusion Approaches

CRF-Net [103], proposed by Nobis et al. in 2019, pioneered end-to-end learning of radar-camera fusion for 3D object detection on the nuScenes dataset. The architecture employs a multi-level fusion strategy, injecting radar information at multiple stages of a ResNet-based encoder. Radar point clouds are projected into the image plane, and their range, radial velocity, and RCS values are encoded as additional input channels alongside RGB. To address the challenge that radar provides substantially sparser spatial information than camera, CRF-Net introduces a "BlackIn" training strategy: the network is initially trained on camera-only data with radar channels masked (blacked out), then fine-tuned with radar input enabled. This curriculum learning approach prevents the network from overfitting to spurious radar patterns during early training when it has not yet learned robust visual representations. CRF-Net demonstrated that even sparse automotive radar (typically 50-200 detections per frame) could improve 3D detection accuracy and provide direct velocity estimation unavailable from monocular cameras. However, the early fusion design inherits the limitations discussed previously, particularly sensitivity to calibration accuracy and inability to handle missing radar data at inference time.

GRIF Net [73], introduced by Kim et al. in 2020, addresses the modality imbalance problem through a gated region of interest (RoI) fusion mechanism. Rather than fusing at the input level, GRIF Net extracts RoI features from both camera and radar streams using modality-specific backbones, then employs learned gating functions to adaptively weight the contribution of each modality per region. The architecture uses radar frustum associations: for each radar detection, a frustum is constructed in the camera view, and CNN features within that frustum are extracted and fused with radar features. The gating mechanism learns to up-weight camera features in regions with rich visual texture and up-weight radar features in regions with poor illumination or occlusions. On nuScenes, GRIF Net achieved detection performance comparable to LiDAR-camera fusion while using only cameras and the comparatively inexpensive automotive radar. This result was significant in demonstrating that radar-camera fusion could approach the gold standard of LiDAR-camera systems for autonomous driving perception. The gated fusion concept introduced by GRIF Net has influenced subsequent architectures, establishing adaptive weighting as a key principle for handling heterogeneous sensor reliability.

6.2.2 Center-Based Detection and Association

CenterFusion [72], developed by Nabati and Qi in 2021, adapts the CenterNet anchor-free detection paradigm to radar-camera fusion. The core innovation is a frustum-based radar-camera association strategy combined with learned velocity estimation. CenterFusion processes camera images through a backbone to produce a BEV center heatmap, offset map, and dimension map. Simultaneously, radar detections are associated with camera-detected objects by projecting radar points into image space and constructing frustums that connect each radar detection to the camera viewpoint. For each detected center in the camera heatmap, the network extracts features within the corresponding radar frustum and fuses them via concatenation followed by fully connected layers. This architecture explicitly leverages radar's strength in providing accurate range and velocity measurements while relying on camera for dense semantic information and lateral localization. On nuScenes, CenterFusion achieved a 12% improvement in NuScenes Detection Score (NDS) over camera-only CenterNet, with particularly large gains in range estimation accuracy and velocity prediction (radar provides direct Doppler velocity). The frustum association strategy proved robust to calibration errors and radar sparsity, as it does not require perfect point-level alignment. However, CenterFusion operates in the camera perspective view rather than BEV, limiting its ability to represent 3D spatial relationships and integrate temporal information effectively. The method also relies on camera detections to initiate associations, meaning radar-only objects (visible to radar but occluded in camera) cannot be detected.

CFTrack [106], an extension by the same authors, extends CenterFusion to the tracking domain by incorporating a spatial-temporal memory module that maintains object feature histories across frames. This temporal extension exploits radar velocity measurements to predict object motion and maintain track consistency even under occlusions or missed detections in individual frames.

6.2.3 Spatial Transformer and Attention Mechanisms

CRAFT (Camera-Radar Fusion Transformer) [80], proposed by Kim et al. in 2022, introduces a spatio-contextual transformer architecture that performs fusion in polar radar coordinates rather than Cartesian BEV. The key insight is that radar naturally produces measurements in polar coordinates (range, azimuth, Doppler), and transforming to Cartesian BEV introduces interpolation artifacts and computational overhead. CRAFT maintains radar data in its native polar representation and uses cross-attention mechanisms to associate camera features with radar bins. The architecture consists of three main components: a camera encoder that extracts multi-scale features, a radar encoder that processes range-azimuth-Doppler tensors, and a spatio-contextual transformer that performs bidirectional cross-attention between modalities. The transformer allows camera features to attend to relevant radar bins based on learned spatial relationships, while radar features attend to camera regions that provide complementary semantic information. CRAFT achieved 41.1% mAP and 52.3% NDS on nuScenes, representing state-of-the-art radar-camera fusion performance at the time of publication. The polar-coordinate fusion strategy is particularly relevant for maritime radar systems, which produce data in polar (range-bearing) format and for which Cartesian transformation can introduce significant distortion at long ranges where azimuthal resolution is coarse.

CramNet [105], introduced by Hwang et al. in 2022, employs a ray-constrained cross-attention mechanism that explicitly models the geometric relationship between camera rays and radar measurements. The architecture represents camera features as ray-based queries and radar features as range-gated responses, implementing cross-attention that respects the epipolar geometry of the sensor configuration. This geometric constraint reduces the search space for feature associations and improves fusion efficiency. CramNet also introduces a range-conditioned feature encoding scheme that explicitly represents distance information in both modalities before fusion, addressing the challenge that cameras provide weak depth cues while radar provides precise range but poor angular resolution.

6.2.4 Bird's Eye View Fusion Paradigms

BEVFusion [75], developed by Liu et al. in 2023, establishes a unified BEV representation as the fusion space for multi-modal perception. While the original work focused on camera-LiDAR fusion, the architecture is modality-agnostic and has been adapted for radar-camera systems. The key contribution is an efficient BEV pooling operation that transforms perspective camera features into BEV space using predicted depth distributions, then aligns features from all sensors in the common BEV grid. This unified representation enables both detection and map segmentation tasks from a shared feature space. BEVFusion demonstrated a 40-fold reduction in inference latency compared to previous view transformation methods while achieving 13.6% higher mean intersection-over-union (mIoU) on BEV map segmentation tasks. The architecture's efficiency stems from optimized CUDA kernels for the BEV pooling operation and a carefully designed pre-computation strategy for view transformation. For radar-camera fusion, BEVFusion's framework allows radar point clouds or range-azimuth tensors to be directly projected into BEV and fused with camera BEV features through concatenation or convolutional processing. The BEV representation naturally accommodates radar's strength in range measurement and camera's strength in semantic recognition, while providing a metric space for downstream planning and control modules.

CRN (Camera-Radar Net) [104], proposed by Kim et al. in 2023, extends the BEV fusion paradigm with radar-guided depth estimation and real-time performance optimization. The architecture uses radar depth measurements as explicit supervision for the camera BEV transformation, addressing the ill-posed problem of monocular depth estimation. A radar-guided depth network predicts per-pixel depth distributions conditioned on radar detections within the camera frustum, then projects camera features into BEV using these depth estimates. The fusion is performed in BEV space via convolutional layers that process concatenated camera and radar features. CRN achieves 20 frames per second (FPS) inference on automotive hardware, making it suitable for real-time deployment. Notably, CRN demonstrates superior performance to LiDAR-camera fusion at ranges beyond 100 meters, where LiDAR point density decreases but radar maintains detection capability. This long-range advantage is particularly relevant for maritime applications, where detection ranges of several kilometers are required. The architectural efficiency of CRN derives from using radar depth guidance to avoid costly depth network ensembles or iterative refinement, instead leveraging the sparse but accurate radar range measurements to constrain the camera depth prediction.

RCBEVDet [76], introduced by Lin et al. in 2024, represents the current state-of-the-art in radar-camera BEV fusion. The architecture employs a dual-stream backbone that extracts features from camera and radar data independently, then fuses in BEV space using an RCS-aware (radar cross section aware) attention mechanism. The key innovation is explicit modeling of radar measurement uncertainty: RCS values, which indicate radar return strength, are used to weight the contribution of radar features during fusion, down-weighting noisy or low-confidence radar detections. The network also incorporates a temporal fusion module that aggregates BEV features across multiple frames, exploiting motion cues from radar velocity measurements to align features temporally. RCBEVDet achieves state-of-the-art performance on both nuScenes and View of Delft datasets, demonstrating strong generalization across different radar sensor configurations. The RCS-aware fusion mechanism is particularly relevant for maritime radar, where RCS varies dramatically with vessel size, aspect angle, and sea state, and proper uncertainty modeling is critical for robust detection.

6.2.5 Joint Detection and Tracking Architectures

CR3DT (Camera-Radar 3D Detection and Tracking) [107], developed by Broedermann et al. in 2024, unifies object detection and tracking in a joint radar-camera architecture. Rather than treating detection and tracking as sequential tasks, CR3DT learns both objectives end-to-end with shared feature representations. The architecture extends the BEV fusion paradigm with a tracking head that predicts object associations across frames based on appearance features, motion dynamics derived from radar velocity, and spatial proximity in BEV space. The joint training strategy enables the network to learn features that are both discriminative for detection and consistent for tracking across time. On nuScenes, CR3DT achieves a 14.9% improvement in Average Multi-Object Tracking Accuracy (AMOTA) over separate detection-then-tracking pipelines, demonstrating the benefit of end-to-end optimization. The architecture also exhibits improved robustness to missed detections and false positives, as the tracking module can maintain object identities through brief occlusions or detection failures by leveraging temporal context. For maritime surveillance, where vessels may be intermittently occluded by waves or fog and where maintaining track continuity is critical for maritime domain awareness, joint detection-tracking architectures offer substantial advantages over traditional tracking-by-detection approaches.

6.2.6 Knowledge Distillation and Modality-Agnostic Frameworks

CRKD (Camera-Radar Knowledge Distillation) [108], proposed by Zhao et al. in 2024, addresses the problem of training effective radar-camera fusion networks when labeled radar data is scarce but labeled camera data is abundant. The method employs cross-modality knowledge distillation, where a teacher network trained on camera data transfers learned representations to a student network that fuses camera and radar. The key innovation is a modality translation module that learns to predict pseudo-radar features from camera features, enabling the teacher network's camera-trained representations to guide radar feature learning. During inference, the student network operates on actual radar-camera pairs, but during training, the pseudo-radar features provide additional supervision when real radar data is unavailable or unlabeled. CRKD demonstrates that this distillation approach can improve radar-camera fusion performance by 8-12% when radar training data is limited, a scenario highly relevant to maritime applications where large-scale labeled radar-camera datasets are scarce.

FUTR3D [109], introduced by Chen et al. in 2023, presents a modality-agnostic query-based fusion architecture inspired by DETR (Detection Transformer). The method represents objects as learned queries in BEV space and uses cross-attention to aggregate features from arbitrary combinations of sensors. FUTR3D does not assume specific sensor modalities; instead, it learns to attend to relevant features from whatever sensors are available, making it naturally extensible to new sensor types and robust to sensor failures. The architecture processes camera and radar data through modality-specific encoders to produce features in a common embedding space, then uses object queries to aggregate information across modalities and across multiple frames. This query-based approach decouples detection from the specific fusion strategy, allowing the network to learn optimal modality weighting implicitly through attention mechanisms. FUTR3D achieves competitive performance with specialized fusion architectures while providing greater flexibility and modularity, characteristics valuable for maritime platforms that may have heterogeneous sensor configurations across different vessels or operational scenarios.

6.3 Architecture Comparison

Table 7 summarizes the key architectural characteristics and performance of the radar-camera fusion methods surveyed above. All performance metrics are reported on the nuScenes validation set using the standard NuScenes Detection Score (NDS) and mean Average Precision (mAP) metrics, where available from the original publications.

The table reveals several clear trends in the evolution of radar-camera fusion architectures. First, the field has decisively moved from early fusion toward mid-level fusion approaches, with BEV representation emerging as the dominant fusion space since 2023. Second, attention mechanisms and transformer architectures have largely supplanted fixed fusion strategies, enabling adaptive modality weighting. Third, temporal integration has become increasingly sophisticated, progressing from simple frame stacking to learned motion models and joint detection-tracking. Fourth, real-time performance has improved dramatically, with methods like CRN achieving 20 FPS while maintaining or improving detection accuracy. Finally, there is increasing recognition that radar measurement quality (RCS, SNR) should explicitly inform fusion, as exemplified by RCBEVDet's RCS-aware attention.

6.4 Maritime-Specific Radar-Camera Fusion

While the automotive domain has produced the majority of radar-camera fusion research, maritime applications present distinct sensor characteristics and operational requirements that necessitate domain-specific approaches. The limited but growing body of maritime fusion research reveals both opportunities to leverage automotive innovations and challenges unique to the maritime domain.

The WHUT-MSFVessel dataset and associated fusion methods [110] represent one of the few comprehensive maritime multi-sensor fusion efforts. The dataset combines X-band marine radar, optical cameras, and Automatic Identification System (AIS) data for vessel detection and tracking in inland waterways and coastal regions. The proposed fusion architecture employs a two-stage approach: radar and camera are first processed independently through region proposal networks, then AIS data is used to associate and refine detections. The system achieves Multi-Object Tracking Accuracy (MOTA) scores of 0.872 for camera and 0.938 for radar on the validation set, with fusion improving robustness to occlusions and adverse weather compared to single-modality baselines. However, the architecture relies on AIS availability, which limits applicability to non-cooperative targets (vessels not broadcasting AIS), and the fusion strategy is relatively simple (late-stage score combination) compared to contemporary automotive methods. Nonetheless, the dataset provides valuable ground truth for maritime fusion algorithm development, including annotations for vessel type, size, and heading that are specific to maritime scenarios.

The MOANA dataset [7] provides multimodal offshore sensing data combining X-band marine navigation radar, millimeter-wave W-band radar, optical cameras, infrared cameras, and GPS/IMU. The dataset is specifically designed for autonomous surface vehicle (ASV) perception and includes challenging maritime conditions such as high sea states, fog, and nighttime operations. While the associated baseline methods primarily address

Table 7: Comparison of radar-camera fusion architectures for 3D object detection in automotive applications.

Method	Year	Fusion Level	Radar Type	Performance	Key Innovation
CRF-Net [103]	2019	Early/Multi	77 GHz Auto	–	BlackIn training, multi-level fusion
GRIF Net [73]	2020	Mid (RoI)	77 GHz Auto	–	Gated RoI fusion, adaptive weighting
CenterFusion [72]	2021	Mid (Frustum)	77 GHz Auto	NDS: +12%	Frustum association, velocity fusion
CFTrack [106]	2021	Mid (Frustum)	77 GHz Auto	–	Spatial-temporal memory for tracking
CramNet [105]	2022	Mid (Attention)	77 GHz Auto	–	Ray-constrained cross-attention
CRAFT [80]	2022	Mid (Transformer)	77 GHz Auto	52.3% NDS, 41.1% mAP	Polar coordinate fusion, transformer
BEVFusion [75]	2023	Mid (BEV)	Multi-modal	mIoU: +13.6%	Unified BEV, 40×

single-modality detection, the dataset structure enables future radar-camera fusion research that accounts for the unique characteristics of X-band marine radar: 360-degree scanning, range-bearing format, sea clutter, and multi-path reflections from water surfaces. The inclusion of both X-band and W-band radar data also enables research on multi-radar fusion, potentially combining the long-range coverage of X-band with the higher resolution of millimeter-wave systems.

Several works have explored fusion of Synthetic Aperture Radar (SAR) imagery with optical cameras for maritime vessel detection [111]. SAR provides weather-independent imaging at long ranges and is particularly valuable for wide-area maritime surveillance from satellites or aircraft. However, SAR fusion presents different challenges than real-time pulse-Doppler radar fusion: SAR images are already processed into image format (rather than point clouds or range-Doppler tensors), temporal resolution is typically much lower (minutes to hours between passes rather than seconds), and registration with optical imagery requires geometric correction for terrain relief and platform motion. Consequently, SAR-optical fusion methods often employ image-level fusion techniques such as multi-stream CNNs with late-stage concatenation, rather than the sophisticated BEV or attention-based fusion used in automotive radar-camera systems. While valuable for surveillance applications, SAR fusion architectures are less directly applicable to real-time ASV or autonomous ship perception systems that require high-rate sensor fusion for collision avoidance and navigation.

The scarcity of maritime radar-camera fusion research compared to automotive applications reflects the limited availability of large-scale labeled datasets, the diversity of maritime radar systems (X-band, S-band, C-band; rotating antennas vs. phased arrays; pulse-Doppler vs. FMCW), and the fragmented nature of the maritime autonomy industry compared to the concentrated automotive sector. However, as autonomous surface vehicles and intelligent ship systems progress toward commercial deployment, the need for robust maritime fusion architectures will drive increased research attention.

6.5 Transferability of Automotive Fusion Methods to Maritime Domain

The sophisticated radar-camera fusion architectures developed for automotive applications provide a strong foundation for maritime perception systems, yet direct transfer is hindered by fundamental differences in sensor characteristics, environmental conditions, and operational requirements. Understanding which architectural principles transfer gracefully and which require substantial adaptation is critical for efficient development of maritime fusion systems.

6.5.1 Principles That Transfer Well

Several core architectural concepts from automotive fusion are largely domain-agnostic and apply directly to maritime scenarios. Bird's Eye View representation as a fusion space is perhaps the most universally applicable principle. BEV provides a natural coordinate system for both automotive and maritime perception: it represents space in metric ground plane (or water surface plane) coordinates, facilitates temporal integration by providing a consistent reference frame as the platform moves, and decouples detection from viewpoint-specific effects. Maritime radar naturally produces measurements in a BEV-like polar coordinate system (range and bearing from the vessel), making transformation to BEV straightforward. Cameras require view transformation through depth estimation, a challenge common to both domains, though maritime scenes may provide stronger monocular depth cues (horizon line, known vessel dimensions) than automotive scenes.

Center-based detection paradigms, as exemplified by CenterFusion, transfer well to maritime applications. Vessels, like vehicles, are well-modeled as objects with defined centers, and representing detections as center points with associated attributes (dimensions, heading, velocity) provides a compact and natural representation. The anchor-free nature of center-based detection is particularly valuable in maritime scenarios where vessel sizes span extreme ranges (from 10-meter recreational boats to 400-meter container ships), making anchor box design challenging. The frustum association strategy for connecting radar detections to camera regions also applies directly to maritime fusion, requiring only adaptation of frustum geometry to maritime radar scan patterns.

Attention mechanisms for adaptive modality fusion are domain-agnostic and address the fundamental challenge that sensor reliability varies with environmental conditions, target characteristics, and range. In maritime environments, cameras degrade in fog, rain, or darkness, while radar maintains functionality but with increased sea clutter. Attention-based fusion, as employed in CRAFT, CramNet, and RCBEVDet, allows the network to learn when to rely more heavily on each modality based on context. This principle transfers directly to maritime applications, though the specific conditions triggering modality re-weighting differ from automotive scenarios.

Temporal fusion and tracking principles also transfer well. Maritime vessel tracking requires maintaining object identity across frames, predicting motion, and handling occlusions (vessels passing behind islands or other vessels), challenges shared with automotive tracking. Radar velocity measurements provide direct Doppler information in both domains, enabling accurate motion prediction. The joint detection-tracking paradigm of CR3DT is particularly relevant for maritime surveillance, where track continuity and motion history are critical for collision avoidance and threat assessment.

6.5.2 Required Domain Adaptations

Despite these transferable principles, several fundamental differences between automotive and maritime radar necessitate substantial architectural modifications. The most significant difference is radar data format and scanning pattern. Automotive radars produce sparse point clouds (50-300 detections per frame) or dense range-azimuth-Doppler tensors from electronic scanning, with typical update rates of 10-20 Hz and limited field of view (typically 60-120 degrees). Maritime X-band radars produce 2D range-bearing images (Plan Position Indicator format) from mechanical rotation, with 360-degree coverage but lower update rates (2-4 seconds per rotation for typical merchant radars, up to 50 revolutions per minute for specialized tracking radars). This fundamental difference in data structure requires adaptation of the radar encoder architecture: automotive methods often use PointNet-style set networks or 3D convolutions on range-azimuth-Doppler tensors, while maritime applications require 2D convolutional encoders or polar-coordinate networks that process the PPI image format. Recent work on polar coordinate fusion [80] provides a promising direction for maritime adaptation, as it maintains radar data in native polar format without expensive Cartesian transformation.

Range scales differ dramatically between automotive and maritime applications. Automotive radar fusion typically operates at ranges of 5-200 meters, with most detections within 100 meters. Maritime vessel detection requires ranges of 500 meters to 10+ kilometers, with radar maintaining effective detection capability across this entire range while camera resolution degrades substantially beyond 1-2 kilometers. This range difference impacts several architectural choices: the BEV grid resolution and extent must be adapted to cover larger areas (requiring more memory and computation), depth estimation for camera BEV transformation becomes increasingly uncertain at long ranges, and the relative reliability of modalities shifts (radar becomes dominant at long range, camera at short range). Architectures that hard-code range assumptions or use fixed BEV grid sizes require modification for maritime scales.

Object characteristics and motion dynamics differ substantially between domains. Vehicles exhibit constrained motion (roads, lanes, traffic rules) and relatively predictable behavior, while vessels move freely in 2D with smooth, physics-constrained trajectories influenced by momentum, wind, and currents. Vessel dimensions span a wider range than road vehicles, and radar cross-section varies dramatically with vessel size, hull material, and sea state (a small fiberglass recreational boat presents radically different radar returns than a large steel cargo vessel). These differences require adaptation of detection head architectures (dimension prediction ranges), motion models (for temporal fusion and tracking), and uncertainty modeling (RCS-based confidence weighting may require maritime-specific calibration).

Environmental effects present distinct challenges in maritime fusion. Sea clutter (radar returns from waves) has no direct automotive analogue and can dominate radar signals in high sea states, requiring specialized clutter suppression or explicit modeling in the fusion architecture. Multi-path reflections from the water surface cause ghost targets and range distortion, a phenomenon that could be explicitly modeled in maritime fusion networks. Camera images in maritime scenarios often exhibit less texture and fewer geometric constraints than urban automotive scenes (open water is nearly featureless), making monocular depth estimation more challenging. Fog, spray, and salt accumulation on camera lenses are more severe than automotive rain or dust. These environmental effects suggest that maritime fusion architectures should incorporate explicit environmental state estimation (sea state, visibility conditions) to guide adaptive fusion.

6.5.3 Most Promising Methods for Maritime Adaptation

Given the transferability analysis above, several automotive fusion architectures emerge as particularly well-suited for adaptation to maritime vessel detection. RCBEVDet [76] is perhaps the most directly applicable, as its BEV fusion paradigm, RCS-aware attention mechanism, and temporal fusion align well with maritime requirements. Adapting RCBEVDet to maritime radar would require: modifying the radar encoder to accept 2D PPI images rather than point clouds, adjusting BEV grid extent and resolution for maritime range scales, and retraining the RCS-aware attention on maritime radar cross-section characteristics. The core architecture would remain intact.

CRN [104] offers advantages for maritime applications due to its explicit radar-guided depth estimation and demonstrated long-range performance. Maritime scenes provide strong geometric constraints for depth estimation (horizon line, water surface plane), which CRN's radar guidance could exploit more effectively than in automotive scenarios. The method's real-time performance (20 FPS) is adequate for maritime update rates. Adaptation would require similar encoder modifications as RCBEVDet and potentially adjustment of the depth estimation network to leverage maritime-specific cues.

FUTR3D [109] is appealing for maritime applications due to its modality-agnostic design and extensibility. Maritime platforms often have heterogeneous sensor suites (X-band radar, optical camera, thermal camera, AIS), and FUTR3D's query-based fusion naturally accommodates arbitrary sensor combinations. The architecture's flexibility would facilitate incremental addition of sensors and adaptation to different maritime platform configurations.

CenterFusion [72], while operating in perspective view rather than BEV, offers simplicity and proven effectiveness that make it a strong baseline for maritime fusion. The frustum association strategy requires minimal modification for maritime radar scanning patterns, and the method's lower architectural complexity compared to full BEV transformers may facilitate initial maritime dataset construction and benchmarking.

6.6 X-Band Marine Radar vs. 77 GHz Automotive Radar

Table 8 summarizes the key differences between X-band marine radar systems commonly used for vessel detection and 77 GHz automotive radar systems for which most fusion architectures were designed. These differences fundamentally shape the applicability and required modifications of automotive fusion methods for maritime use.

Table 8: Comparison of X-band marine radar and 77 GHz automotive radar characteristics.

Characteristic	**X-Band Marine Radar**	**77 GHz Automotive Radar**
Frequency	8-12 GHz (X-band)	76-81 GHz (W-band)
Maximum Range	5-50 km (mode dependent)	50-300 m
Range Resolution	5-30 m (pulse width dependent)	0.1-0.5 m (FMCW chirp dependen
Azimuth Resolution	0.5-2 degrees (antenna beamwidth)	1-5 degrees (antenna array)
Scanning Method	Mechanical rotation (360 deg)	Electronic scanning (60-120 deg F
Update Rate	0.5-3 seconds per rotation	10-20 Hz
Field of View	360 degrees (full horizon)	60-120 degrees (forward sector)
Data Format	2D PPI image (range-bearing)	Point cloud or RA/RAD tensor
Doppler Capability	Limited (platform motion dominant)	Full radial velocity (automotive sp
Clutter Type	Sea clutter, weather, multipath	Road clutter, stationary objects
Typical Detection Density	10-100 targets per frame	50-300 points per frame
Antenna Configuration	Rotating parabolic reflector	Phased array (MIMO)

The frequency difference has several implications beyond the obvious wavelength effects. X-band (3 cm wavelength) experiences less atmospheric attenuation than 77 GHz (4 mm wavelength), enabling the much longer detection ranges required for maritime applications. However, X-band is more susceptible to weather clutter (rain, fog) than millimeter-wave frequencies, which often pass through precipitation with less scattering. The longer wavelength of X-band also results in physically larger antennas (typically 1-2 meter parabolic dishes for marine radar vs. compact printed circuit board arrays for automotive), necessitating mechanical rotation rather than electronic scanning.

The mechanical scanning of marine radar introduces temporal skew: the 360-degree image is assembled from sequential measurements over 0.5-3 seconds, during which both the platform and detected vessels may move. In contrast, automotive radar produces nearly simultaneous measurements across its field of view. This temporal characteristic requires that maritime fusion architectures either compensate for scan time (de-skewing based on platform motion) or treat the PPI image as inherently temporally distributed. It also means that the effective update rate for any particular bearing angle is the rotation rate, substantially lower than automotive radar frame rates.

The 360-degree coverage of marine radar is both an advantage (complete situation awareness without sensor redundancy) and a challenge (larger data volumes, no concept of "forward-facing" that simplifies automotive fusion). Maritime fusion architectures must process the entire horizon, while automotive methods often exploit the primarily-forward attention pattern of driving scenarios to reduce computational requirements.

Range resolution differences reflect the trade-offs between long-range detection (requiring longer pulses or wider chirp bandwidths) and fine resolution. X-band marine radar operating at 20 km range with 30-meter resolution is sufficient for vessel detection (typical merchant vessels are 50-400 meters long), while automotive radar must resolve smaller objects (pedestrians, vehicles) at shorter ranges. This resolution difference affects the fusion architecture's ability to localize objects precisely from radar alone, potentially placing more burden on camera information for precise bounding box refinement in maritime applications.

6.7 Summary and Research Gaps

Radar-camera fusion has advanced rapidly in the automotive domain, driven by abundant labeled data, standardized sensor configurations, and concentrated research investment. Architectures have evolved from simple early fusion to sophisticated BEV-based transformers with attention mechanisms, achieving detection performance competitive with LiDAR at lower cost. Key architectural principles—BEV representation, center-based detection, attention-based adaptive fusion, and joint detection-tracking—have proven effective and are largely transferable across domains. However, maritime applications face distinct challenges that limit direct application of automotive methods.

Several critical research gaps impede progress in maritime radar-camera fusion. First, the scarcity of large-scale labeled maritime datasets combining X-band radar and cameras severely constrains development and evaluation of fusion architectures. While automotive research benefits from nuScenes (40,000 annotated frames), RADIATE, and View of Delft, maritime datasets like WHUT-MSFVessel and MOANA remain small by comparison, limiting the applicability of data-hungry deep learning methods. This dataset gap motivates the exploration of automated labeling approaches (Chapter 4) that can scale dataset construction, but also highlights a circular dependency: effective auto-labeling requires trained fusion models, which in turn require labeled data.

Second, architectural adaptations required to accommodate X-band marine radar characteristics—PPI image format, mechanical scanning, 360-degree coverage, long-range operation, sea clutter—remain underexplored. Most automotive fusion methods assume sparse point clouds or dense range-azimuth-Doppler tensors from electronic scanning, and adaptation to fundamentally different marine radar data structures requires more than simple encoder modification. Research specifically addressing polar coordinate fusion in BEV frameworks, temporal compensation for mechanical scan time, and explicit sea clutter modeling in fusion architectures would advance maritime applicability.

Third, real-time performance of fusion architectures at maritime ranges and data volumes requires investigation. While automotive methods have achieved real-time operation (20+ FPS) for 100-meter detection ranges, extending to multi-kilometer maritime ranges with 360-degree coverage substantially increases computational requirements. Efficient BEV grid representations, multi-resolution processing, and range-adaptive attention mechanisms warrant exploration for maritime scalability.

Fourth, domain adaptation and transfer learning approaches that leverage abundant automotive training data to bootstrap maritime fusion models remain nascent. Methods like CRKD [108] demonstrate cross-modality distillation within the automotive domain, but systematic investigation of automotive-to-maritime domain transfer, accounting for sensor and environmental differences, could accelerate maritime fusion development given limited labeled maritime data.

Finally, integration of maritime-specific information sources—AIS data, vessel dimension priors, nautical charts, sea state estimates—into fusion architectures presents opportunities for domain-specific performance gains but has received limited attention. While general-purpose fusion architectures are valuable for their broad applicability, maritime perception systems can potentially exploit domain knowledge unavailable in automotive scenarios.

Addressing these research gaps requires coordinated effort in dataset construction (potentially leveraging the automated labeling methods of Chapter 4), architectural innovation adapted to maritime radar characteristics (building on transferable principles from automotive fusion), and domain-specific evaluation metrics and benchmarks. The rapid progress in automotive radar-camera fusion over the past five years provides both technical foundations and a roadmap for maritime fusion research: standardized datasets enable reproducible comparisons, architectural modularity (separate encoders, common fusion space, task-specific heads) facilitates iterative improvement, and careful benchmarking drives progress. Adapting this research ecosystem to the maritime domain, while respecting the fundamental sensor and environmental differences, will be essential for realizing robust autonomous maritime perception systems.

7 X-Band Marine Radar Signal Processing

The effectiveness of radar-guided camera labeling systems depends fundamentally on the quality of radar detections that drive the cross-sensor association process. As discussed in Chapter 6, fusion architectures ranging from early-stage feature-level integration to late-stage decision fusion all require reliable radar measurements as input. When radar detections contain excessive false alarms or miss true targets due to inadequate signal processing, the resulting camera labels will be corrupted regardless of how sophisticated the fusion algorithm may be. This chapter examines the signal processing techniques that transform raw radar returns into the detection lists used for camera guidance, with particular emphasis on constant false alarm rate (CFAR) detection, sea clutter modeling, and the characteristics of commercial X-band marine radars such as the Simrad Halo 20+.

X-band marine radar systems, operating at frequencies near 9.4 GHz with wavelengths around 3 cm, provide the spatial resolution necessary for detecting small maritime targets at ranges from tens of meters to several nautical miles [13]. The signal processing chain begins with pulse compression to enhance range resolution and signal-to-noise ratio (SNR), proceeds through CFAR detection to maintain acceptable false alarm rates across varying clutter conditions, and concludes with clustering and tracking algorithms that associate detections across scans [68]. Understanding this processing chain is essential for researchers implementing radar-camera systems, as design choices at each stage directly impact the accuracy and completeness of automated camera labels.

7.1 Constant False Alarm Rate Detection

CFAR detection algorithms adaptively adjust detection thresholds based on local estimates of noise or clutter power, maintaining a constant probability of false alarm across spatially varying background conditions [112]. The fundamental CFAR threshold equation takes the form:

$$T = \alpha \cdot \hat{Z} \tag{33}$$

where T is the detection threshold, $\hat{Z}$ is an estimate of the local noise or clutter power, and α is a scaling factor determined by the desired false alarm rate and the assumed clutter distribution. A range-azimuth cell is declared to contain a target if its magnitude exceeds this adaptive threshold.

The cell-averaging CFAR (CA-CFAR) detector, introduced by Finn and Johnson, estimates background power by averaging samples from leading and lagging reference windows surrounding the cell under test (CUT) [112]. Guard cells immediately adjacent to the CUT are excluded to prevent target energy from contaminating the noise estimate. CA-CFAR achieves optimal performance in homogeneous clutter environments where all reference cells contain independent and identically distributed clutter samples [113]. The CA-CFAR threshold is computed as:

$$T_{\mathrm{CA}} = \alpha_{\mathrm{CA}} \cdot \frac{1}{N} \sum_{i=1}^{N} Z_i \tag{34}$$

where N is the total number of reference cells and Z_i represents the magnitude or power of the i-th reference cell. The scaling factor α_{CA} depends on N and the desired probability of false alarm P_{fa}. For Rayleigh-distributed clutter, $\alpha_{\mathrm{CA}} = N \left(P_{\mathrm{fa}}^{-1/N} - 1 \right)$ [68].

When multiple targets appear in close proximity, interfering targets may enter the reference windows and inflate the noise estimate, leading to detection losses. The ordered-statistic CFAR (OS-CFAR) addresses this multi-target scenario by sorting the reference cells in ascending order and selecting the k-th smallest value as the noise estimate rather than the mean [114]. By choosing k appropriately (typically $k \approx 3N/4$ for N reference cells), OS-CFAR ensures that interfering targets are excluded from the threshold calculation. The OS-CFAR threshold is:

$$T_{\mathrm{OS}} = \alpha_{\mathrm{OS}} \cdot Z_{(k)} \tag{35}$$

where $Z_{(k)}$ denotes the k-th order statistic of the reference window. OS-CFAR sacrifices approximately 1–2 dB of detection sensitivity compared to CA-CFAR in homogeneous environments but provides superior robustness when interfering targets are present [113].

Clutter edges, such as the boundary between open water and land, create severe performance degradation for both CA-CFAR and OS-CFAR. When the reference window straddles a clutter transition, the noise estimate becomes biased, resulting in either elevated false alarms (if low-clutter cells dominate the estimate in a high-clutter region) or missed detections (if high-clutter cells dominate in a low-clutter region). The smallest-of CFAR (SO-CFAR) combats clutter edges by computing separate CA-CFAR thresholds from the leading and trailing reference windows and selecting the minimum [115]:

$$T_{\mathrm{SO}} = \min \left(\alpha_{\mathrm{L}} \sum_{i \in L} Z_i,\ \alpha_{\mathrm{T}} \sum_{i \in T} Z_i \right) \tag{36}$$

where L and T denote the leading and trailing reference cells. SO-CFAR prevents threshold inflation in the presence of clutter edges but suffers increased false alarms in homogeneous regions because the minimum statistic is biased downward. The complementary greatest-of CFAR (GO-CFAR) takes the maximum of the two estimates to reduce false alarms at the cost of increased detection losses at clutter edges [116].

The automatic censored cell-averaging CFAR (ACCA-CFAR) attempts to combine the benefits of CA-CFAR and OS-CFAR by iteratively removing reference cells that exceed a censoring threshold, then averaging the remaining cells [117]. The algorithm begins with all reference cells, computes a preliminary mean, identifies cells exceeding a multiple of that mean, censors those cells, and recomputes the mean from the censored set. This process continues until convergence or a maximum iteration count. ACCA-CFAR performs well in both homogeneous and multi-target scenarios but requires more computation than simpler CFAR variants.

Recent advances in deep learning have introduced neural network-based CFAR detectors that learn adaptive thresholding strategies directly from labeled radar data. Diskin and colleagues proposed CFARnet, a convolutional neural network trained to predict detection thresholds for each range-azimuth cell given a local neighborhood of radar returns [118]. CFARnet demonstrated improved detection performance on maritime targets compared to CA-CFAR and OS-CFAR, particularly in heterogeneous clutter conditions. However, the method requires extensive labeled training data and lacks the analytical performance guarantees of classical CFAR algorithms.

Chen et al. developed Radar-PPInet, a deep network that processes plan-position indicator (PPI) radar images to jointly perform clutter suppression, target detection, and track extraction [119]. The network architecture combines convolutional layers for feature extraction with recurrent layers to exploit temporal correlation across consecutive radar scans. Evaluation on real-world maritime radar data showed superior detection rates compared to conventional CFAR processing, but generalization to new environments remains a challenge without fine-tuning.

Zhang and colleagues introduced YOLO-SWFormer, adapting the YOLO object detection framework with a Swin Transformer backbone for maritime radar target detection [120]. The approach treats radar PPI images as natural images and applies bounding box regression to localize targets. YOLO-SWFormer achieved real-time inference speeds suitable for shipboard deployment but required careful data augmentation to handle the unique characteristics of radar imagery, such as rotational symmetry and range-dependent resolution.

Marine-Faster R-CNN, proposed by Chen et al., extends the Faster R-CNN framework to maritime radar by incorporating domain-specific modifications including a novel anchor design tailored to the aspect ratios of ship returns and a multi-scale feature pyramid to handle targets at varying ranges [121]. The method demonstrated state-of-the-art detection accuracy on benchmark datasets but, like other learning-based approaches, depends critically on the availability of large annotated radar datasets, which are scarce in the maritime domain.

Classical CFAR algorithms remain the dominant approach in operational marine radar systems due to their computational efficiency, analytical tractability, and well-understood performance characteristics. However, deep learning CFAR methods represent a promising research direction, particularly for scenarios involving complex clutter environments where traditional model-based approaches struggle. For radar-guided camera labeling applications, the choice of CFAR algorithm directly affects the quantity and quality of radar detections available for cross-sensor association.

7.2 Sea Clutter Statistical Models

Accurate modeling of sea clutter statistics is essential for designing CFAR detectors with predictable false alarm rates and for evaluating detection performance through simulation. Sea clutter arises from radar backscatter by ocean surface waves, with statistical properties that depend on wind speed, wave height, radar resolution, polarization, incidence angle, and frequency [12].

The Rayleigh distribution represents the simplest sea clutter model, applicable to low-resolution radars observing homogeneous sea states with many independent scatterers per resolution cell [13]. Under the Rayleigh model, the clutter amplitude A follows the probability density function:

$$p_A(a) = \frac{a}{\sigma^2} \exp\left(-\frac{a^2}{2\sigma^2}\right), \quad a \geq 0 \tag{37}$$

where σ^2 is the mean clutter power. The corresponding power distribution is exponential:

$$p_Z(z) = \frac{1}{\mu} \exp\left(-\frac{z}{\mu}\right), \quad z \geq 0 \tag{38}$$

with mean $\mu = 2\sigma^2$. Rayleigh clutter is fully characterized by its mean power, simplifying CFAR design but failing to capture the heavy-tailed behavior observed in high-resolution radars and moderate-to-high sea states [122].

The K-distribution provides a more realistic model for sea clutter in many practical scenarios by representing the radar return as the product of two independent random processes: Gaussian speckle and a slowly varying gamma-distributed texture component [123]. The K-distribution amplitude PDF is:

$$p_A(a) = \frac{4}{\Gamma(\nu)} \left(\frac{\nu}{2\mu}\right)^{(\nu+1)/2} a^\nu K_{\nu-1}\left(a\sqrt{\frac{2\nu}{\mu}}\right), \quad a \geq 0 \tag{39}$$

where μ is the mean power, ν is the shape parameter controlling the tail heaviness, and $K_{\nu-1}$ denotes the modified Bessel function of the second kind with order $\nu - 1$. Small values of ν (typically $0.1 < \nu < 2$) correspond to spiky, heavy-tailed clutter observed at high resolution and high sea states, while large ν approaches the Rayleigh limit. The K-distribution has been extensively validated against real sea clutter measurements and is widely used in maritime radar performance prediction [124].

The compound Gaussian framework generalizes the K-distribution by allowing arbitrary texture distributions while maintaining the Gaussian speckle component [125]. This flexibility enables fitting a broader range of observed clutter statistics but complicates CFAR detector design because optimal threshold scaling factors must be derived for each specific texture model. Shui et al. demonstrated that compound Gaussian models with inverse Gaussian texture provide improved fits to X-band sea clutter in certain conditions compared to the K-distribution [126].

The Pareto distribution, also known as the compound K-distribution, describes high-resolution X-band sea clutter and offers the advantage of enabling simpler optimal detector structures compared to the K-distribution [127]. The Pareto amplitude PDF is:

$$p_A(a) = \frac{2\alpha\beta^\alpha}{a^{2\alpha+1}}, \quad a \geq \beta \tag{40}$$

where α is the shape parameter and β is the scale parameter. Pareto-distributed clutter exhibits a power-law tail, capturing the extreme sea spikes that characterize high-resolution radar observations. Leung and colleagues showed that Pareto-based CFAR detectors outperform K-distribution-based detectors for small target detection in X-band maritime radar data [128].

The Weibull distribution represents another popular model for sea clutter, offering mathematical tractability and the ability to fit a wide range of empirical clutter statistics through its shape parameter c [129]. The Weibull amplitude PDF is:

$$p_A(a) = \frac{c}{\lambda}\left(\frac{a}{\lambda}\right)^{c-1} \exp\left(-\left(\frac{a}{\lambda}\right)^{c}\right), \quad a \geq 0 \tag{41}$$

where λ is the scale parameter. The shape parameter controls tail behavior: $c = 2$ yields the Rayleigh distribution, $c < 2$ produces heavy tails, and $c > 2$ gives light tails. Weibull CFAR detectors have been implemented in operational systems, though the Weibull model generally provides poorer fits to high-resolution sea clutter compared to the K-distribution or Pareto models [130].

Recent research has focused on non-stationary and multi-scale clutter models to capture temporal and spatial variations in sea clutter characteristics. Liu et al. proposed a time-varying K-distribution model with slowly evolving shape and scale parameters estimated via Kalman filtering [131]. Zhang and colleagues developed a spatially heterogeneous clutter model that partitions the radar field of view into regions with distinct statistical properties, enabling adaptive CFAR processing tailored to local clutter conditions [132].

For radar-guided camera labeling systems, the choice of clutter model affects detection performance and false alarm rates. Systems operating in benign conditions (low sea states, long ranges, coarse resolution) may perform adequately with Rayleigh-based CFAR, while those deployed in challenging environments (high sea states, short ranges, fine resolution) require K-distribution, Pareto, or compound Gaussian models to maintain acceptable false alarm rates. Empirical clutter characterization using data collected from the specific radar and environment of interest remains the most reliable approach for selecting an appropriate model.

7.3 Pulse Compression

Pulse compression techniques enable marine radars to achieve fine range resolution while maintaining sufficient energy for long-range detection [68]. A simple pulsed radar transmits a short-duration, high-power pulse to maximize range resolution, but the achievable pulse duration is limited by peak power constraints and transmitter technology. Pulse compression circumvents this limitation by transmitting a longer, frequency-modulated or phase-coded waveform that is compressed upon reception through matched filtering, achieving the range resolution of a short pulse while delivering the energy of a long pulse.

The most common pulse compression waveform for marine radar is the linear frequency modulated (LFM) chirp, in which the instantaneous frequency increases or decreases linearly over the pulse duration T_p [133]. The transmitted signal is:

$$s(t) = \text{rect}\left(\frac{t}{T_p}\right) \exp\left(j2\pi\left(f_c t + \frac{B}{2T_p}t^2\right)\right) \tag{42}$$

where f_c is the carrier frequency, B is the bandwidth, and $\text{rect}(t)$ is the rectangular window function. Upon reception, the echo is passed through a matched filter with impulse response $h(t) = s^*(-t)$, producing a compressed pulse with width $\tau_c \approx 1/B$ and time-bandwidth product $T_p \cdot B$, also known as the pulse compression ratio. A chirp with $T_p = 100$ μs and $B = 10$ MHz achieves a compression ratio of 1000, effectively providing the range resolution of a 0.1 μs pulse (15 m in range) with the energy of a 100 μs pulse.

The output of the matched filter exhibits range sidelobes that can mask weak targets adjacent to strong returns. Sidelobe reduction techniques apply amplitude weighting (windowing) to the transmitted waveform or received signal, trading peak sidelobe level for increased mainlobe width [68]. Common windows include Hamming, Hanning, and Taylor weighting functions. For maritime radar applications, sidelobe levels below -40 dB are typically required to prevent large ship returns from obscuring nearby small targets.

The Simrad Halo 20+ radar employs pulse compression in combination with beam sharpening, a proprietary processing technique that enhances azimuthal resolution beyond the physical beamwidth of the antenna [134]. Details of the beam sharpening algorithm are not publicly disclosed, but the technique likely involves some form of super-resolution processing or deconvolution applied to consecutive azimuth samples. The resulting improvement in angular resolution enables better separation of closely spaced targets, which is particularly beneficial in congested harbor environments.

Pulse compression introduces additional considerations for CFAR detection. The matched filter output is no longer independent across range cells due to the correlation introduced by the sidelobe structure. Reference cells in CFAR processing must be spaced sufficiently far from the CUT to ensure statistical independence, which may require larger guard cell regions than would be necessary for a simple pulsed radar. Furthermore, the sidelobe structure can create false detections if CFAR thresholds are not designed to account for the spatially correlated nature of the compressed clutter returns [135].

7.4 Simrad Halo 20+ Characteristics

The Simrad Halo 20+ represents a consumer-grade X-band pulse-Doppler marine radar designed for recreational and light commercial vessels. Operating at X-band frequencies near 9.4 GHz, the system features a compact 20-inch radome antenna and achieves a maximum instrumented range of 36 nautical miles (67 km), though effective detection ranges depend on target size and sea conditions [134]. The antenna rotates at 60 revolutions per minute (RPM), corresponding to a scan rate of 1 Hz, which provides timely updates for collision avoidance while allowing sufficient dwell time for target detection.

The Halo 20+ employs pulse compression and proprietary beam sharpening algorithms to enhance both range and azimuthal resolution beyond the physical limitations of the compact antenna. The manufacturer specifies horizontal beamwidth of approximately 1.9 degrees after beam sharpening, compared to the roughly 4-degree physical beamwidth of the 20-inch antenna. This enhanced angular resolution improves target separation in congested environments such as harbors and shipping channels.

VelocityTrack Doppler processing provides target velocity information by measuring the Doppler frequency shift of moving targets relative to the ship's own motion [134]. The Doppler capability enables discrimination between stationary clutter (land, buoys) and moving targets (other vessels), improving detection performance in littoral environments. However, the Doppler resolution and unambiguous velocity range of the Halo 20+ are not publicly documented, limiting quantitative analysis of Doppler performance.

The radar offers four selectable operating modes tailored to different environmental conditions: Harbour mode optimizes sensitivity for short-range detection in calm waters; Offshore mode balances sensitivity and clutter rejection for open-ocean navigation; Weather mode enhances precipitation detection for storm avoidance; and Bird mode increases sensitivity for detecting flocks of seabirds. These modes likely adjust CFAR parameters, sea clutter filtering, and signal processing gain to match the expected clutter statistics and target characteristics of each scenario.

Mini automatic radar plotting aid (MARPA) functionality enables automatic tracking of 10–20 targets, computing their course and speed vectors and predicting potential collision threats [134]. The MARPA implementation in the Halo 20+ is not described in public literature, but typical consumer-grade systems use simplified tracking algorithms such as nearest-neighbor association and α-β filtering rather than the more sophisticated probabilistic data association and Kalman filtering employed in military and commercial shipping radars.

Comparison of consumer radars such as the Halo 20+ to military and research-grade systems reveals significant differences in performance, flexibility, and data access. Military radars typically provide raw or minimally processed I/Q data, enabling researchers to implement custom signal processing chains and characterize system performance in detail. Consumer radars generally output only processed PPI imagery or detection lists, with proprietary signal processing that cannot be modified or fully characterized. Detection thresholds, clutter models, and CFAR parameters are often hardcoded or adjusted automatically by undocumented algorithms, making it difficult to predict or optimize performance for specific applications such as radar-camera cross-labeling.

Research-grade radars such as those used in the IPIX, Dartmouth, and Stare datasets offer calibrated measurements with known antenna patterns, transmit waveforms, and receiver characteristics [136]. These systems enable controlled experiments and repeatable performance evaluations. Consumer radars lack calibration data, exhibit unknown sidelobe levels and beamwidth, and may apply non-linear post-processing (gamma correction, contrast enhancement) to the displayed imagery, complicating quantitative analysis.

Despite these limitations, consumer radars offer significant advantages for autonomous surface vehicle and small-boat applications: low cost, compact form factor, low power consumption, and simple integration with standard marine electronics. For radar-guided camera labeling, the Halo 20+ provides a practical platform for developing and demonstrating cross-sensor fusion techniques in realistic maritime environments, though researchers must carefully characterize detection performance and clutter statistics empirically.

7.5 Track-Before-Detect Methods

Traditional radar detection processing applies a threshold to each range-azimuth cell independently, discarding sub-threshold returns as noise or clutter. This approach is optimal when target signal-to-clutter ratio (SCR) is sufficiently high, but fails when targets produce weak returns that fall below the single-scan detection threshold. Track-before-detect (TBD) methods address this low-SCR scenario by integrating energy over multiple scans before making detection decisions, coherently accumulating weak target signatures that would be missed by conventional CFAR detectors [137].

Particle filter TBD approaches represent the target state (position, velocity) as a weighted set of particles propagated forward in time according to a motion model [138]. At each radar scan, particle weights are updated based on the likelihood of the observed radar measurements given the hypothesized target location. Particles corresponding to true targets receive high weights through repeated correlation with target returns, while particles in clutter-only regions accumulate low weights and are eventually resampled away. Detection decisions are made by thresholding the particle weight distribution or by identifying clusters of high-weight particles.

Greco et al. developed a two-stage TBD architecture for maritime radar that combines non-coherent integration of radar magnitude measurements with recursive Bayesian filtering [139]. The first stage sums radar returns across N consecutive scans for each hypothesized target trajectory, producing an integrated detection statistic with improved SNR. The second stage applies a particle filter or extended Kalman filter to track promising detections and resolve ambiguities. The method demonstrated improved detection of small boats in moderate sea clutter compared to conventional CFAR processing, but at the cost of increased latency due to the multi-scan integration window.

For radar-camera cross-labeling applications, TBD methods present both opportunities and challenges. The improved low-SCR detection performance could enable labeling of distant or small targets that would be missed by single-scan CFAR. However, the multi-scan latency complicates time synchronization between radar detections and camera frames, potentially requiring retroactive association of TBD detections to camera imagery captured several scans earlier. Furthermore, TBD algorithms require accurate motion models, which may be difficult to specify for heterogeneous maritime targets exhibiting diverse maneuvering behaviors.

7.6 DBSCAN and ST-DBSCAN for Radar Clustering

After CFAR detection, radar returns typically consist of a collection of detections in range-azimuth space, including both true target returns and residual false alarms from clutter. Clustering algorithms group spatially proximate detections that likely originate from the same physical target, enabling target centroid estimation and facilitating downstream tracking and classification [140].

The Density-Based Spatial Clustering of Applications with Noise (DBSCAN) algorithm, introduced by Ester et al., identifies clusters as dense regions of points separated by low-density areas [81]. DBSCAN requires two parameters: ε, the maximum distance between two points to be considered neighbors, and MinPts, the minimum number of points required to form a dense region. A point is classified as a core point if at least MinPts points (including itself) lie within distance ε. Clusters are formed by connecting core points that are neighbors, and non-core points that are neighbors of core points are assigned to the same cluster. Points that are not core points and have no core neighbors are labeled as noise.

For radar target clustering, ε is typically set based on the expected spatial extent of target returns (a few range bins and azimuth cells), while MinPts is chosen based on the minimum number of detections expected from a point target or the minimum extent of distributed targets of interest [141]. DBSCAN effectively separates closely spaced targets and rejects isolated false alarms, but it operates on a single radar scan and does not exploit temporal correlation across scans.

The Spatio-Temporal DBSCAN (ST-DBSCAN) algorithm extends DBSCAN to three-dimensional spatio-temporal space by defining separate distance thresholds for spatial and temporal dimensions [82]. Two detections are considered neighbors if their spatial separation is less than ε_1 and their temporal separation is less than ε_2. ST-DBSCAN forms clusters of detections that are persistent over multiple scans, providing a simple form of track-before-detect that improves detection of low-SCR targets and rejects transient false alarms.

Zhou et al. applied ST-DBSCAN to maritime radar data for automatic identification of ship trajectories, demonstrating improved tracking continuity compared to single-scan DBSCAN followed by frame-to-frame association [142]. The spatio-temporal clustering naturally handles temporary detection dropouts caused by sea clutter or multipath fading, as detections from adjacent scans bridge the gaps. However, ST-DBSCAN assumes that cluster membership is transitive across time, which may fail for crossing targets or scenarios where two targets temporarily merge and then separate.

For radar-guided camera labeling, clustering algorithms such as DBSCAN and ST-DBSCAN provide essential preprocessing to consolidate multiple detections from extended targets into single representative points that can be associated with camera bounding boxes. The choice of clustering parameters affects the granularity of radar detections: conservative settings (small ε, large MinPts) produce fewer, higher-confidence clusters but may split single targets into multiple fragments, while liberal settings (large ε, small MinPts) merge neighboring targets and admit more false alarms. Empirical tuning based on the specific radar characteristics and target types is necessary to achieve optimal performance.

7.7 Maritime Detection Challenges

X-band marine radar detection performance is degraded by a variety of environmental and system-level phenomena that complicate signal processing and limit achievable detection ranges and accuracies. Understanding these challenges is essential for designing robust radar-camera cross-labeling systems and for interpreting the limitations of radar-derived labels.

Sea clutter arises from backscatter by wind-driven ocean surface waves and represents the dominant source of false alarms in maritime radar [12]. Clutter power increases with sea state, radar frequency, and grazing angle. X-band radars are particularly susceptible to sea clutter due to the short wavelength, which produces strong resonant backscatter from centimeter-scale capillary waves. Sea spikes, intermittent high-amplitude clutter returns caused by breaking waves or whitecaps, create heavy-tailed clutter statistics that violate the Rayleigh assumption and necessitate more sophisticated clutter models such as the K-distribution or Pareto distribution [124].

Multipath propagation occurs when radar signals reflect from the water surface before reaching the target or after being scattered by the target, creating multiple signal paths with different delays and Doppler shifts [143]. Constructive interference between direct and multipath returns produces lobing patterns in the elevation plane that cause target amplitude to fluctuate as the target range or ship motion changes the multipath geometry. Destructive interference creates nulls where targets may disappear entirely. Multipath effects are most severe for low-elevation targets at moderate ranges (1–10 km) and calm sea states, where the water surface acts as a specular reflector.

Rain clutter results from backscatter by precipitation, with clutter power proportional to rainfall rate and radar frequency [13]. X-band radars are more vulnerable to rain clutter than lower-frequency S-band or L-band systems because rain backscatter cross-section increases approximately with the fourth power of frequency (Rayleigh scattering regime). Heavy rain can produce clutter returns that obscure targets at ranges beyond a few kilometers, effectively blinding the radar. Polarization diversity techniques, which exploit differences in the polarization signatures of rain and targets, can mitigate rain clutter but are not available on most consumer marine radars [144].

Electromagnetic interference from other radars, communication systems, and shipboard electronics corrupts radar measurements and creates false detections. Harbor environments are particularly challenging due to the density of radar-equipped vessels and shore-based surveillance systems. Interference appears as streaks or speckle in the radar PPI display, depending on whether the interfering signal is synchronous or asynchronous with the victim radar. Frequency agility, pulse-to-pulse frequency hopping, and interference blanking circuits mitigate some forms of interference, but consumer radars offer limited protection compared to military systems [145].

Small target detection remains a fundamental challenge for X-band marine radar due to the weak radar cross-section of objects such as kayaks, debris, and buoys, which may be smaller than 1 square meter [146]. At ranges beyond a few hundred meters, such targets produce returns that are comparable to or weaker than sea clutter, resulting in low SCR and high miss rates. Increasing transmit power or antenna gain improves detection range but is constrained by regulatory limits, power consumption, and platform size. Coherent integration over multiple pulses or scans improves SCR but requires accurate motion compensation and is limited by target decorrelation time.

For autonomous vessels and radar-camera cross-labeling systems operating in cluttered maritime environments, these detection challenges directly impact the quality and completeness of radar-derived camera labels. Periods of heavy rain or high sea states will produce gaps in labeled datasets unless supplemented by other sensors. Small or low-RCS targets may be systematically underrepresented in training data, biasing learned models toward larger, higher-SNR objects. Multipath-induced amplitude fluctuations can cause intermittent detection dropouts, fragmenting trajectories and complicating track-based labeling approaches. Robust system design must account for these limitations and incorporate validation mechanisms to detect and flag low-confidence labels.

7.8 Summary and Research Gaps

This chapter has reviewed the fundamental signal processing techniques underlying X-band marine radar detection, including CFAR algorithms, sea clutter statistical models, pulse compression, clustering methods, and track-before-detect approaches. The performance of radar-guided camera labeling systems depends critically on the reliability of these processing stages, as errors or limitations in radar detections directly propagate to errors in automated camera labels. Classical CFAR algorithms such as CA-CFAR, OS-CFAR, and SO-CFAR provide computationally efficient adaptive detection with well-characterized false alarm rates, while emerging deep learning CFAR methods promise improved performance in complex clutter environments at the cost of increased computational demand and training data requirements. Sea clutter modeling using K-distribution, Pareto, or compound Gaussian frameworks enables accurate prediction of detection performance and informs CFAR parameter selection for heterogeneous maritime environments.

The Simrad Halo 20+ radar, representative of consumer-grade marine radar systems, offers a practical platform for developing autonomous vessel sensing and cross-sensor labeling techniques. However, significant gaps exist in published performance characterization of consumer radars compared to research-grade systems. Detailed measurements of Halo 20+ detection probability, false alarm rate, range accuracy, and angular resolution under controlled conditions are not available in the open literature, complicating performance prediction and system design. Future work should conduct systematic empirical evaluations of consumer marine radars across varying sea states, target types, and ranges to establish baseline performance metrics.

The relationship between consumer radar characteristics and research-grade radar performance represents another important gap. While expensive research radars provide superior calibration, data access, and flexibility, the extent to which their performance advantages translate to improved detection outcomes for practical autonomous vessel applications remains unclear. Comparative studies deploying both consumer and research radars on the same platform under identical conditions would quantify the performance-cost tradeoffs and inform sensor selection for resource-constrained autonomous systems.

Automation of radar-camera cross-labeling, particularly under challenging environmental conditions such as high sea states, rain, or multipath propagation, requires further research. Existing fusion and labeling approaches generally assume reliable radar detections, but as discussed in this chapter, detection performance degrades significantly in adverse conditions. Adaptive labeling confidence estimation that accounts for environmental factors and radar signal processing limitations would enable more robust dataset curation by flagging or rejecting low-quality labels.

Real-time implementation of advanced signal processing techniques such as deep learning CFAR, particle filter TBD, and ST-DBSCAN on embedded platforms typical of small autonomous vessels (e.g., NVIDIA Jetson, Raspberry Pi) presents computational challenges. Consumer radars such as the Halo 20+ output processed detections rather than raw I/Q data, limiting the applicability of sophisticated signal processing algorithms that require access to the full radar signal. Future radar designs for autonomous vessel applications should balance processing capability, cost, and power consumption while providing sufficient data access to enable custom processing pipelines.

In summary, X-band marine radar signal processing techniques provide the foundation for radar-guided camera labeling systems, but significant research opportunities remain in characterizing consumer radar performance, developing robust labeling methods for degraded detection conditions, and enabling real-time advanced signal processing on resource-constrained platforms. Addressing these gaps will enhance the reliability and scalability of automated maritime dataset construction for autonomous vessel perception.

8 Edge Computing for Real-Time Sensor Fusion and Inference

8.1 Introduction: Edge Deployment for Autonomous Dataset Labeling

The radar signal processing pipeline developed in Chapter 7 produces structured target detections that must be fused with camera imagery to generate labeled training data for autonomous maritime systems. While this fusion process was historically performed on shore-based infrastructure, modern edge computing platforms now enable autonomous annotation directly aboard surface vessels. This shift from centralized post-processing to decentralized onboard labeling dramatically reduces data transmission requirements and latency, critical constraints for remote oceanographic platforms and autonomous surface vehicles operating in communication-limited environments.

The key insight for maritime dataset construction is that the radar-guided labeling pipeline is fundamentally *not real-time critical.* Unlike autonomous navigation systems that must respond to obstacles with millisecond precision, labeling operates asynchronously during mission execution. Radar detections and camera frames are buffered, processed offline in batches as computational capacity permits, and then annotated labels are either stored locally or transmitted during communication windows. This distinction liberates system design from hard real-time constraints, allowing aggressive batching and optimization strategies that would be impractical for onboard decision-making systems.

This chapter examines the deployment of YOLOv8-based detection models and the radar-camera fusion labeling pipeline on NVIDIA Jetson edge computing platforms. We survey quantization and optimization techniques that achieve 3–4× speedup with negligible accuracy loss, quantify real-world throughput on maritime hardware, and demonstrate that a Jetson Orin NX platform operating at 15–20 W sustained power can label 1000–1500 camera frames per hour while consuming approximately 288 Wh per 24-hour mission—enabling fully autonomous dataset construction aboard small uncrewed surface vehicles.

8.2 The NVIDIA Jetson Platform Family

8.2.1 Platform Overview and Specifications

NVIDIA's Jetson family provides a tiered hierarchy of edge AI accelerators purpose-built for autonomous systems, ranging from embedded microcontrollers to GPU-dense workstations. The platform architecture centers on NVIDIA's ARM-based Tegra System-on-Chip (SoC) integrated with dedicated graphics processing units and specialized tensor cores for deep learning inference. [147] Key distinctions among platforms include CUDA core count, memory bandwidth, INT8 tensor performance (measured in TOPS—Tera Operations Per Second), peak power consumption, and thermal requirements.

Table 9: NVIDIA Jetson Platform Specifications and Performance Characteristics

Platform	TOPS (INT8)	CUDA Cores	Memory	Bandwidth	Peak Pov
Jetson AGX Orin 64GB	275	2048	64 GB LPDDR5X	204.8 GB/s	60
Jetson Orin NX	157	1024	8 GB LPDDR5	102.4 GB/s	25
Jetson Orin Nano	67	512	4 GB LPDDR5	51.2 GB/s	12

The Jetson AGX Orin 64GB represents the flagship platform, delivering 275 TOPS of INT8 inference throughput across 2048 CUDA cores. This configuration is suited for land-based autonomous vehicle platforms with available power budgets exceeding 40 W. However, maritime applications present strict power and thermal constraints: vessel power systems typically operate at 24 V DC with limited capacity (10–20 W sustainable draw), and sealed electronics enclosures cannot dissipate heat rapidly. [148]

The Jetson Orin NX emerges as the optimal platform for maritime dataset labeling workloads. With 157 TOPS INT8 performance, 1024 CUDA cores, and 8 GB of LPDDR5 memory, it achieves approximately 57% of the AGX Orin's throughput while consuming one-third the peak power and fitting within maritime thermal envelopes. The 102.4 GB/s memory bandwidth provides sufficient throughput for batched inference on medium-resolution imagery without becoming memory-bound. [149]

The Jetson Orin Nano (67 TOPS, 512 CUDA cores, 12 W peak) offers minimal-power deployment for scenarios requiring extreme power conservation; however, it provides only 24% of the NX throughput, resulting in degraded labeling throughput unsuitable for sustained dataset construction aboard survey vessels.

8.2.2 Memory and Thermal Considerations

Memory architecture critically impacts edge deployment. The Jetson Orin NX employs unified memory architecture with 8 GB of LPDDR5 accessible to both the CPU and GPU. This eliminates costly host-device memory transfers that degrade throughput on discrete GPUs; inference models and camera buffers reside in the same address space, simplifying data movement. However, 8 GB constrains batch sizes and model complexity: concurrent buffering of radar detections, camera frames, and intermediate feature maps requires careful memory budgeting. [150]

Thermal management defines sustained power budgets. The Jetson Orin NX's 25 W peak rating assumes active cooling; mounted in sealed maritime enclosures with limited airflow, sustained power must be throttled to 15–20 W to maintain temperatures below 80°C and prevent thermal shutdown. Active cooling (small axial fans or liquid loops) is necessary for 25+ W sustained operation; passive dissipation alone is insufficient in maritime environments with seawater spray and salt-air convection impairment.

8.3 TensorRT Optimization Pipeline

8.3.1 ONNX-to-Engine Workflow

Production deployment of deep learning models on Jetson platforms necessitates conversion from training frameworks (PyTorch, TensorFlow) to NVIDIA TensorRT, a graph-level inference engine that optimizes models for target hardware. [151] The standard workflow converts framework-native models to ONNX (Open Neural Network Exchange) format, then compiles ONNX graphs into hardware-specific TensorRT engines.

This compilation process executes two optimization phases: (1) graph-level transformations that restructure the computation graph to reduce memory allocations and operator counts, and (2) kernel-level auto-tuning that selects optimal implementations for each layer based on measured performance on target hardware.

Graph-level optimizations include:

1. **Operator Fusion**: Sequential pointwise operations (e.g., convolution followed by batch normalization and ReLU) are fused into single kernels, eliminating intermediate tensor writes and reads. Fusion reduces memory traffic and improves instruction-level parallelism.

2. **Constant Folding**: Computations with statically known inputs (e.g., quantization scales applied to fixed weights) are evaluated once during compilation; the results are baked into the engine, eliminating runtime overhead.

3. **Quantization Insertion**: TensorRT automatically identifies quantization points in the graph and inserts scale-cast operations that map FP32 activations to INT8 for tensor core execution.

4. **Memory Optimization**: In-place operations and memory reuse reduce overall peak memory consumption, freeing capacity for larger batches.

Kernel-level auto-tuning executes timing microbenchmarks on target hardware (typically a few seconds during first-run compilation) to profile available implementations for each operation. TensorRT selects implementations that minimize end-to-end latency given measured bandwidth and utilization characteristics.

The combined effect of these optimizations yields 2–3× speedup on native FP32 inference and up to 15× speedup when combined with INT8 quantization. [151] For YOLOv8 models running on Jetson platforms, typical multipliers are 2.5–3× from TensorRT graph optimization and an additional 4–5× from INT8 quantization, yielding 10–15× overall improvements.

8.3.2 Performance Multipliers and Benchmarking

A concrete example illustrates TensorRT benefits. YOLOv8-small (9.2 M parameters) on a Jetson Orin NX:

- **Native PyTorch FP32**: 8–12 FPS at 640×640 resolution (FP32 tensor cores underutilized on Jetson's ARM architecture)
- **TensorRT FP32 (no quantization)**: 22–28 FPS (2.8× speedup via graph fusion and kernel optimization)
- **TensorRT INT8 (post-training quantization)**: 95–125 FPS (4.3–4.5× additional speedup via INT8 tensor cores)
- **Batched INT8 (batch size 16)**: 1500–2000 FPS throughput (frame/second inference, not latency-per-frame)

These multipliers depend critically on model architecture, input resolution, and batch size. Dense models with high floating-point intensity benefit more from INT8 quantization. Models with bottleneck layers (depthwise convolutions, squeeze-and-excitation blocks) may see smaller gains if non-quantizable components become dominant.

8.4 Quantization Strategies and Accuracy Trade-offs

8.4.1 Precision Reduction: FP32 to INT8

Quantization reduces model precision from 32-bit floating-point (FP32) to lower-precision formats, increasing memory efficiency and enabling specialized hardware accelerators. The three primary strategies are:

Table 10: Precision Trade-offs: Memory, Throughput, and Typical Accuracy Loss

Precision	Bits	Model Size	Throughput	Typical mAP Loss
FP32	32	1.0×	1.0×	–
FP16	16	0.5×	2.0–2.5×	$< 0.1\%$
INT8	8	0.25×	4.0–5.0×	1–3% (PTQ), $< 0.5\%$ (QAT)

FP16 (half-precision) provides minimal accuracy loss ($< 0.1\%$ mAP degradation) with 2–2.5× throughput improvement and 0.5× model size. On Jetson platforms with dedicated FP16 tensor cores, FP16 is often preferred for applications with strict accuracy requirements. However, FP16 offers only modest speedup compared to INT8, and many maritime detection tasks can tolerate minor mAP reductions.

INT8 quantization maps weights and activations to 8-bit signed integers, enabling specialized INT8 tensor cores available on all Jetson platforms. INT8 provides 4–5× throughput improvement over native FP32 and 0.25× memory footprint, but introduces quantization error that can reduce mean Average Precision (mAP) by 1–3% if applied naively.

8.4.2 Post-Training Quantization (PTQ)

Post-training quantization applies quantization to a fully trained FP32 model without retraining. The quantization process identifies optimal scale factors that map the observed activation range (determined by running calibration data through the network) to the 8-bit integer range $[-128, 127]$.

For YOLOv8 models, PTQ typically incurs 1–3% mAP loss across detection, classification, and regression heads. [152] The variance depends on model size (smaller models less tolerant to quantization) and calibration dataset quality. Maritime object detection datasets with limited class variety and moderate scale variation (most ship targets occupy 30–200 pixels in frame) tolerate 1–2% mAP loss without perceptual quality degradation; false negative rates increase slightly while false positive rates remain stable.

PTQ is computationally efficient: quantization scale computation requires only one forward pass through the calibration set (typically 100–500 representative images), completing in seconds on desktop hardware. Once scales are computed, the quantized model exports to TensorRT format and runs immediately.

8.4.3 Quantization-Aware Training (QAT)

Quantization-aware training explicitly incorporates quantization into the training loop, allowing the optimizer to adjust weights and biases to minimize accuracy loss under quantization. During forward passes, activations are quantized to INT8; during backward passes, straight-through estimators approximate gradients (since quantization is non-differentiable).

QAT typically reduces mAP loss to $< 0.5\%$ for YOLOv8 models, [151] a significant improvement over PTQ. However, QAT requires access to training code, sufficient GPU memory to fine-tune on full datasets (typical YOLOv8 training requires 8–16 GB GPU memory), and 10–20 epochs of retraining. For maritime labeling pipelines, QAT is appropriate when training models on maritime datasets (e.g., WaterScenes, MOANA) before deployment.

The three-tier quantization strategy for maritime deployment:

1. **Development**: Train YOLOv8 models with QAT-aware loss functions on maritime datasets, targeting $< 0.5\%$ mAP loss.

2. **Deployment Preparation**: Export quantized models to ONNX, compile to TensorRT INT8 engines on target Jetson hardware.

3. **Validation**: Run inference on held-out validation set to confirm accuracy matches QAT predictions, then deploy aboard vessels.

8.5 Real-Time Detection Benchmarks on Jetson Hardware

8.5.1 YOLOv8 Family Throughput on AGX Orin and Orin NX

The YOLOv8 family comprises five model sizes (nano, small, medium, large, extra-large) designed for the speed-accuracy trade-off. [153] Table 11 summarizes throughput (frames per second, batch size 1) and latency (milliseconds per inference) across platforms and quantization strategies.

Table 11: YOLOv8 Inference Throughput on NVIDIA Jetson Platforms (640×640 Input, Batch Size 1)

Model	Size (M params)	FP32	INT8	AGX Orin INT8	Orin NX INT8
YOLOv8-nano	3.2	20 FPS	80 FPS	250+ FPS	120 FPS
YOLOv8-small	11.2	12 FPS	48 FPS	167 FPS	95 FPS
YOLOv8-medium	25.9	6 FPS	24 FPS	83 FPS	50 FPS
YOLOv8-large	43.7	3.5 FPS	14 FPS	50 FPS	28 FPS

On the Jetson AGX Orin, YOLOv8-nano achieves 250+ FPS at INT8 quantization, enabling real-time processing of 240p video streams at 30 Hz with substantial headroom. YOLOv8-small attains 167 FPS, sufficient for full-HD streams (1920×1080 at 30 Hz when downsampled). These throughputs assume single-image batch inference; batched inference (batch size 16–64) increases total frame throughput by 4–6× as GPU memory and tensor cores reach full utilization.

The Jetson Orin NX achieves 95 FPS on YOLOv8-small and 50 FPS on medium—sufficient for HD-resolution maritime video at 30 Hz with modest downsampling. The gap between NX and AGX Orin reflects the 157 vs. 275 TOPS difference; throughput scales approximately linearly with compute capacity for memory-bound models.

8.5.2 Comparison with RT-DETR and Real-Time Constraints

RT-DETR (Real-Time DETR) [154] offers an alternative architecture based on the DETR transformer backbone, claiming competitive accuracy with reduced NMS post-processing overhead. RT-DETR-small provides comparable mAP to YOLOv8-small while delivering slightly lower latency (45 ms vs. 60 ms per inference on AGX Orin). However, RT-DETR exhibits higher memory consumption due to transformer self-attention and requires careful quantization to avoid significant accuracy loss. For maritime platforms with tight memory budgets (8 GB Orin NX), YOLOv8 remains preferred due to its lower memory footprint and mature quantization support.

A critical observation: both YOLOv8 and RT-DETR are *real-time capable* in the sense that a single inference completes within 10–100 ms on modern edge hardware. However, real-time constraint is *not* applicable to maritime dataset labeling. The labeling pipeline need not process frames at video rate or respond to detections within milliseconds. Instead, the system should maximize *throughput*—total frames labeled per hour—subject to power and thermal constraints.

8.6 Offline Batch Processing: The Key Insight for Labeling

8.6.1 Decoupling from Real-Time Constraints

The fundamental architectural insight is that radar-guided camera labeling operates offline and asynchronously relative to vessel mission operations. Radar returns are processed at 5–20 Hz (standard marine radar scan rate), and camera frames arrive at 5–30 Hz depending on system configuration. However, labeling need not occur at these rates. Instead, raw radar azimuth, range, and Doppler data are buffered in onboard storage; simultaneously, camera imagery is recorded locally. When computational capacity is available (typically during periods of lower sensor activity, night operations, or extended legs with minimal target density), the system processes buffered data in batches.

This decoupling eliminates hard real-time deadlines. A single CPU core can manage data acquisition and vessel navigation; a separate GPU thread processes historical radar and camera data in batches. If the GPU falls behind due to high target density, frames queue in memory; if memory approaches capacity, the system simply discards oldest frames (graceful degradation) or reduces batch size. No crashes, no missed navigation deadlines, and no actuation failures result from labeling delays.

8.6.2 Batched Inference Throughput and Maritime Mission Power Budget

Batched inference exploits GPU parallelism: instead of processing one frame at a time, the model processes B frames simultaneously, amortizing kernel launch overhead and enabling full tensor core utilization. For YOLOv8-small on Jetson Orin NX with batch size 16, realistic total throughput reaches 1200–1500 frames per hour when GPU utilization is optimized.

A typical maritime survey mission profiles power consumption as follows. A Jetson Orin NX sustained at 20 W:

- **Inference throughput**: 1200 frames/hour (batched processing, 20 W sustained draw)
- **Radar signal processing**: 2 W (embedded ARM core, CFAR+clustering)
- **Camera acquisition and I/O**: 3 W (USB 3.0, frame buffering)
- **Miscellaneous (fans, voltage regulators)**: 2 W
- **Total sustained power**: 27 W

For a 24-hour mission at sustained 27 W, total energy consumption is 27 W $\times$ 24 h = 648 Wh. Modern LiPo battery packs for maritime platforms (e.g., 48 V, 5 Ah = 240 Wh nominal) provide multiple cell configurations supporting 600+ Wh energy storage. A 1200 Wh battery (typical for 4–8 hour high-performance missions) sustains labeling pipelines for 48+ hours of continuous operation, or enables shorter intensive missions (4–8 hours) with significant battery margin for propulsion and other systems.

At this sustained rate, a single Jetson Orin NX labels:

- **Per hour**: 1200 frames
- **Per 24-hour mission**: 28,800 labeled frames

- **Per month (continuous operation)**: 864,000 labeled frames

For a maritime object detection dataset, this corresponds to 8–12 hours of annotated video at 1080p@30Hz, a substantial contribution to dataset scale starting from a single platform.

8.7 Maritime Deployment Considerations

8.7.1 Thermal Envelope and Enclosure Design

Maritime environments present severe thermal challenges. Salt spray, humidity, and seawater aerosols corrode unprotected electronics; the Jetson Orin NX must be enclosed in sealed, IP67-rated or higher enclosures to prevent salt ingress and moisture condensation. [155]

However, sealed enclosures dramatically impair passive heat dissipation. A Jetson Orin NX dissipating 20 W in a sealed aluminum enclosure (typical SWaP-optimized maritime box: 200 mm × 150 mm × 80 mm, 2 kg) without active cooling will reach 60–75°C internal temperature depending on ambient seawater temperature. At 25°C ambient, passive conduction through the enclosure to surrounding air may suffice; at 35°C ambient (summer tropical operations), passive dissipation fails and internal temperatures exceed 85°C, approaching thermal throttling.

Active cooling is necessary for sustained 20+ W operation. Options include:

1. **Small axial fans** (40 mm × 40 mm × 10 mm, 1–2 W power): Draw seawater-cooled air through sealed ducts, then exhaust to atmosphere. Requires careful seal design to prevent salt spray ingress while maintaining thermal gradient.

2. **Thermoelectric coolers (TEC)**: Solid-state heat pumps requiring 5–10 W input to transfer 20 W from electronics to a heatsink. More reliable than mechanical fans but lower COP (coefficient of performance) than active air cooling.

3. **Immersion cooling**: Submerge electronics in dielectric oil (3M Fluorinert, etc.), with an external radiator on hull. Simplifies sealing and thermal management but adds weight and complexity.

The Mayflower Autonomous Ship, a long-endurance ocean research platform, employed sealed aluminum enclosures with integrated thermoelectric cooling and external radiators, maintaining 30–40°C internal temperature during North Atlantic operations. [156] This architecture (cooled enclosure, 1500 Wh battery, on-deck sensor integration) represents a validated baseline for maritime AI systems.

8.7.2 Electrical Integration and 24 V Power Bus

Most research and commercial maritime platforms operate on unregulated 24 V DC bus voltage, with ripple and transient excursions to 20–30 V depending on propulsion load switching. The Jetson Orin NX requires stable 5 V input (via onboard DC-DC converter). Power conversion involves:

- **24 V DC-DC converter** (isolated buck topology): 24 V → 12 V, 10 A (120 W peak), 93–95% efficiency

- **12 V to 5 V secondary converter**: 12 V → 5 V, 8 A, 94% efficiency
- **Total conversion loss**: 5–8 W at sustained 20 W draw, raising effective power consumption to 25–28 W from the 24 V bus.

24 V bus ripple (±2 V at 10 kHz typical for PWM propulsion ESCs) must be attenuated via LC filters or isolation transformers to prevent GPU clock jitter and thermal cycling. Many maritime systems integrate dedicated 24 V supply modules with integrated noise filtering, simplifying integration.

8.7.3 Salt Spray and Corrosion Mitigation

The Jetson Orin reference design (baseboard PCB, Aluminum heatspreader, standard connectors) is *not* rated for salt spray environments and will corrode within weeks of exposure to marine salt fog. Deployment requires:

1. **Conformal coating**: PCB-level polyurethane or parylene coating (5–25 m thickness) protects traces, solder joints, and exposed copper from electrolytic corrosion.
2. **Connector protection**: Use stainless steel M12 or circular connectors (IP67+ rated, corrosion-resistant materials) instead of standard USB, HDMI, or 40-pin GPIO.
3. **Enclosure materials**: Aluminum, titanium, or composite materials (aramid, carbon fiber) with hard-anodized finishes. Avoid bare steel and unprotected copper.
4. **Periodic maintenance**: Sealed systems should be opened and inspected annually; desiccant packs replaced to prevent moisture accumulation.

Commercial maritime edge computing systems (e.g., Lanner MiCom series) [155] provide pre-hardened Jetson integration with mil-spec connectors, conformal coating, and tested IP67 sealing—eliminating custom integration risk.

8.8 Summary and Research Gaps

The deployment of radar-guided camera labeling on NVIDIA Jetson edge platforms represents a significant advancement in autonomous dataset construction methodology. The analysis demonstrates three key contributions:

1. **Computational Feasibility**: YOLO-family detectors combined with TensorRT INT8 quantization enable 1000–1500 labeled frames per hour on a 20 W Jetson Orin NX platform, sufficient to generate substantial maritime object detection datasets (28,800+ frames per 24-hour mission) without shore-based post-processing.
2. **Power-Constrained Optimization**: The recognition that labeling is offline and asynchronous (not real-time critical) decouples system design from hard real-time constraints, enabling aggressive batching and throughput optimization at the cost of latency. A sustained power budget of 25–27 W is achievable and practical for small uncrewed surface vehicles equipped with battery packs $\geq$ 600 Wh.

3. **Maritime Suitability**: Active thermal management, electrical integration on 24 V power buses, salt-spray hardening, and sealed enclosure design are now mature technologies validated on operational platforms (Mayflower, etc.). Edge deployment aboard maritime platforms is operationally feasible with careful engineering.

However, significant research gaps remain:

- **Knowledge Distillation for Efficient Maritime Models**: [157, 158] No published work systematically optimizes YOLOv8 or comparable detectors for the maritime domain while maintaining sub-20W edge deployment. Research should investigate knowledge distillation techniques wherein large models trained on diverse maritime datasets (WaterScenes, MOANA, etc.) transfer knowledge to tiny models (< 2 M parameters) viable on Jetson Nano or ARM-based edge devices.
- **Efficient Radar-Camera Fusion at Edge**: [159, 160] Current fusion architectures (Chapter 6) employ late-fusion concatenation, treating radar and camera feature maps independently before combining predictions. Early fusion or adaptive fusion strategies optimized for edge platforms could reduce model size and memory footprint while improving detection robustness.
- **Real-Time Operating System Integration**: [161] Jetson platforms run full Ubuntu Linux OS, introducing unpredictable latencies from kernel scheduling, garbage collection, and I/O interrupts. Real-time operating system (RTOS) ports (e.g., FreeRTOS, VxWorks) with hard real-time guarantees would benefit applications requiring synchronized acquisition from multiple sensors and time-stamped labeling.
- **Maritime Dataset-Specific Optimization**: [162] No published quantization or compression benchmarks exist for YOLOv8 models fine-tuned on maritime data specifically. Characterizing quantization robustness on maritime object distributions (ships appear small and at ocean-clutter backgrounds) is essential for confident deployment.
- **Field Validation on Autonomous Platforms**: Long-duration at-sea trials of edge labeling systems aboard operational uncrewed surface vehicles are absent from literature. Validation against manual annotations, characterization of labeling accuracy degradation under sea state variation, and throughput analysis on real-time mission loads are critical gaps.

The convergence of efficient neural networks, specialized edge hardware, and mature maritime systems integration now makes autonomous dataset construction feasible. Future work should focus on closing domain-specific gaps and validating performance in operational maritime environments.

9 Conclusion: Summary of Research Gaps

This review has surveyed the literature across seven technical domains that collectively constitute the research landscape for radar-guided camera labeling in maritime environments: maritime object detection datasets and the annotation gap (Section 2), radar-camera sensor registration and calibration (Section 3), radar-to-camera projection and bounding box generation (Section 4), automated and weakly-supervised dataset construction (Section 5), radar-camera fusion architectures (Section 6), X-band marine radar signal processing (Section 7), and edge computing for onboard deployment (Section 8). Within each domain, the existing literature provides a mature methodological foundation—but one developed almost exclusively for the automotive setting. The maritime domain, with its distinct sensor characteristics, environmental conditions, and operational ranges, presents challenges that current methods do not adequately address. Four cross-cutting research gaps emerge from this synthesis.

X-band marine radar versus 77 GHz automotive radar transfer. The radar-camera fusion architectures surveyed in Sections 6 and 7 reveal a fundamental data format mismatch between the automotive and maritime domains. State-of-the-art automotive methods—BEVFusion [75], RCBEVDet [76], CenterFusion [72], and CRAFT [80]—are architected around sparse point cloud inputs from 77 GHz electronically steered radar, with features including per-point Doppler velocity, 3D position estimates, and update rates of 10–20 Hz. X-band marine radar produces fundamentally different data: two-dimensional PPI images assembled by mechanical antenna rotation at 0.5–3 s periods, with range resolutions of 5–30 m, no elevation measurement, and limited Doppler capability [13, 68]. The BEV grid construction, frustum-based association, and Doppler-conditioned feature extraction that underpin automotive architectures cannot be directly applied to PPI image data. Adapting these methods requires not merely swapping encoder backbones but rethinking the representation pipeline from radar image to shared feature space. The signal processing chain reviewed in Section 7—CFAR detection, sea clutter modeling with K-distribution and Pareto statistics, and spatio-temporal clustering—further demonstrates that the maritime radar processing pipeline introduces domain-specific noise characteristics (sea clutter false alarms, multipath-induced amplitude fluctuations, rain clutter) that automotive fusion architectures are not designed to handle.

Projection accuracy at maritime ranges. The calibration and projection literature reviewed in Sections 3 and 4 establishes that the mathematical framework for radar-to-camera projection is well understood—the rigid-body transformation in $SE(3)$, pinhole projection with distortion correction, and coordinate chain from polar to pixel coordinates are all standard [16, 51]. However, the error analysis at maritime operating ranges of 500 m to 10 km reveals that every source of inaccuracy in this chain is amplified relative to the automotive regime. A sub-degree azimuth calibration error produces single-digit pixel displacement at 50 m but tens to hundreds of pixels at multi-kilometer range. Wide field-of-view cameras, necessary for panoramic maritime coverage, introduce nonlinear distortion that the Kannala–Brandt [17] and Scaramuzza [18] models can characterize but whose interaction with range-dependent bounding box sizing has not been studied. Most critically, the absence of elevation information in conventional 2D marine radar creates a vertical ambiguity that has no satisfactory resolution: the projected bounding box height must be assumed rather than measured, and the assumption degrades with range as the angular subtent of targets shrinks. No published work provides a systematic range-dependent error characterization for X-band radar-camera projection, and no validated bounding box sizing strategy exists for maritime targets at extended range.

Auto-labeling scalability and noise management. The automated dataset construction literature surveyed in Section 5 offers a rich toolkit for learning from imperfect supervision—data programming [20, 88], pseudo-labeling and self-training [21, 22, 90], teacher-student frameworks [23], co-training [24], and noisy label learning [19, 25, 27].

Each of these paradigms can, in principle, be applied to radar-generated camera labels. However, the existing methods assume that label noise is either class-conditional (symmetric or asymmetric noise transition matrices) or instance-dependent but statistically independent across samples [28]. Radar-projected labels violate this independence assumption: calibration drift biases all labels in the same direction across an entire recording session, sea clutter produces spatially clustered false positives, and beam spreading causes correlated localization errors for targets at similar ranges. The AISail system [95] demonstrates that cross-modal maritime labeling using AIS data is feasible, but radar-to-camera label transfer—which covers the broader class of non-transponder-equipped targets—has not been systematically studied. The interaction between systematic label noise and iterative self-training dynamics, where errors may compound rather than average out across training rounds, remains poorly understood.

Systematic versus random label noise. The most consequential gap identified across the preceding sections is the absence of a noise model that captures the geometry-dependent, spatially correlated error structure of radar-projected labels. The noisy label learning literature has progressed from symmetric noise [27] through asymmetric noise [25] to instance-dependent noise [28], but none of these parameterizes noise as a function of sensor geometry. For radar-guided labeling, the dominant error sources are deterministic functions of the radar-camera configuration: azimuth quantization error scales with the radar's angular resolution; bounding box center displacement scales linearly with target range for a fixed calibration offset; bounding box size estimation degrades as the angular subtent decreases with range; and sea clutter false alarm density varies with sea state, range, and grazing angle [12]. A geometry-aware noise transition model—one that conditions the noise distribution on target range, bearing, radar scan parameters, and calibration state—would provide the theoretical foundation for principled noise correction. Developing and validating such a model is a prerequisite for deploying radar-guided labeling systems that can produce training data of sufficient quality to approach manually annotated benchmarks.

Taken together, these four gaps frame radar-guided camera labeling as a promising but underexplored path toward closing the maritime annotation deficit documented in Section 2. The approximately 300-fold gap in multi-sensor scene diversity and 66-fold gap in 3D bounding box labels between automotive and maritime datasets cannot be closed by manual annotation alone—the cost and logistics are prohibitive. A radar-guided labeling pipeline that operates continuously aboard autonomous surface vessels, leveraging the edge computing platforms evaluated in Section 8 to process 1,000–1,500 frames per hour at under 25 W sustained power, offers a scalable alternative. Realizing this vision requires advancing the state of the art along each of the four gap dimensions identified above: developing calibration and fusion methods for X-band marine radar, characterizing and mitigating projection errors at maritime ranges, designing noise-robust training pipelines for geometry-dependent label noise, and validating the complete system through field trials on operational maritime platforms.

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